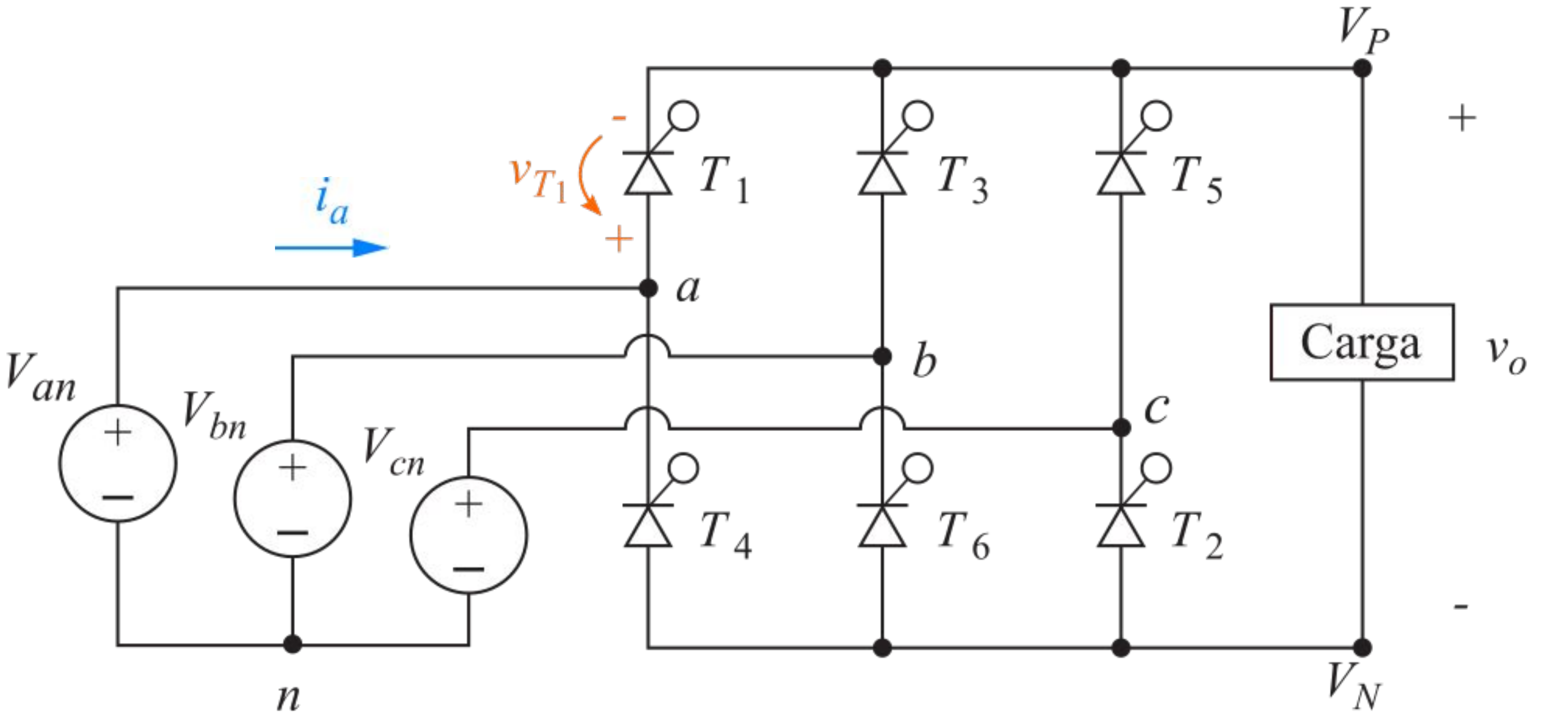
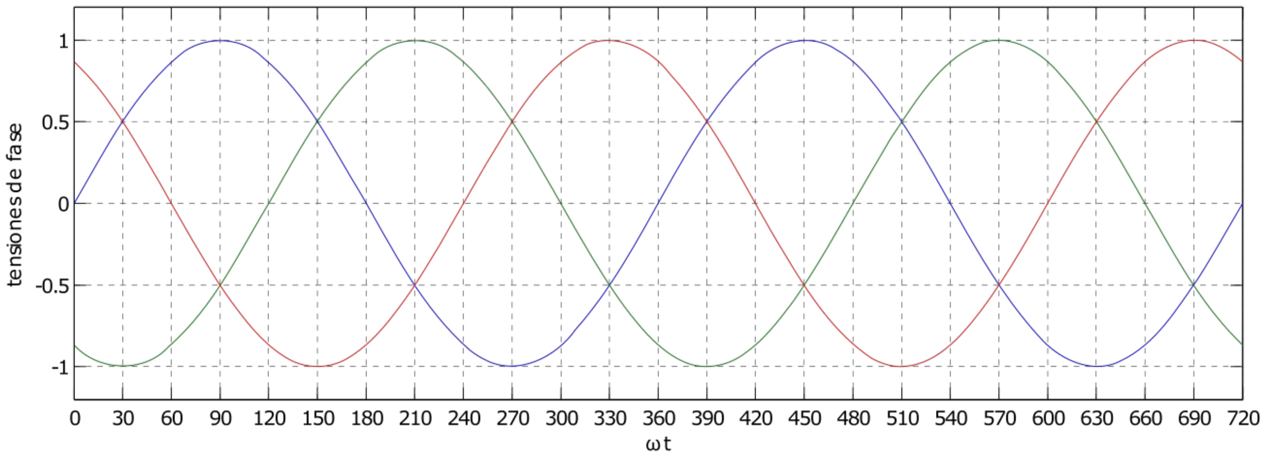
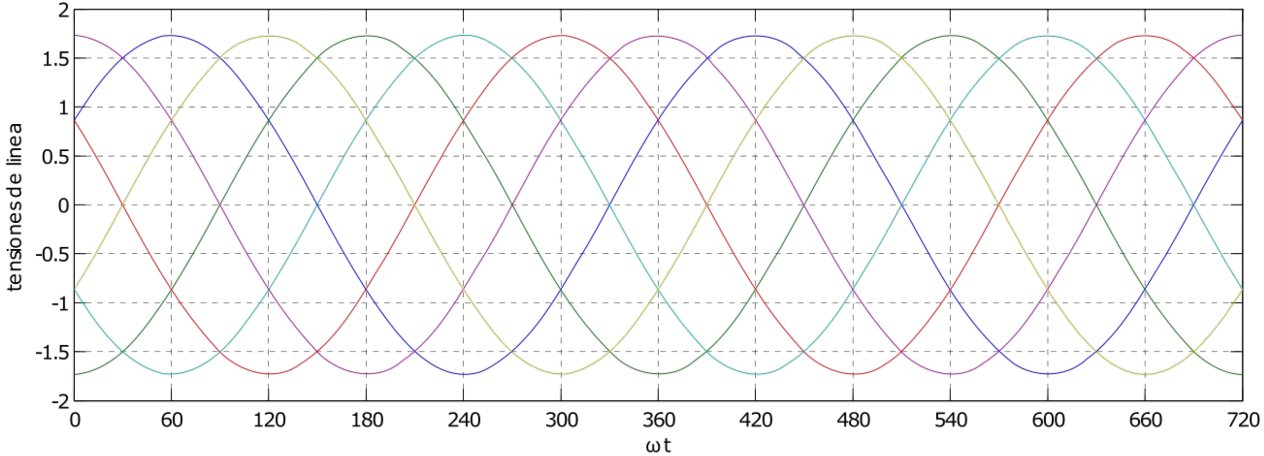
Conversión CA/CC

Rectificación Controlada
Trifásica

Rectificador Controlado Trifásico




tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt


tensiones de linea
2
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1
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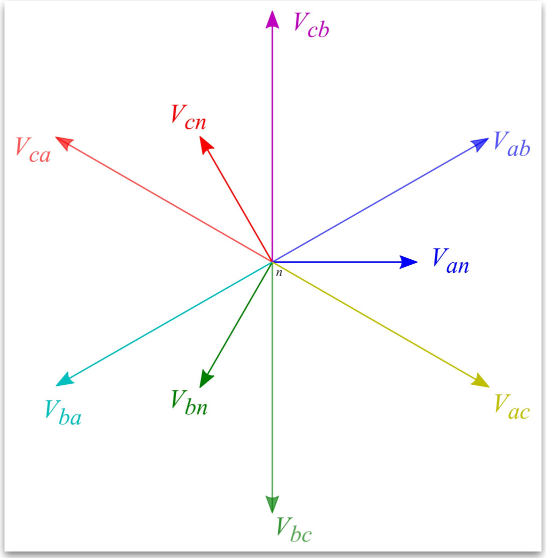
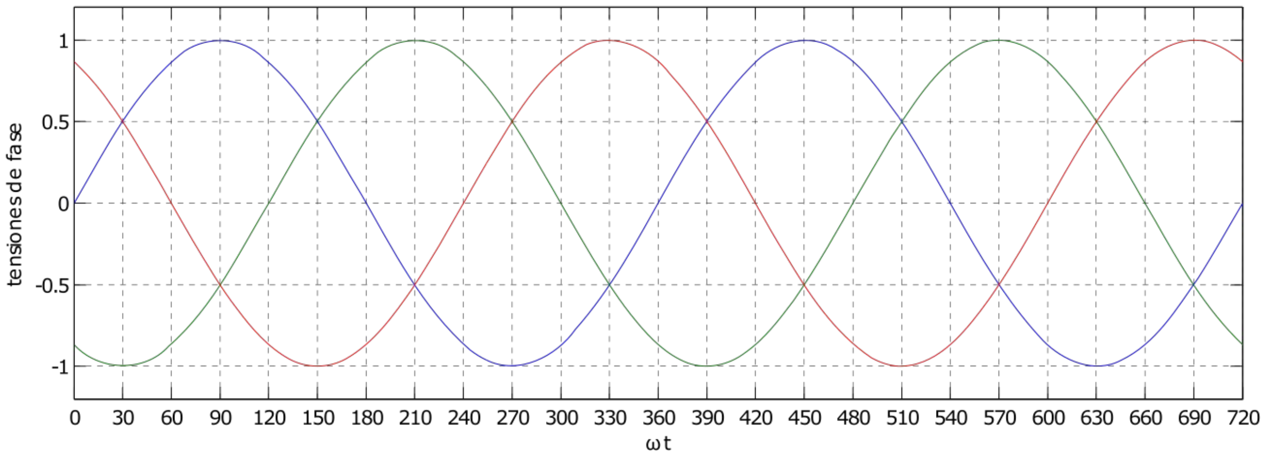
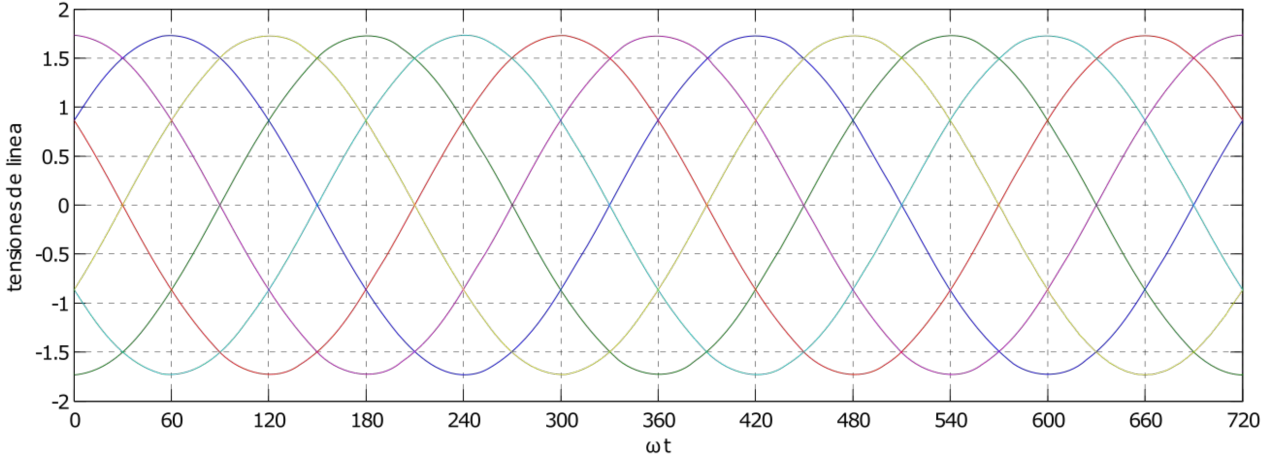
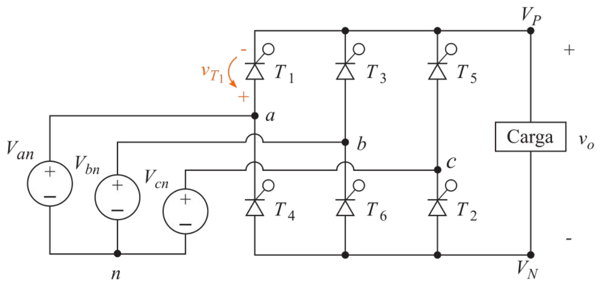
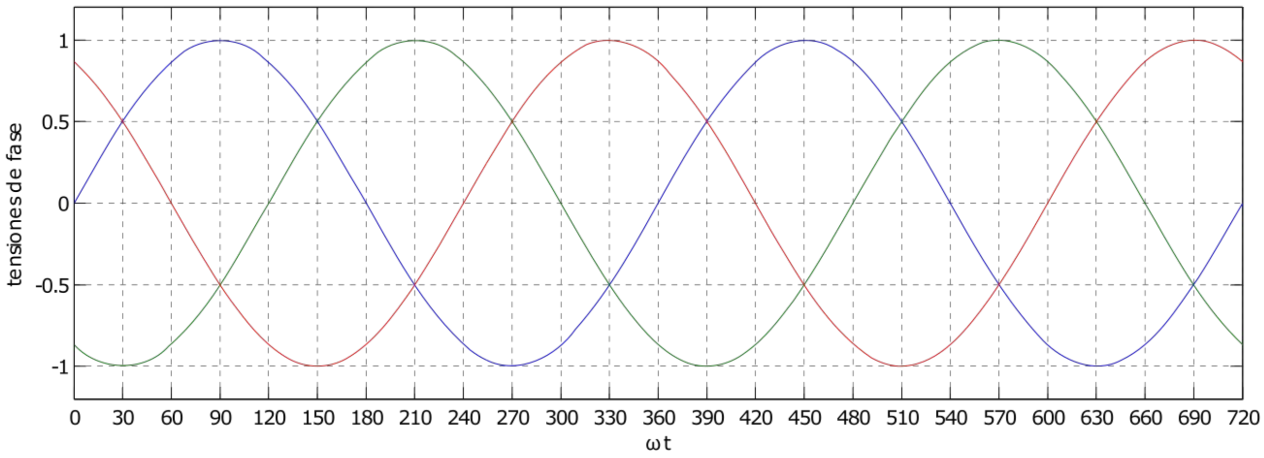
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt


V_{cb}
V_{cn}
V_{ca}
V_{ab}
V_{an}
n
V_{ba}
V_{bn}
V_{ac}
V_{bc}

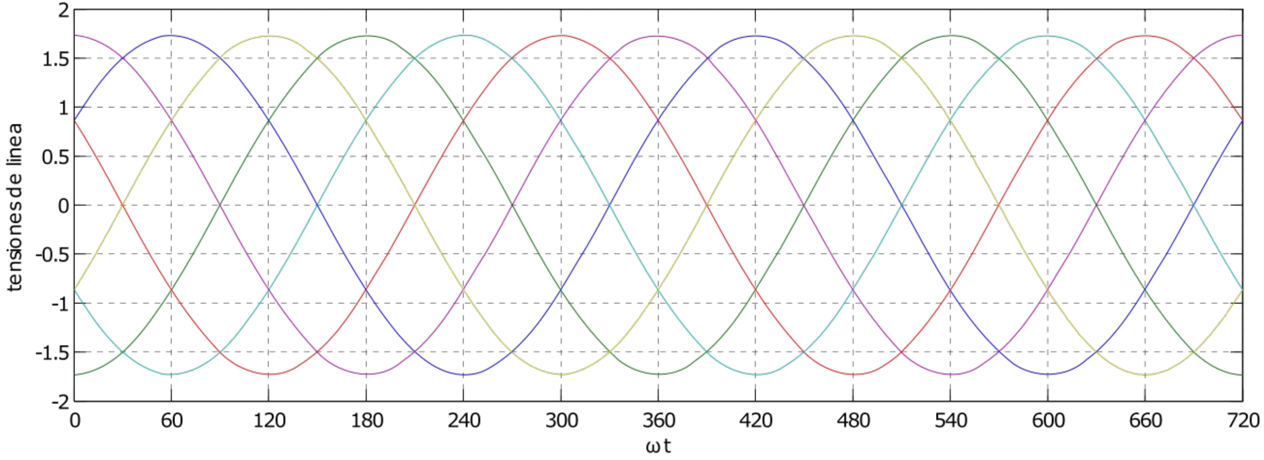

tensiones de fase
ωt
0
30
60
90
120
150
180
210
240
270
300
330
360
390
420
450
480
510
540
570
600
630
660
690
720
1
0.5
0
-0.5
-1


tensiones de linea
ωt
0
60
120
180
240
300
360
420
480
540
600
660
720
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2

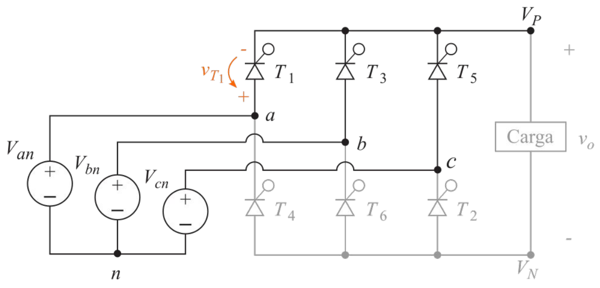

V_P
V_N
v_{T_1}
T_1
T_3
T_5
T_4
T_6
T_2
a
b
c
n
V_{an}
V_{bn}
V_{cn}
Carga
v_o



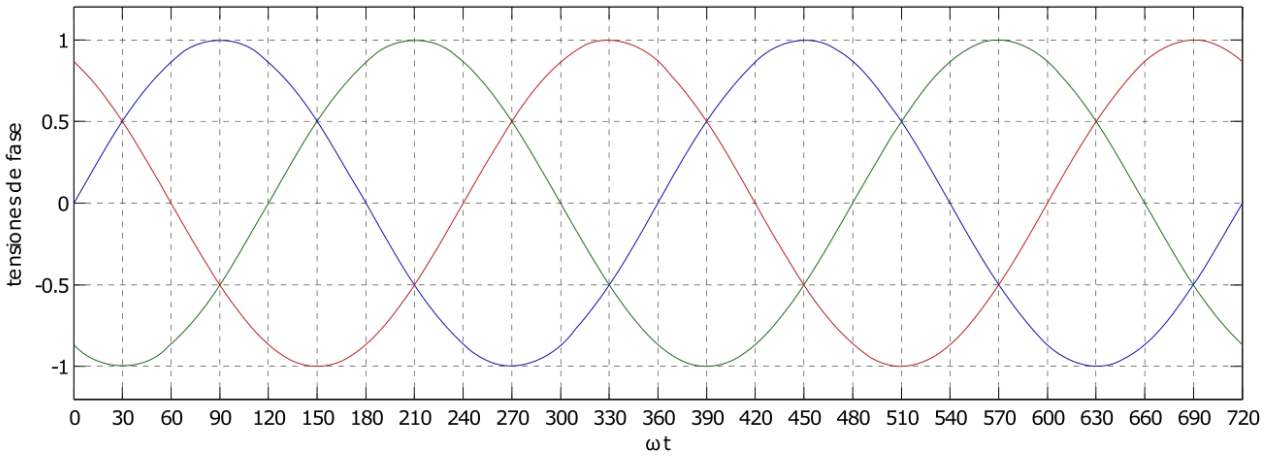
tensiones de fase
ωt



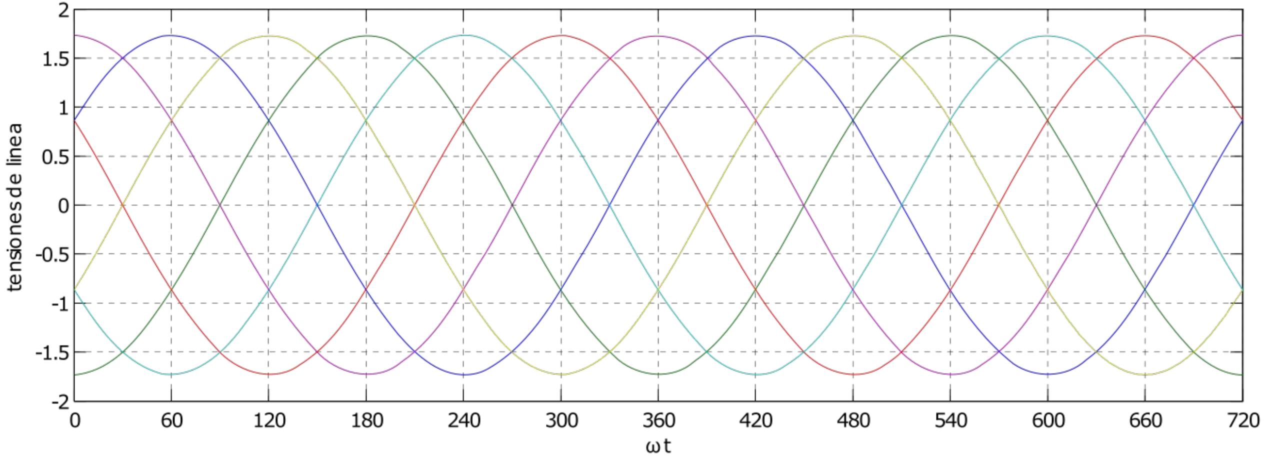
tensiones de linea
ωt



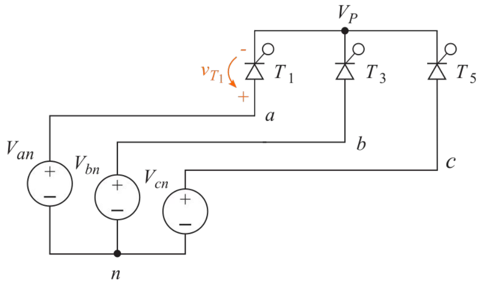
V_P
T_1
T_3
T_5
v_{T_1}
a
b
c
V_{an}
V_{bn}
V_{cn}
Carga
v_o
T_4
T_6
T_2
n
V_N



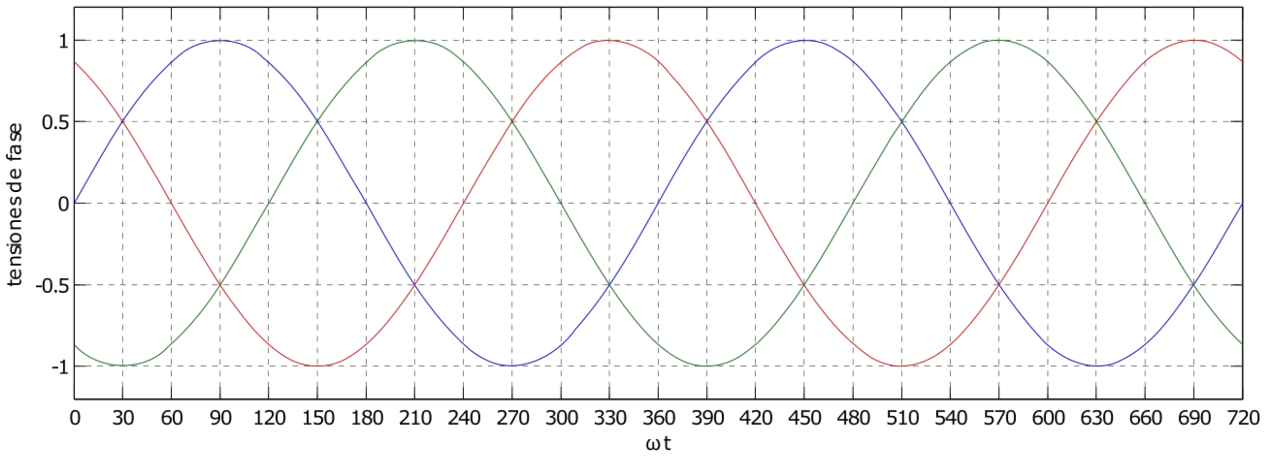
tensiones de fase
ωt

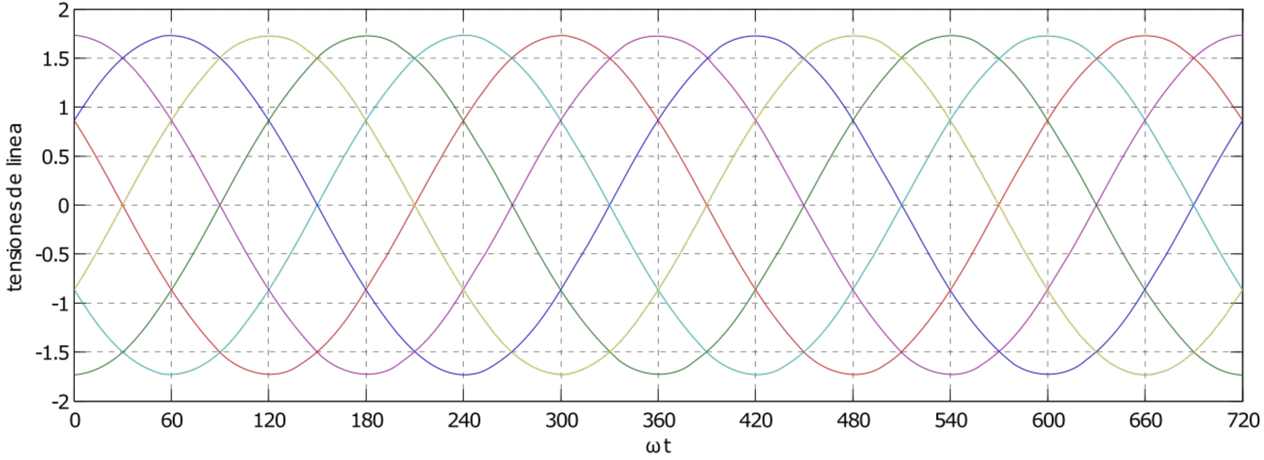


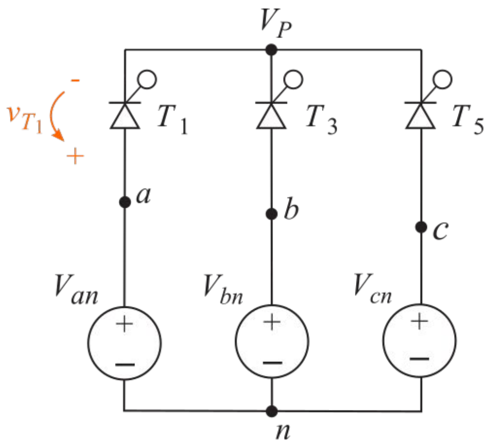
tensiones de linea
ωt

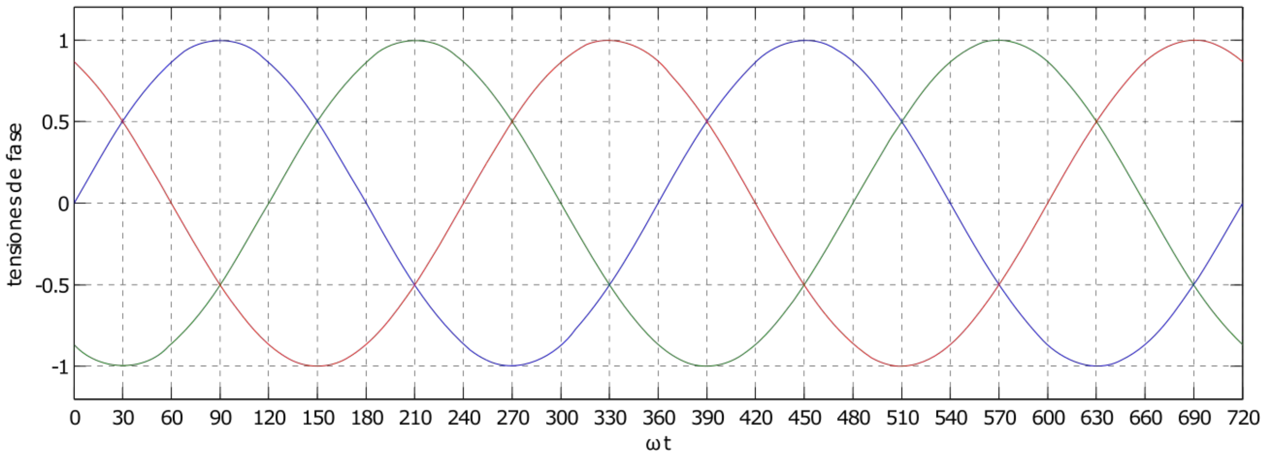


V_P
v_{T_1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
n

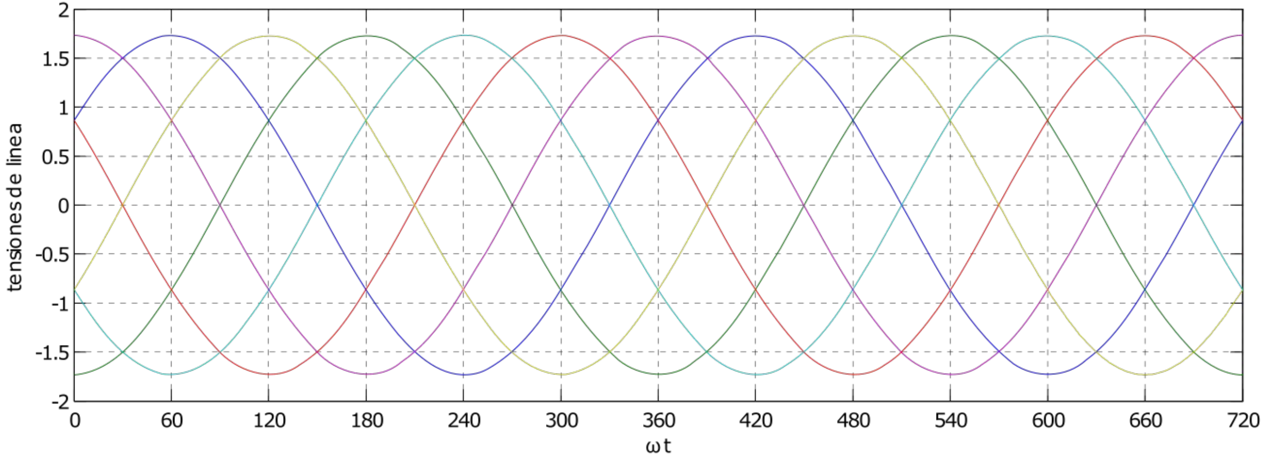

tensiones de fase
ωt


tensiones de linea
ωt

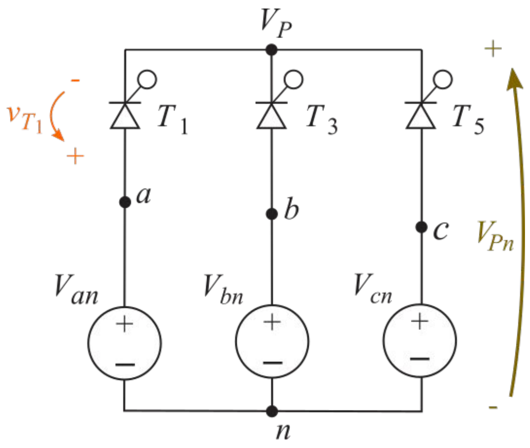

V_P
v_{T1}
-
+
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
n



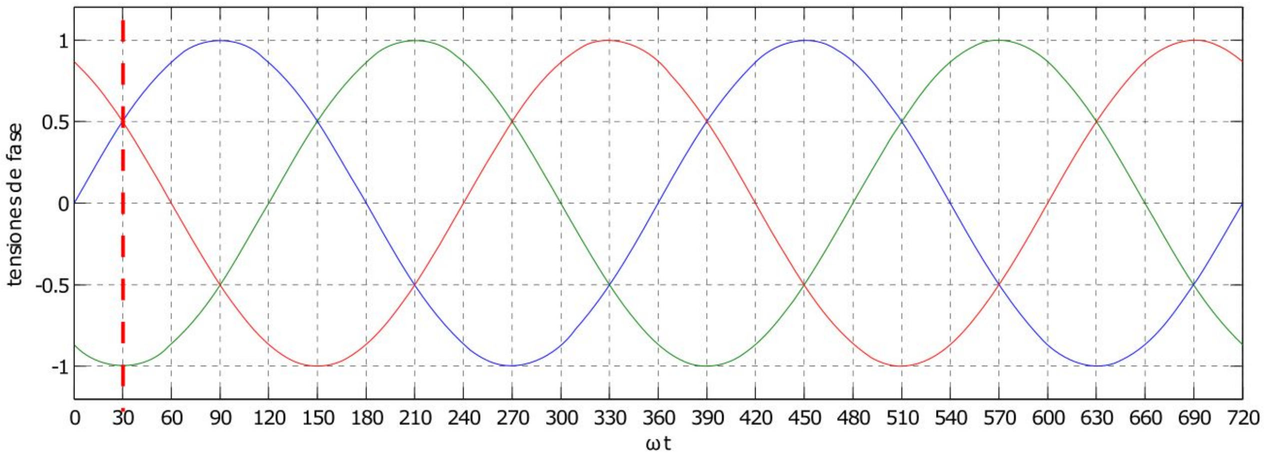
tensiones de fase
ωt

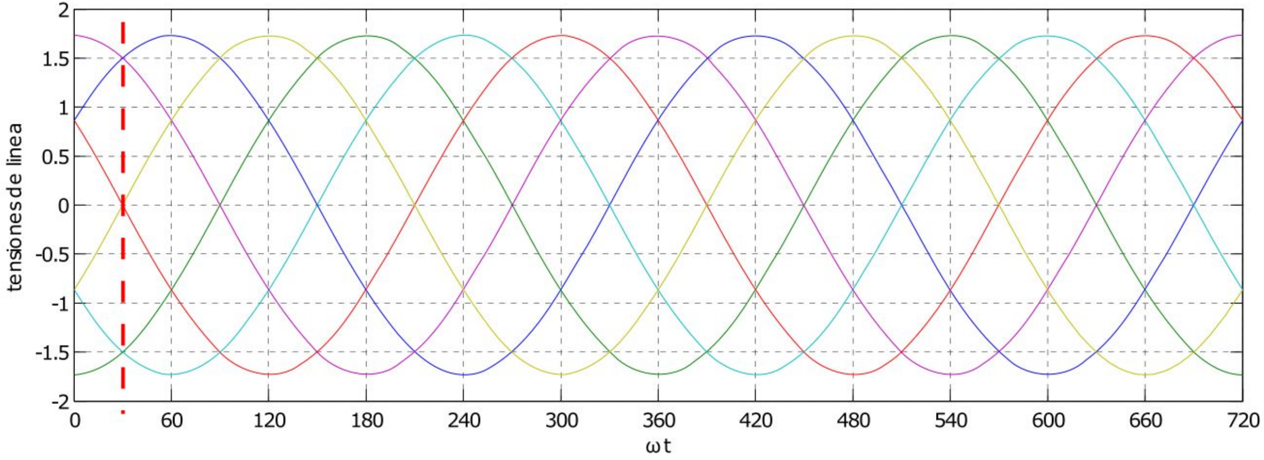


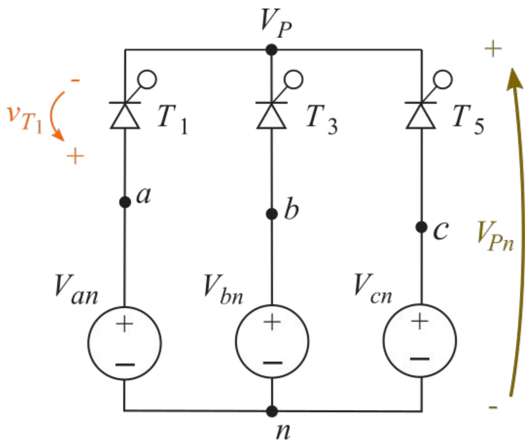
tensiones de linea
ωt

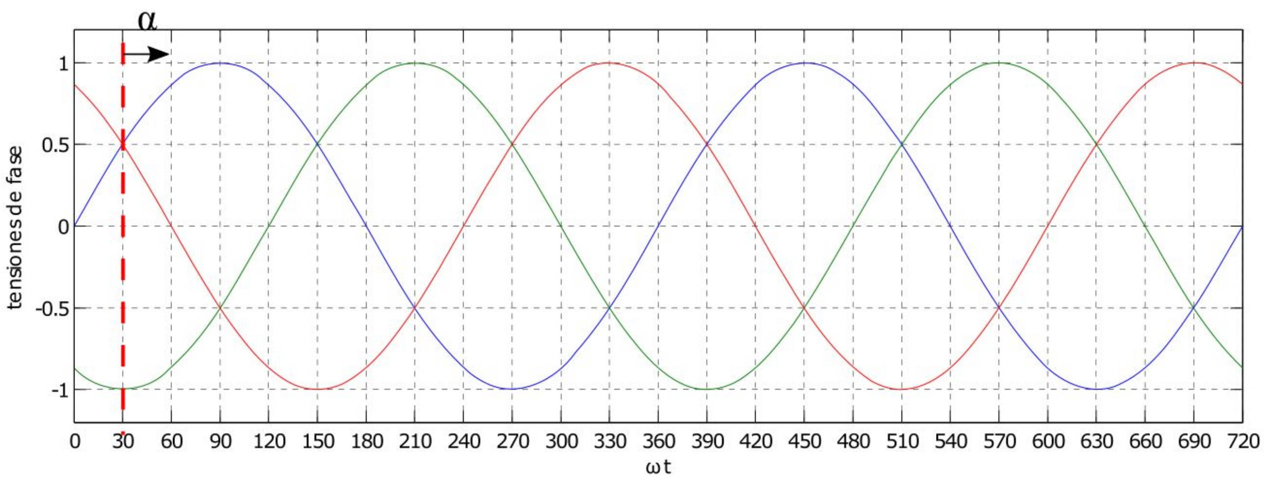


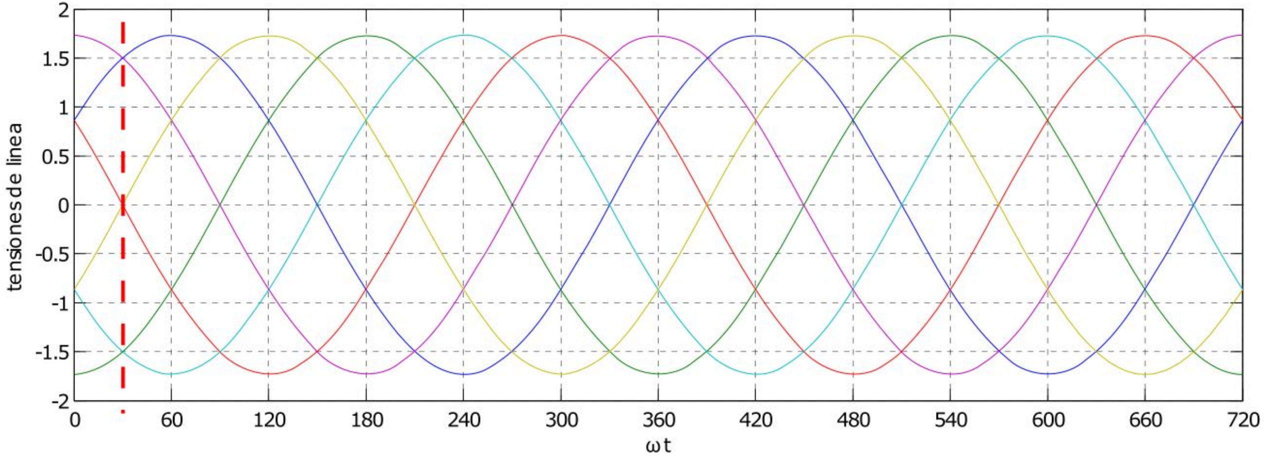
V_P
T_1
T_3
T_5
v_{T1}
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n

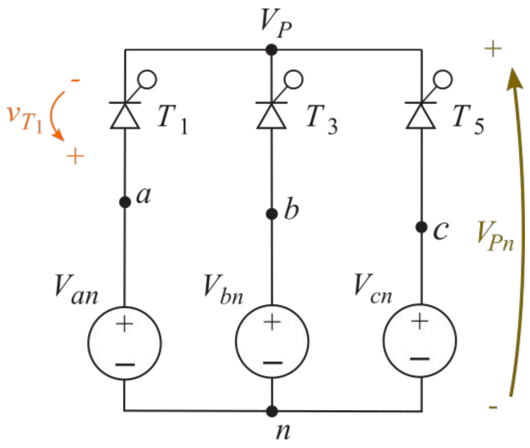

tensiones de fase
ωt

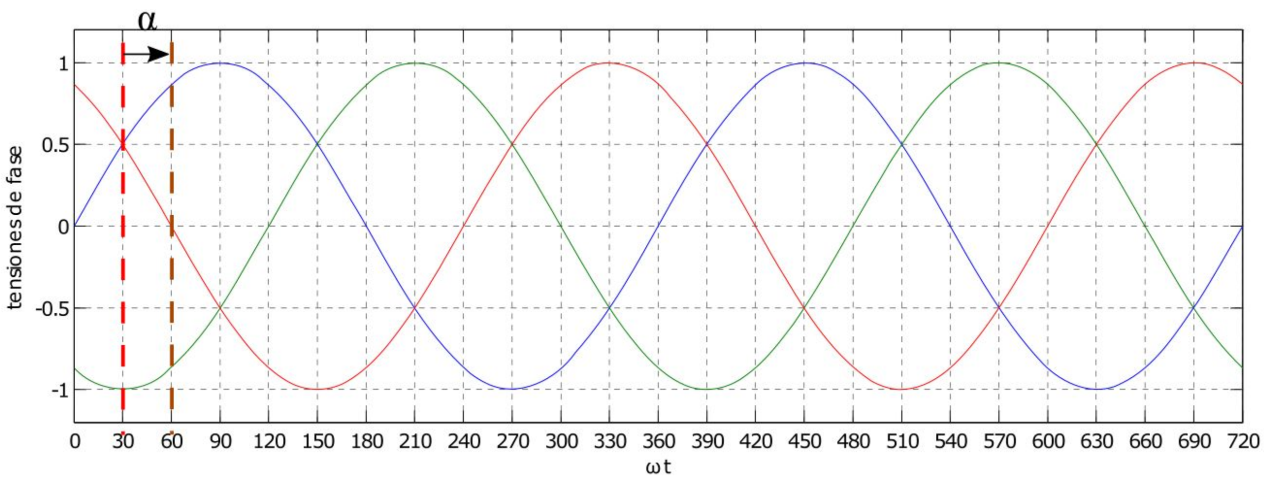

tensiones de linea
ωt


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n

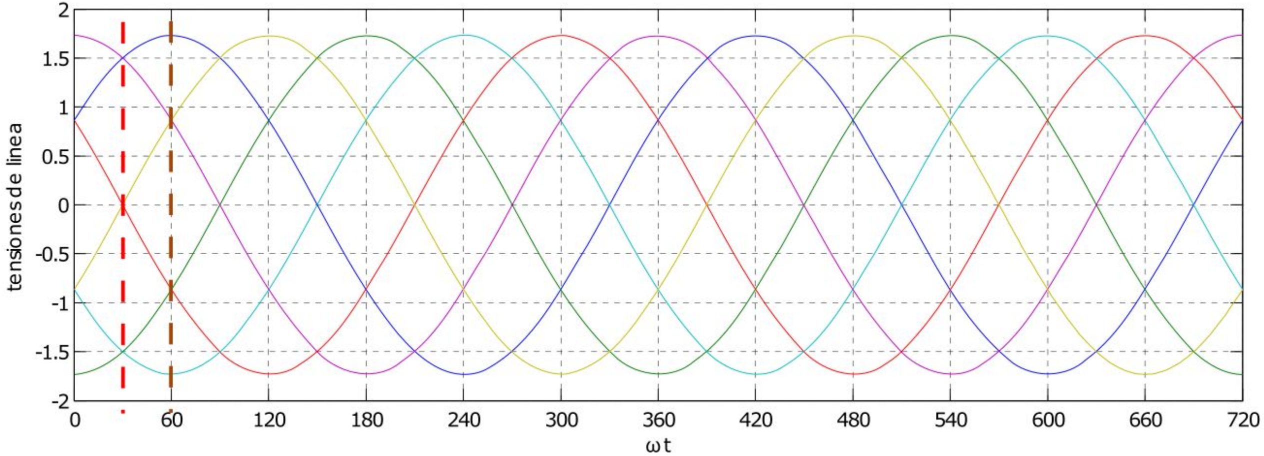

α
tensiones de fase
ωt


tensiones de linea
ωt

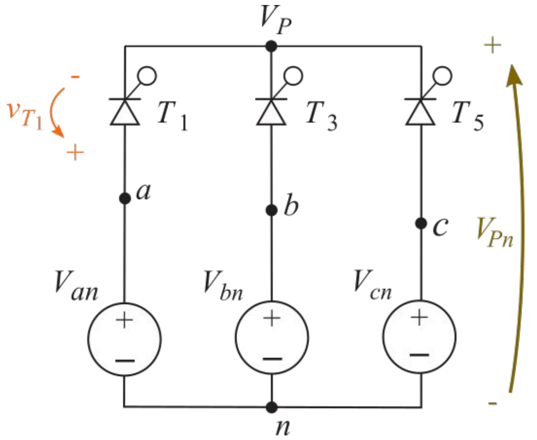

V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n



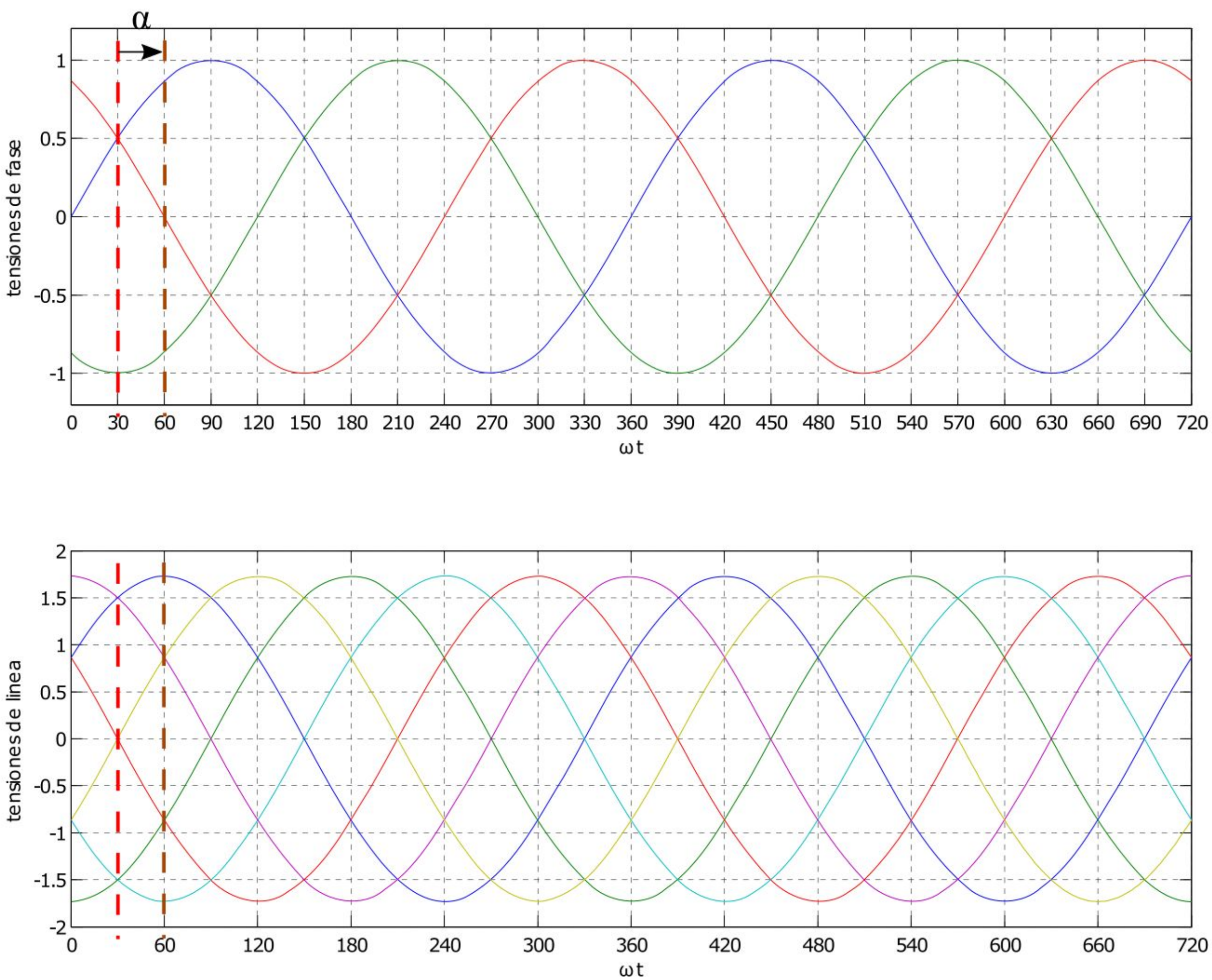
α
tensiones de fase
ωt

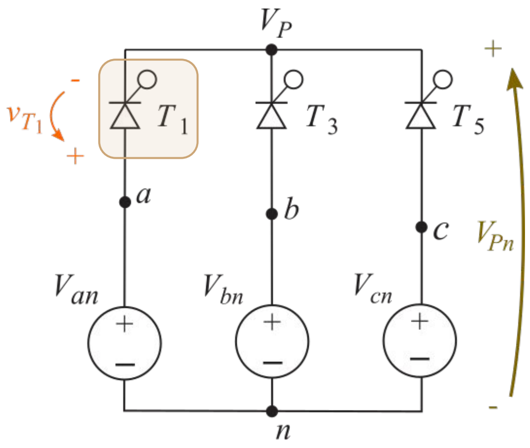


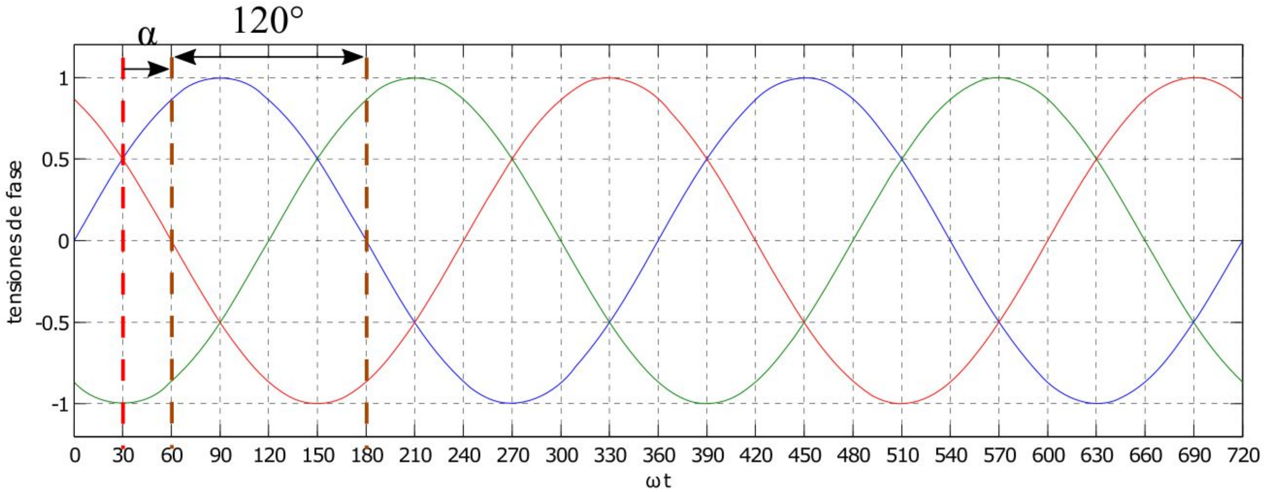
tensiones de linea
ωt



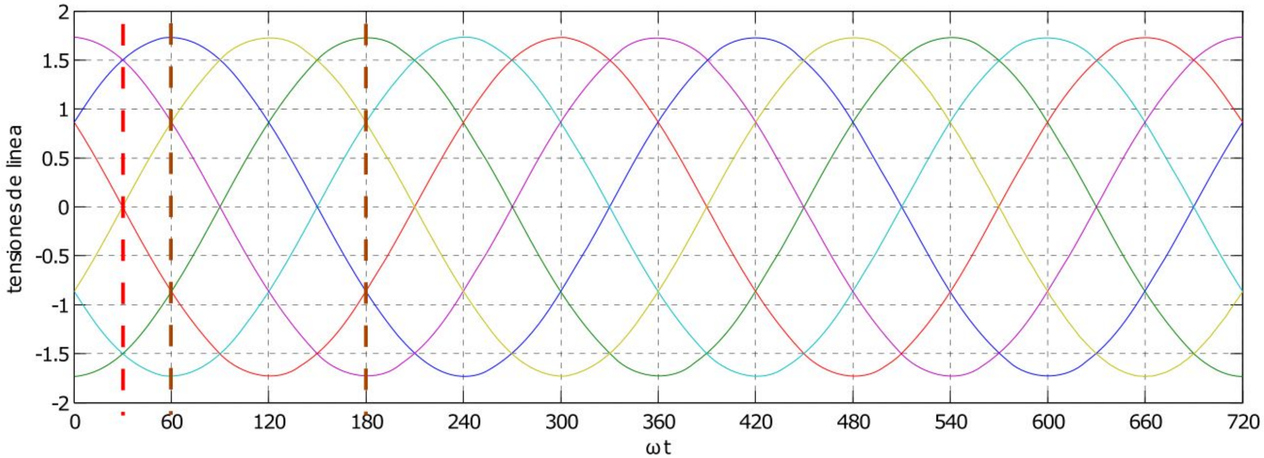
V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n


α
tensiones de fase
ωt
tensiones de linea
ωt

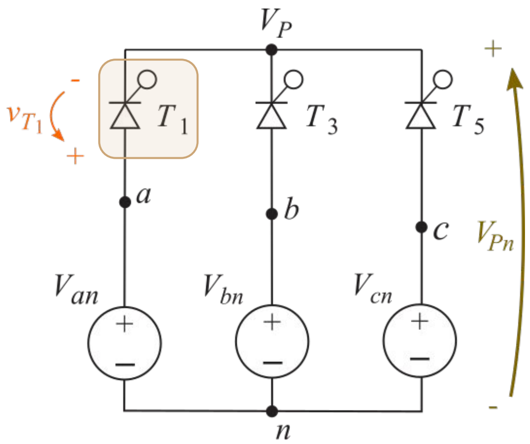

V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n



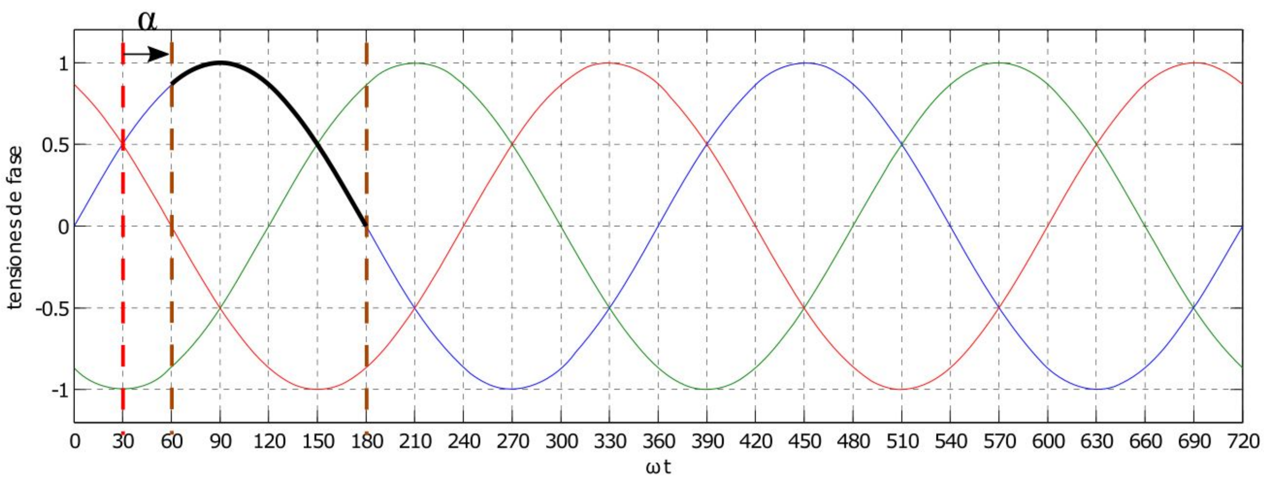
α
120°
tensiones de fase
ωt

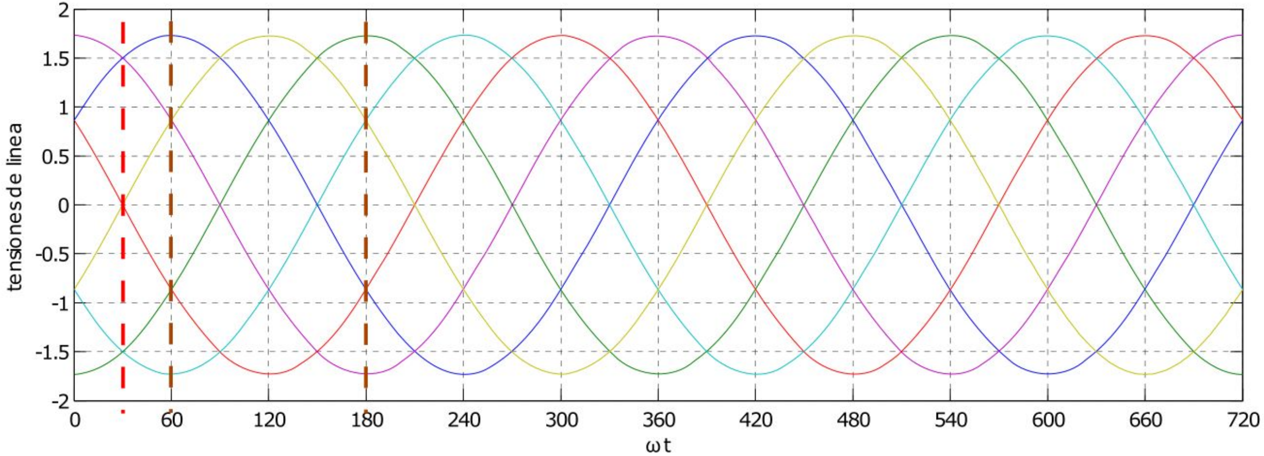


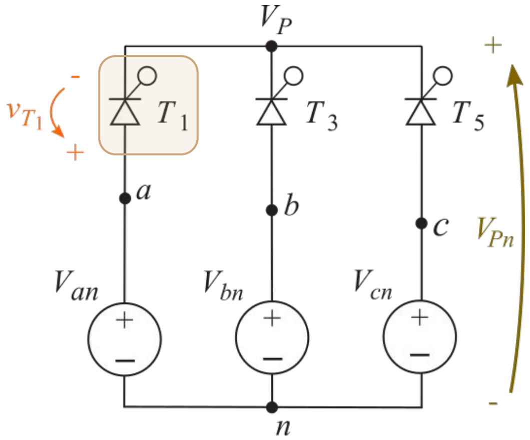
tensiones de linea
ωt

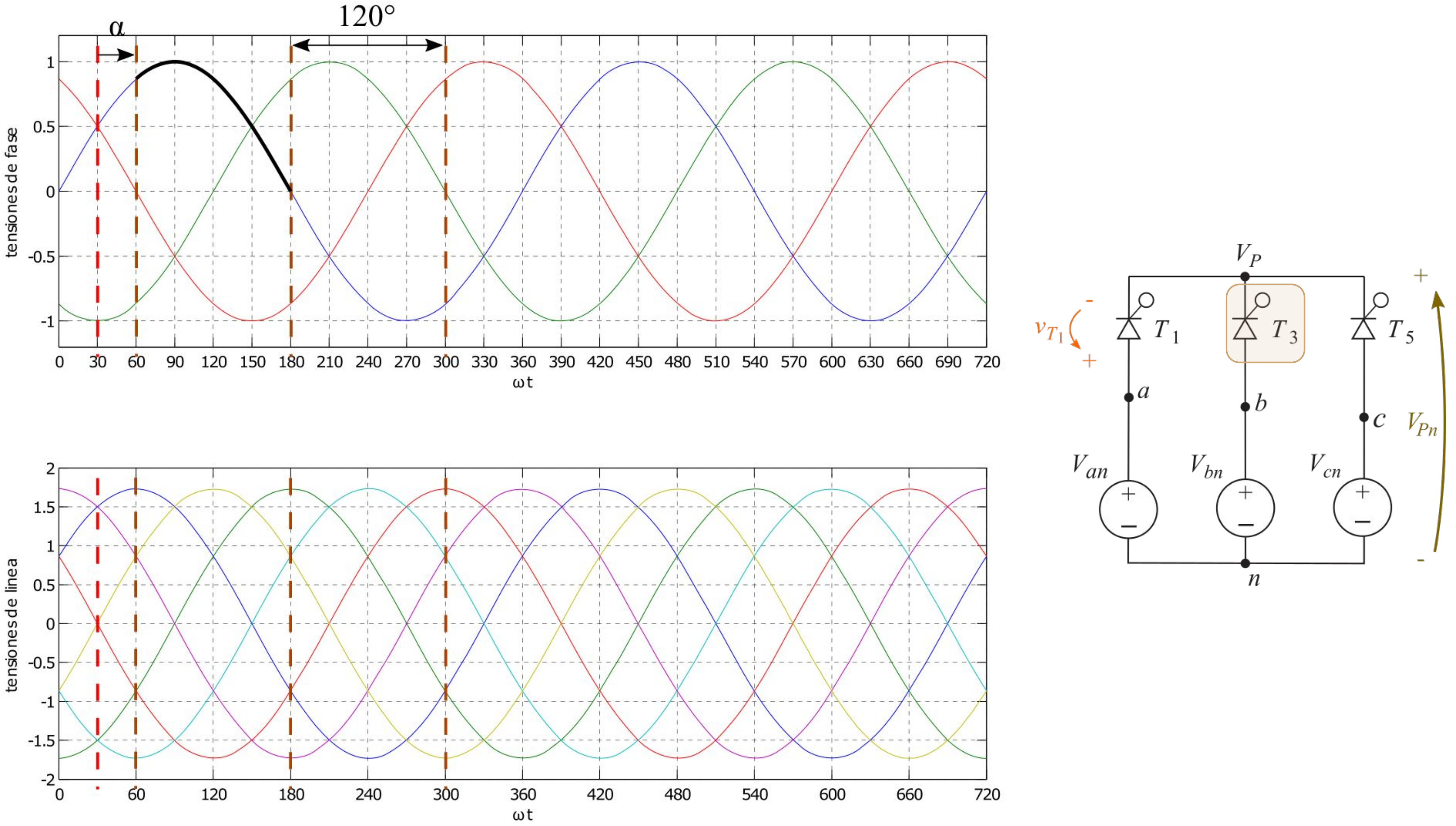


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n

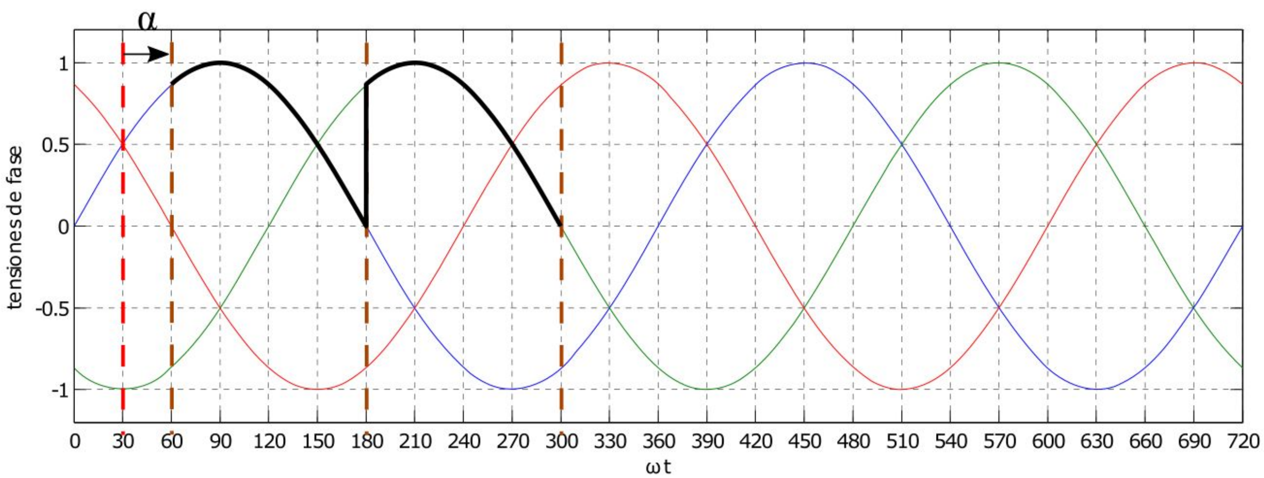

α
tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt

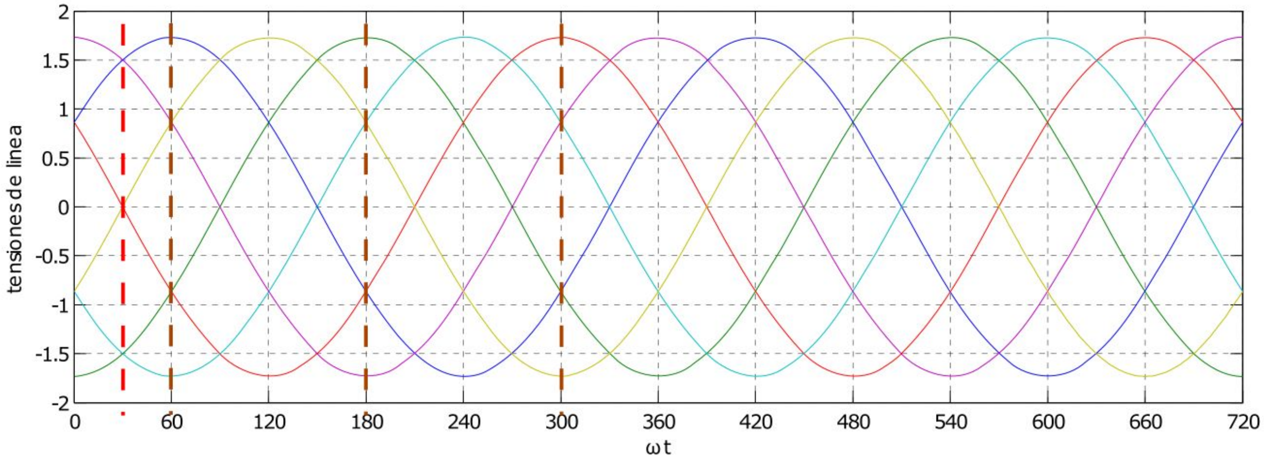

tensiones de linea
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt

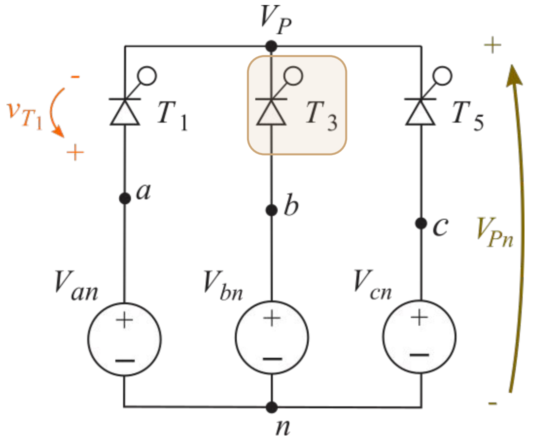

V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
V_{Pn}
n

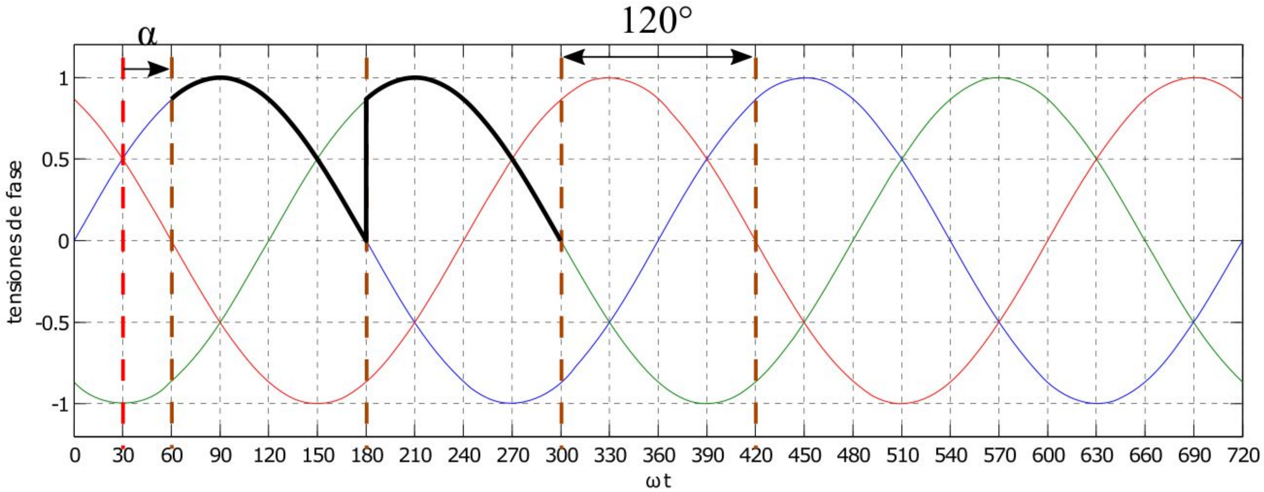


α
120°
tensiones de fase
ωt
tensiones de linea
ωt
V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n


α
tensiones de fase
ωt


tensiones de linea
ωt


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n


α
120°
tensiones de fase
ωt
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
1
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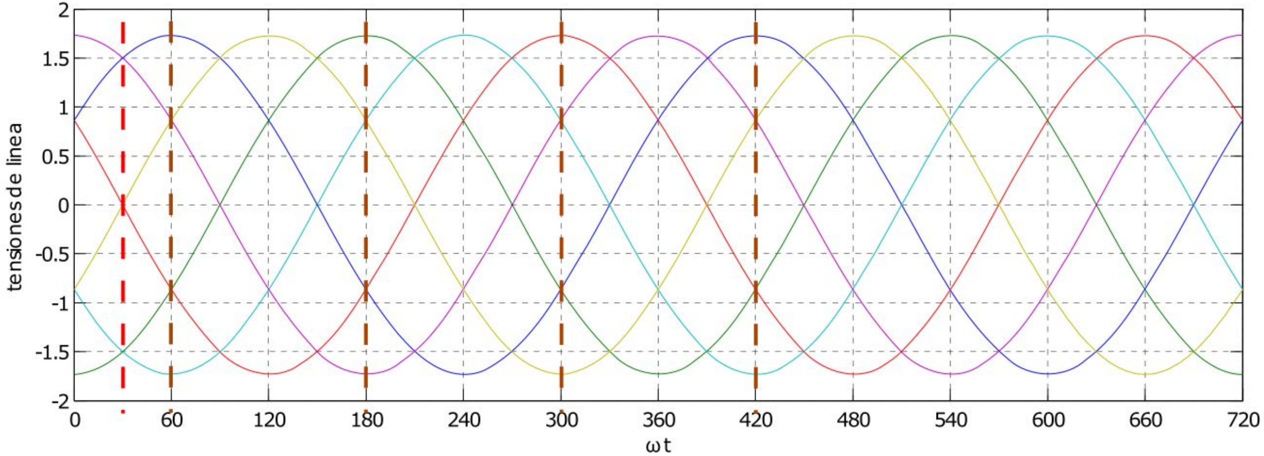
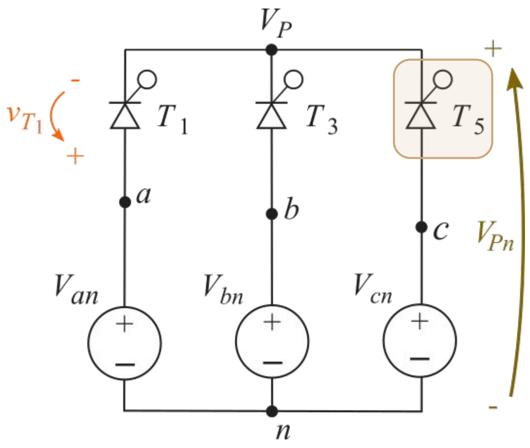
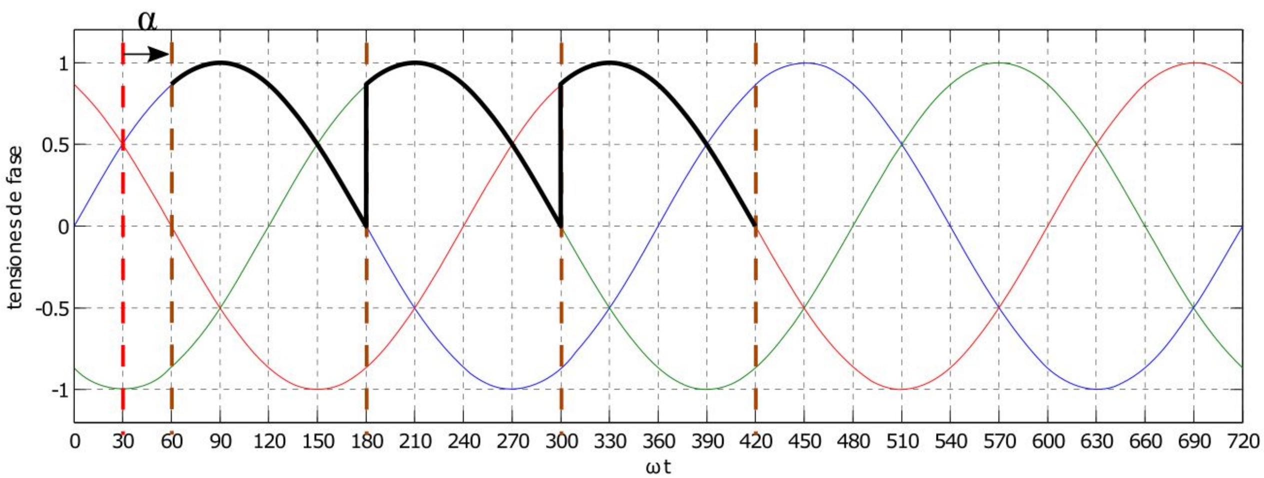
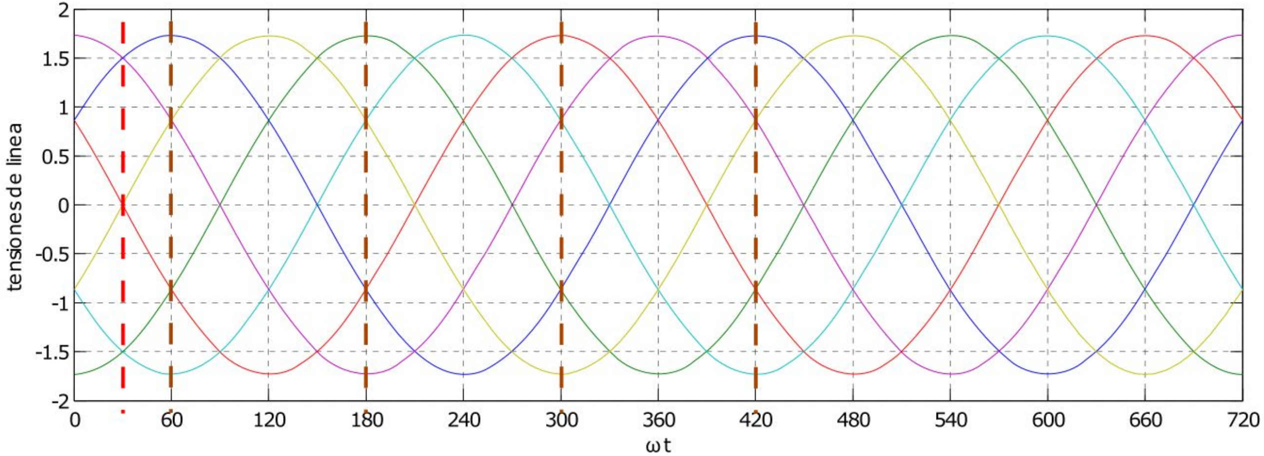
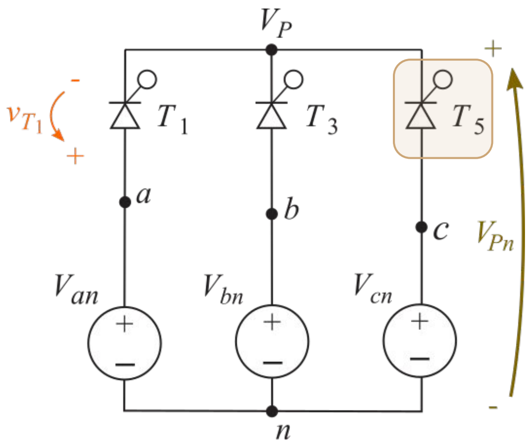
0
-0.5
-1

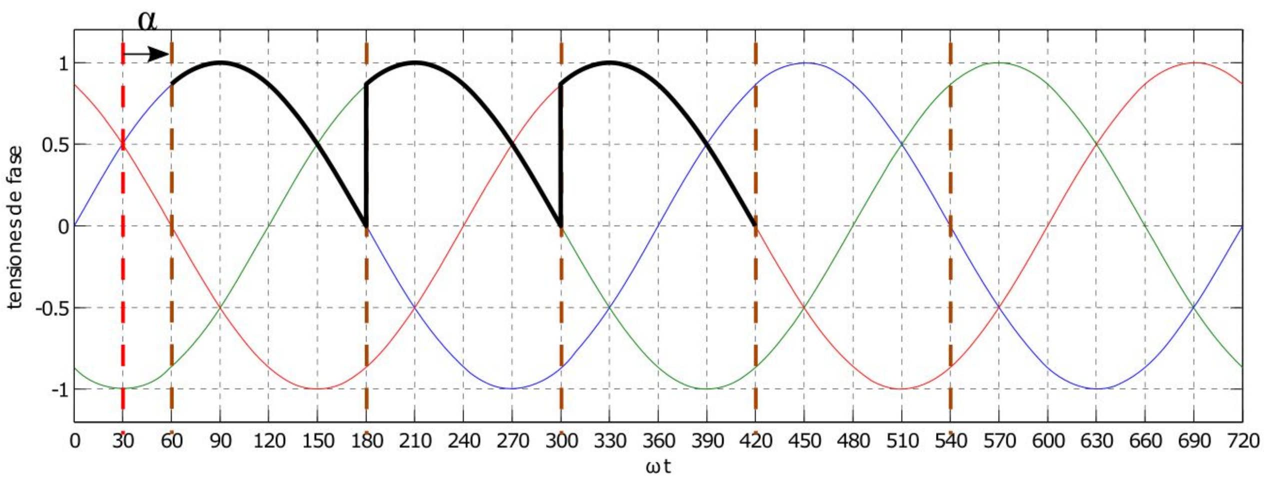

tensiones de linea
ωt
0 60 120 180 240 300 360 420 480 540 600 660 720
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
V_{Pn}
n

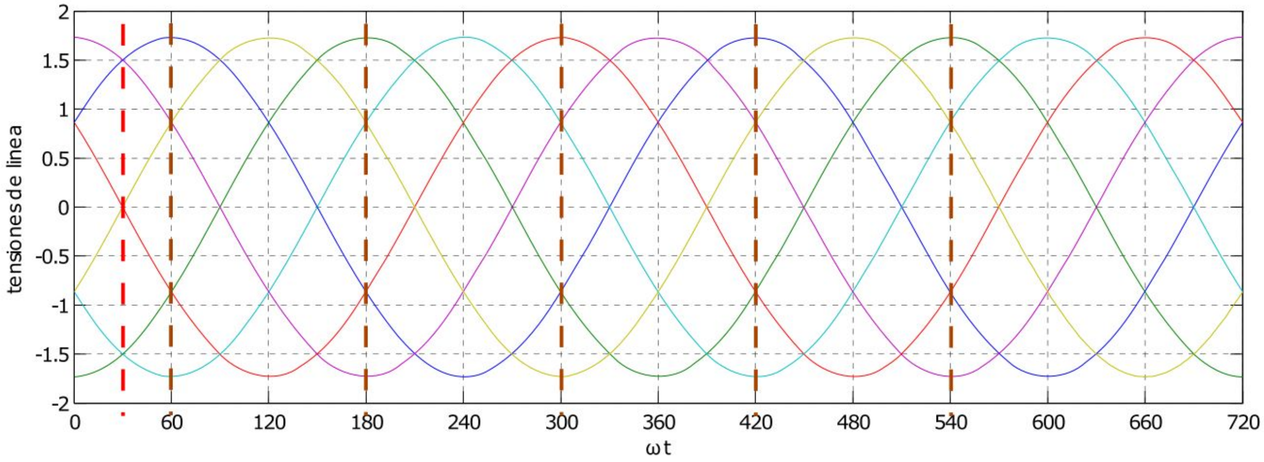

α
tensiones de fase
ωt


tensiones de linea
ωt

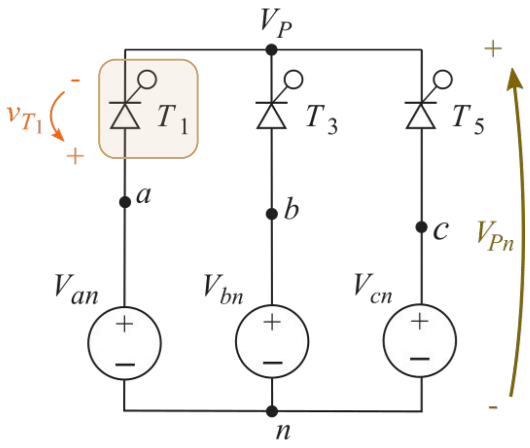

V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n



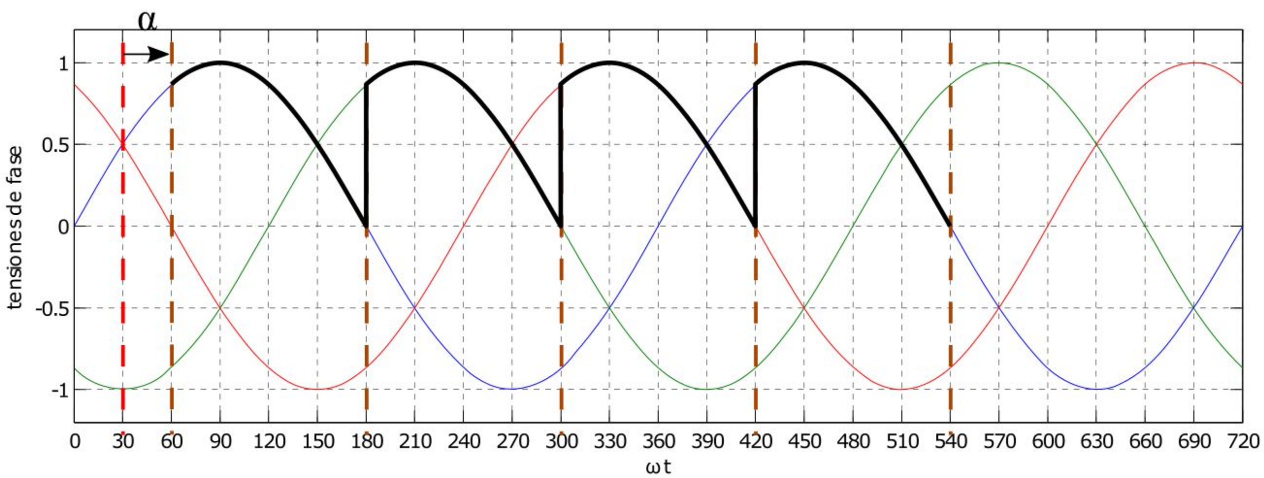
α
tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt

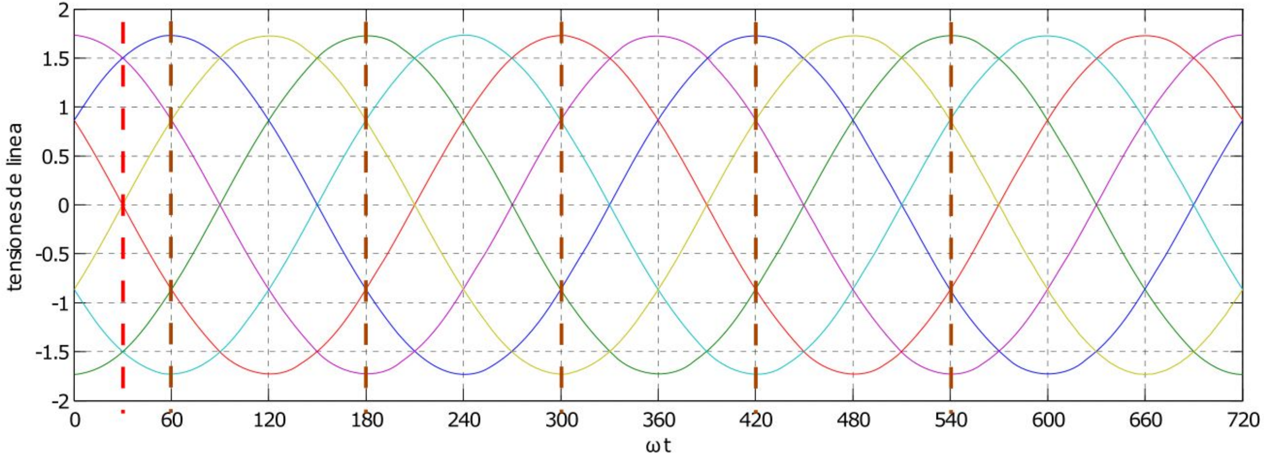


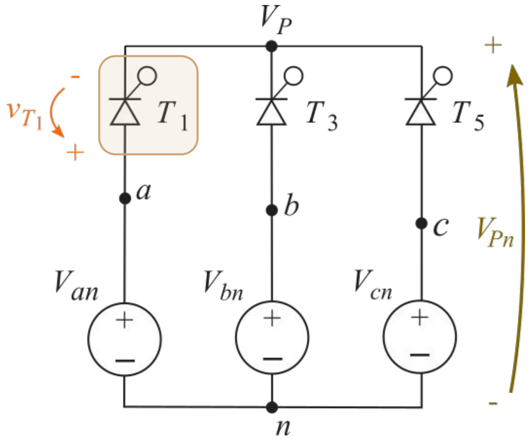
tensiones de linea
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt

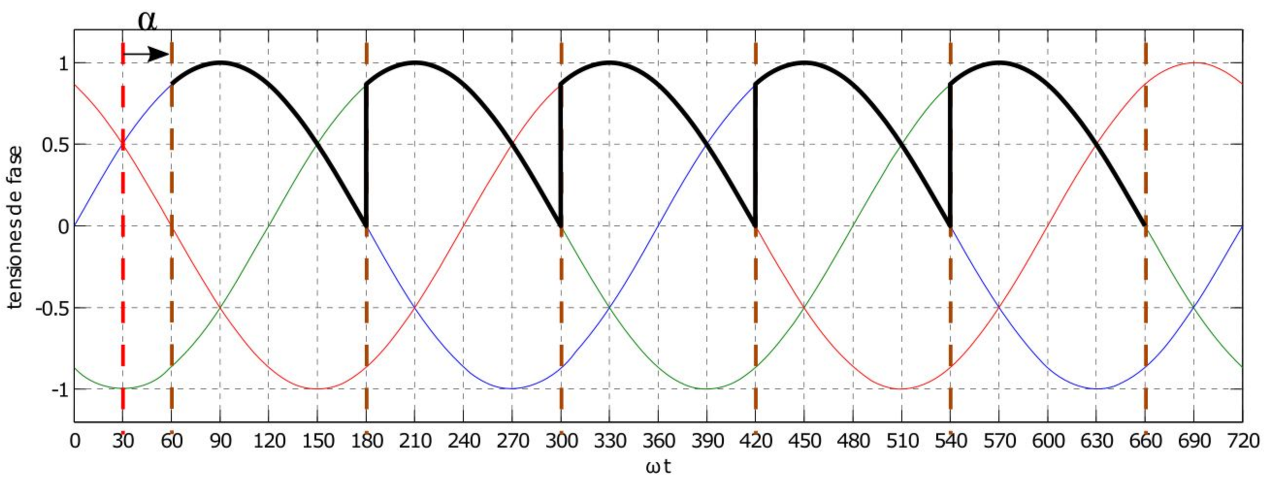


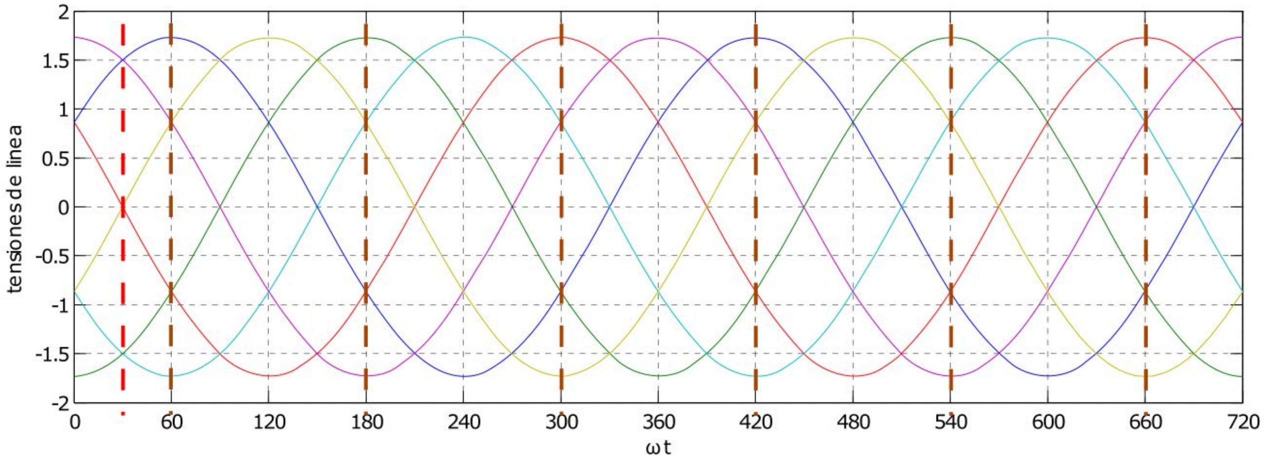
V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
V_{Pn}
n
+
-

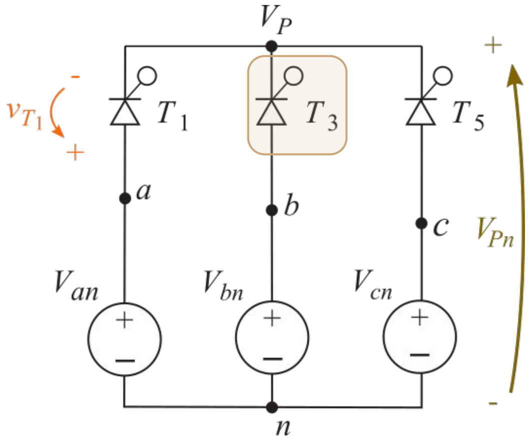

α
tensiones de fase
ωt
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720

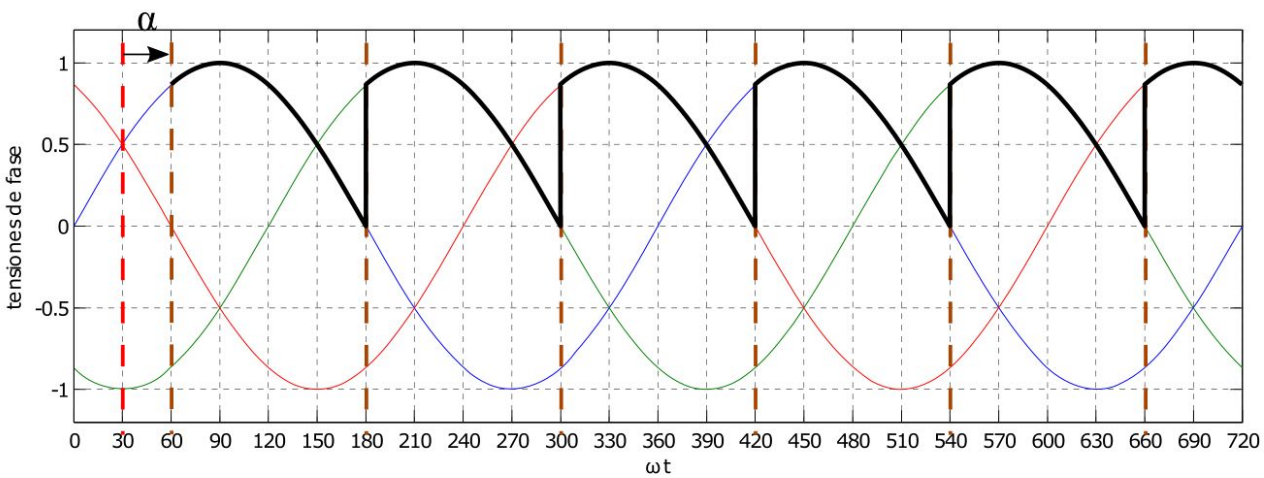

tensiones de linea
ωt
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n

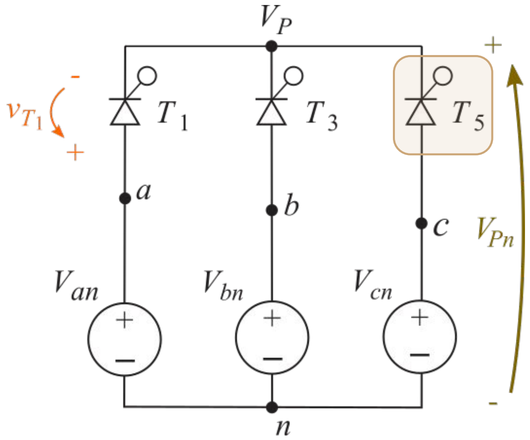

α
tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt


tensiones de linea
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt

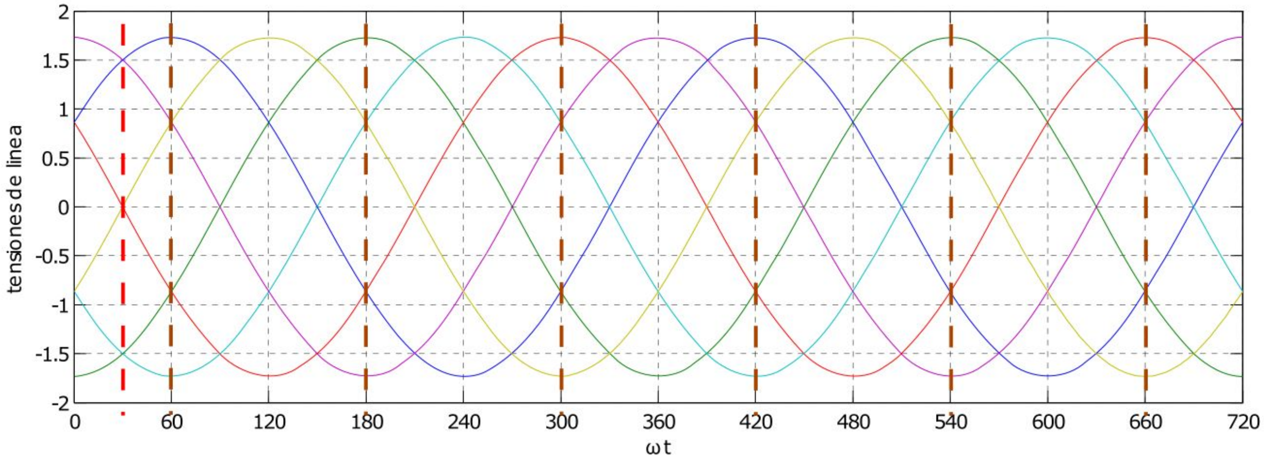

V_P
v_{T1}
- +
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
V_{Pn}
+ -
n



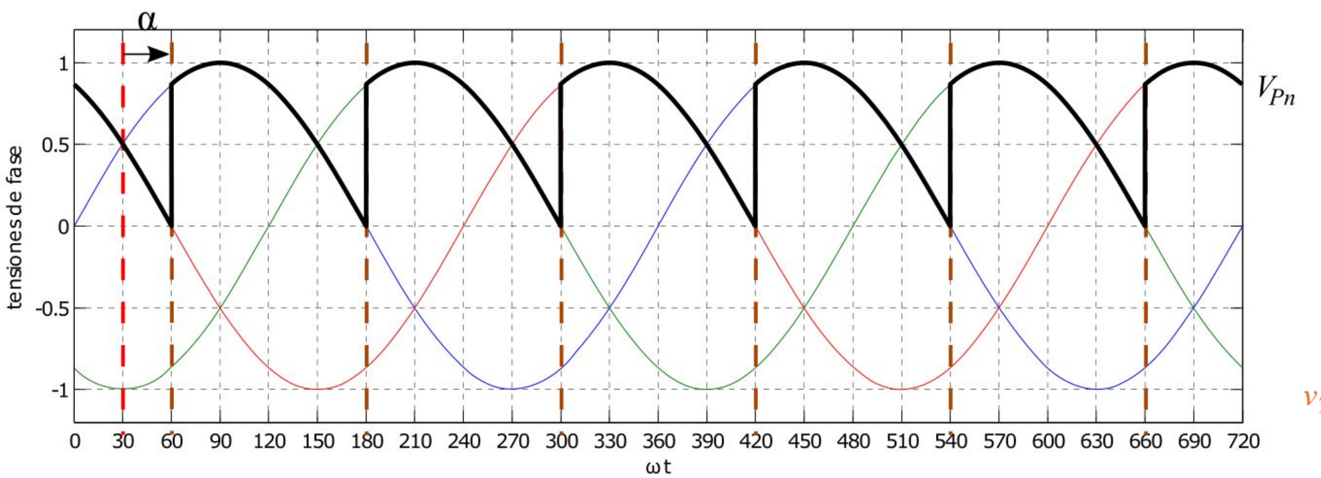
α
tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt

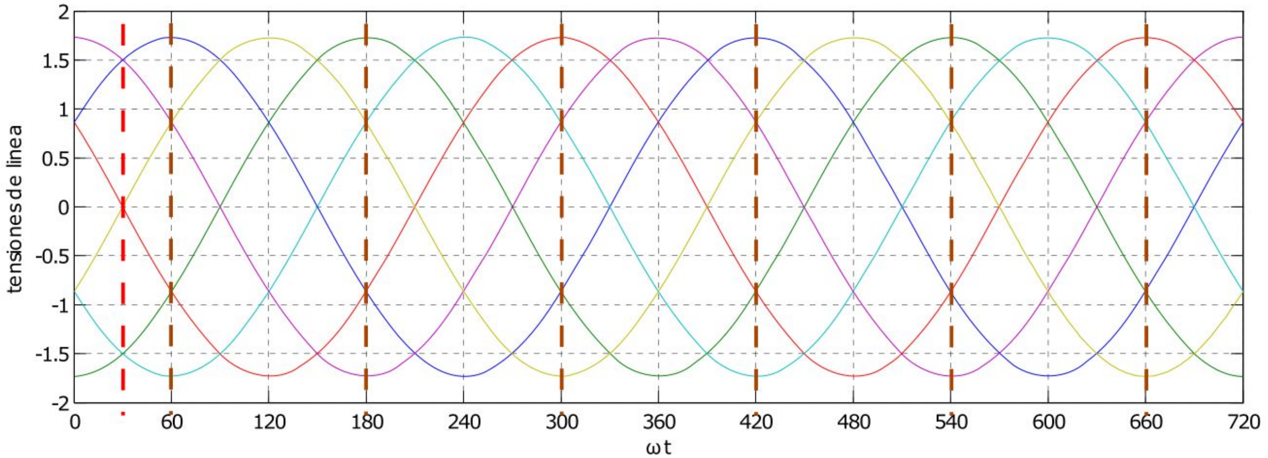


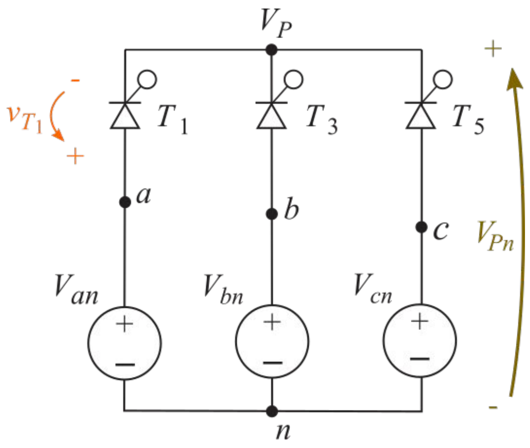
V_P
v_{T1}
-
+
T_1
T_3
T_5
+
-
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n

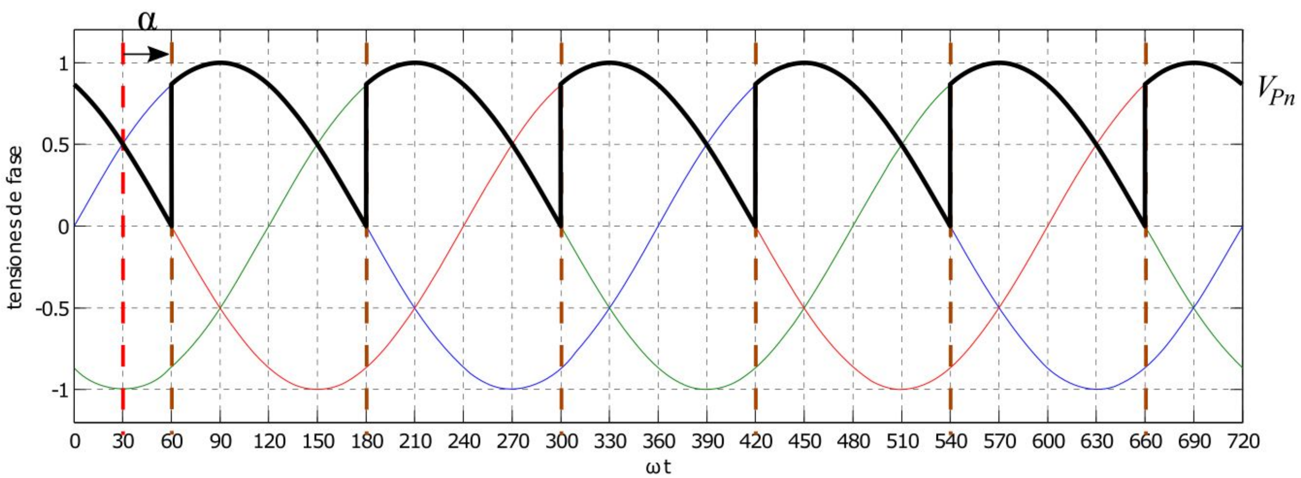


tensiones de linea
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt

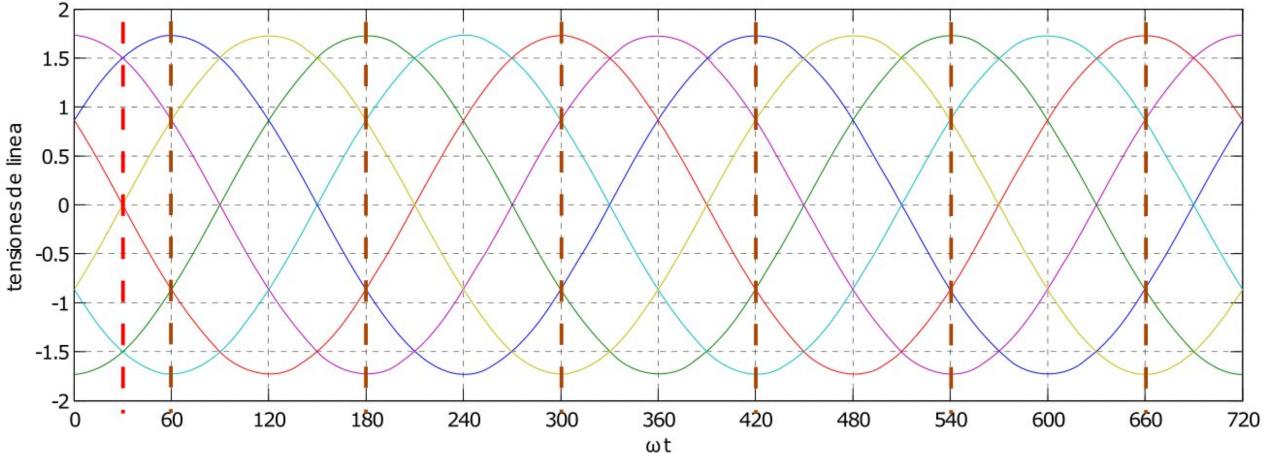

α
V_{Pn}
tensiones de fase
ωt


tensiones de linea
ωt

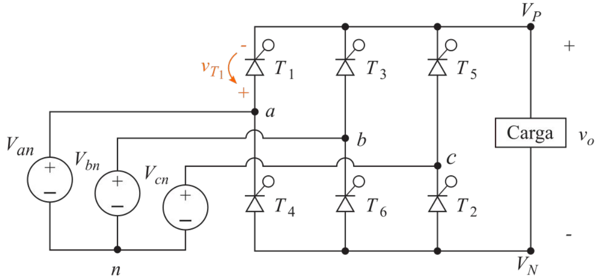

V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{Pn}
V_{an}
V_{bn}
V_{cn}
n



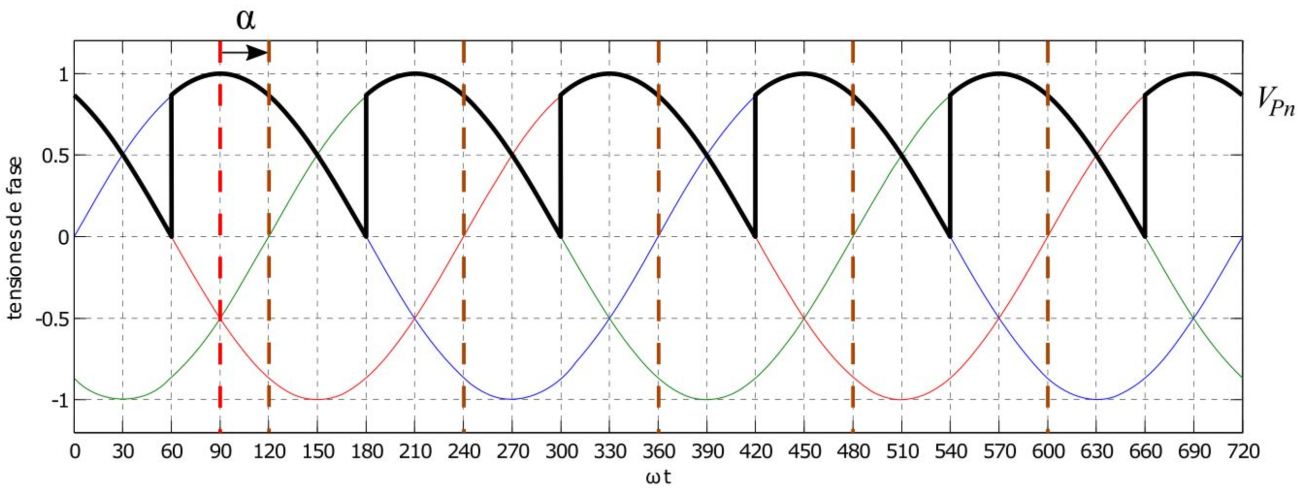
α
V_{Pn}
tensiones de fase
ωt



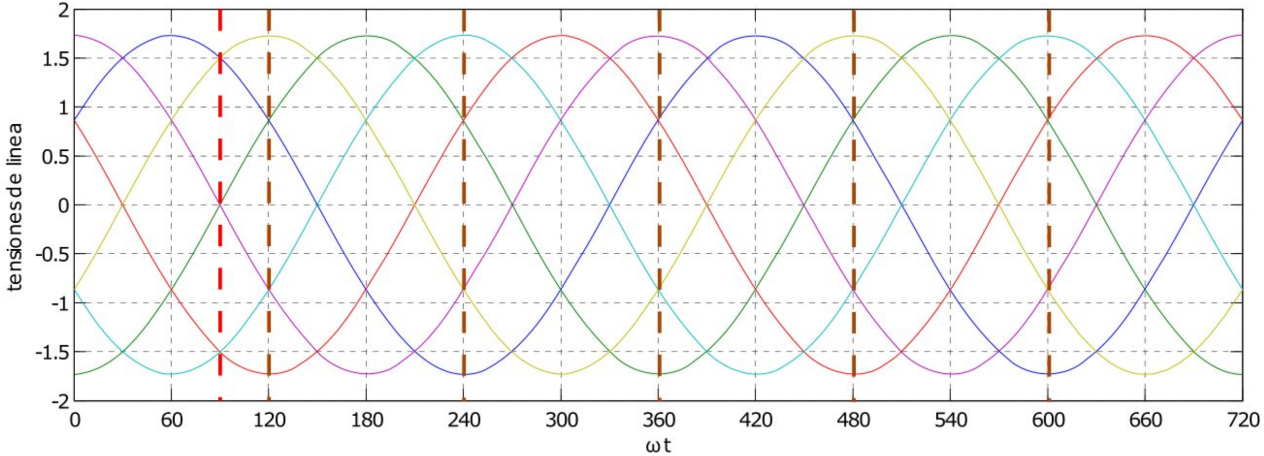
tensiones de linea
ωt



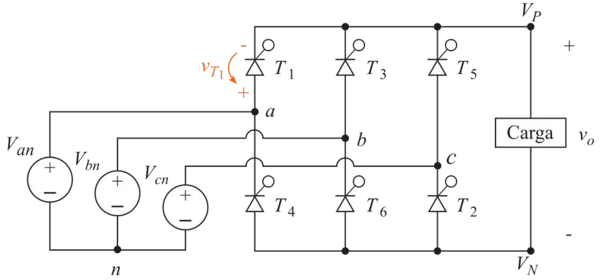
V_P
V_N
v_{T_1}
T_1
T_3
T_5
T_4
T_6
T_2
a
b
c
n
V_{an}
V_{bn}
V_{cn}
Carga
v_o



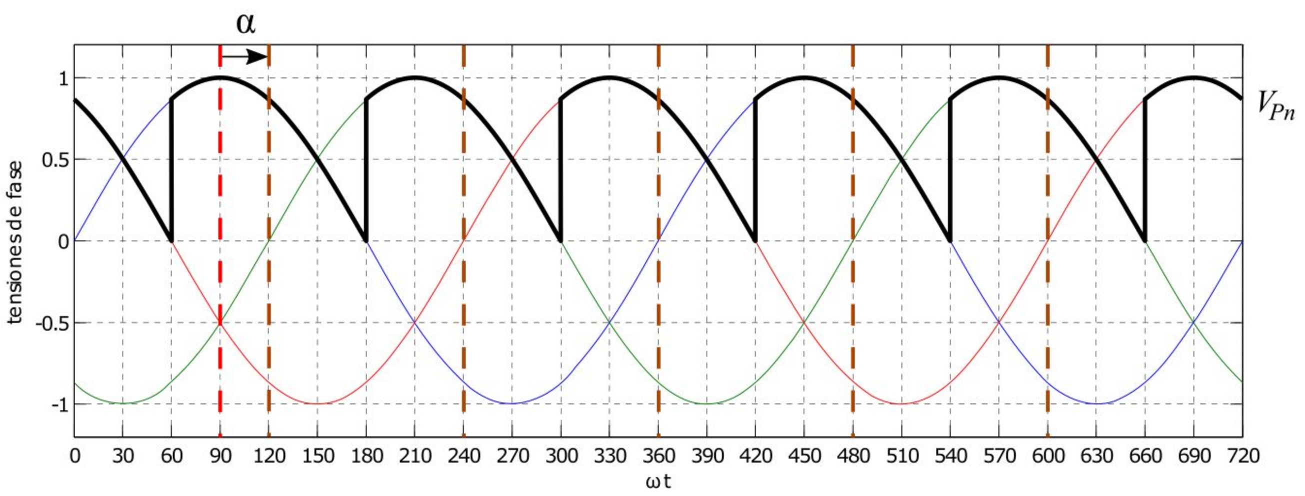
α
V_{Pn}
tensiones de fase
ωt



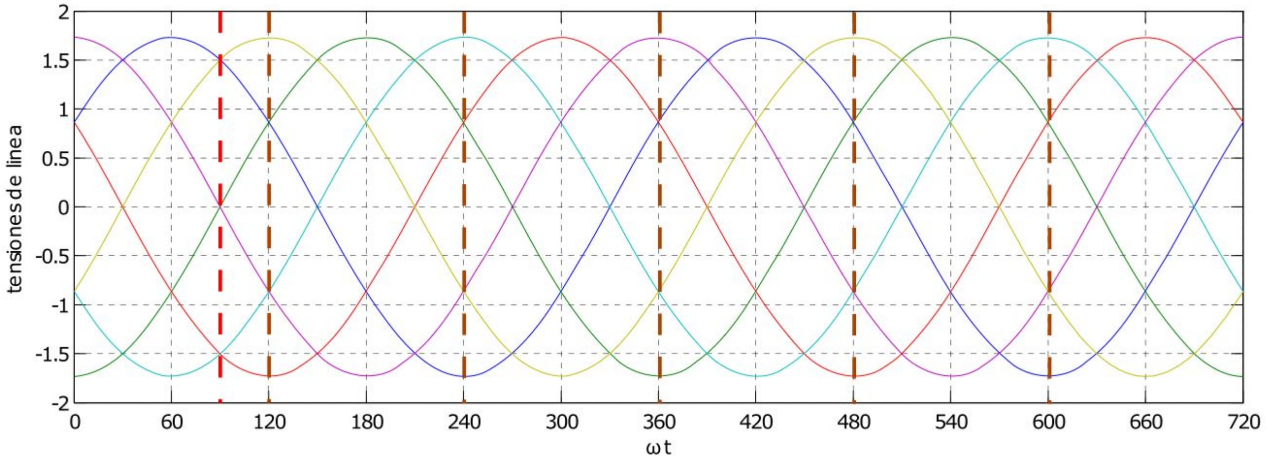
tensiones de linea
ωt



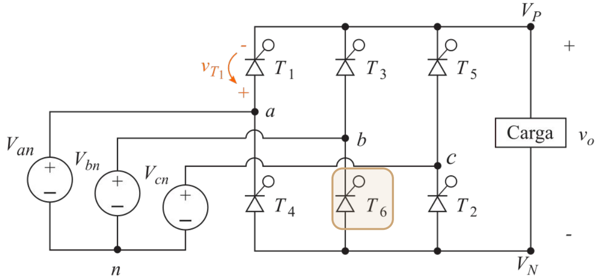
V_P
V_N
v_{T_1}
T_1
T_3
T_5
T_4
T_6
T_2
a
b
c
n
V_{an}
V_{bn}
V_{cn}
Carga
v_o



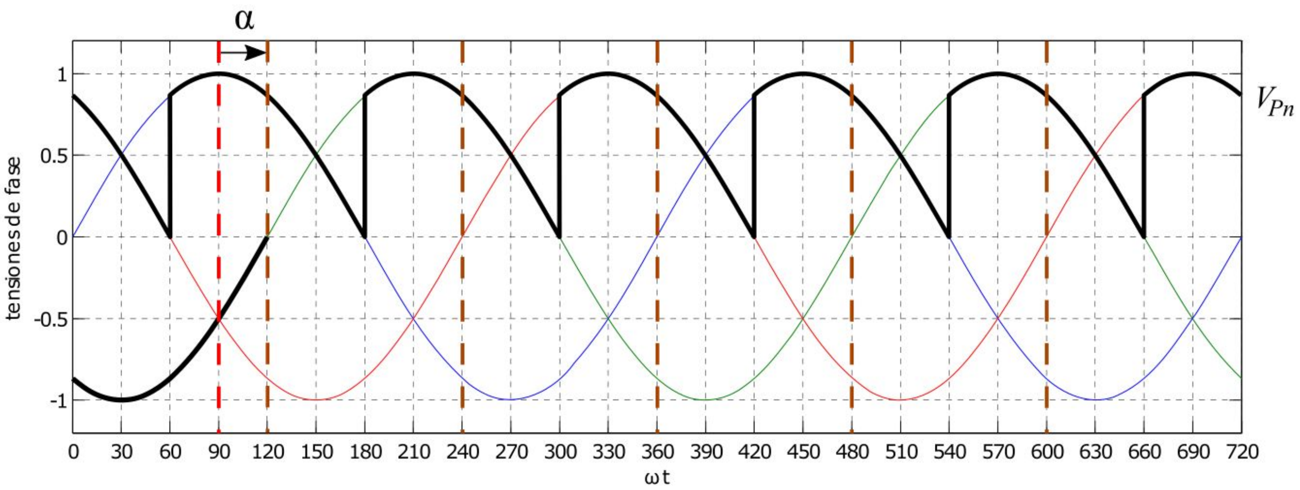
α
V_{Pn}
tensiones de fase
ωt

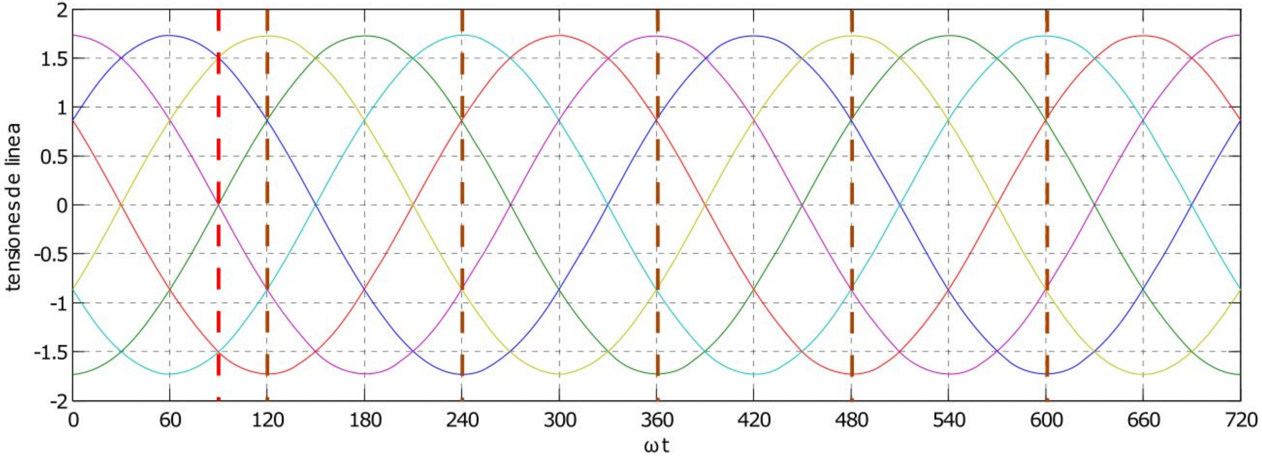


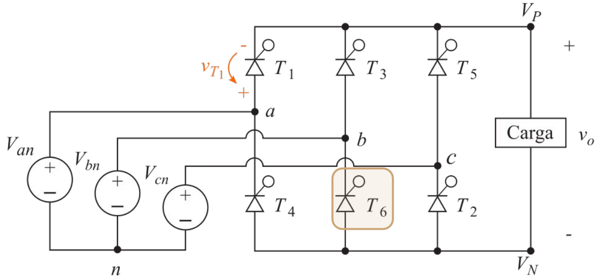
tensiones de linea
ωt

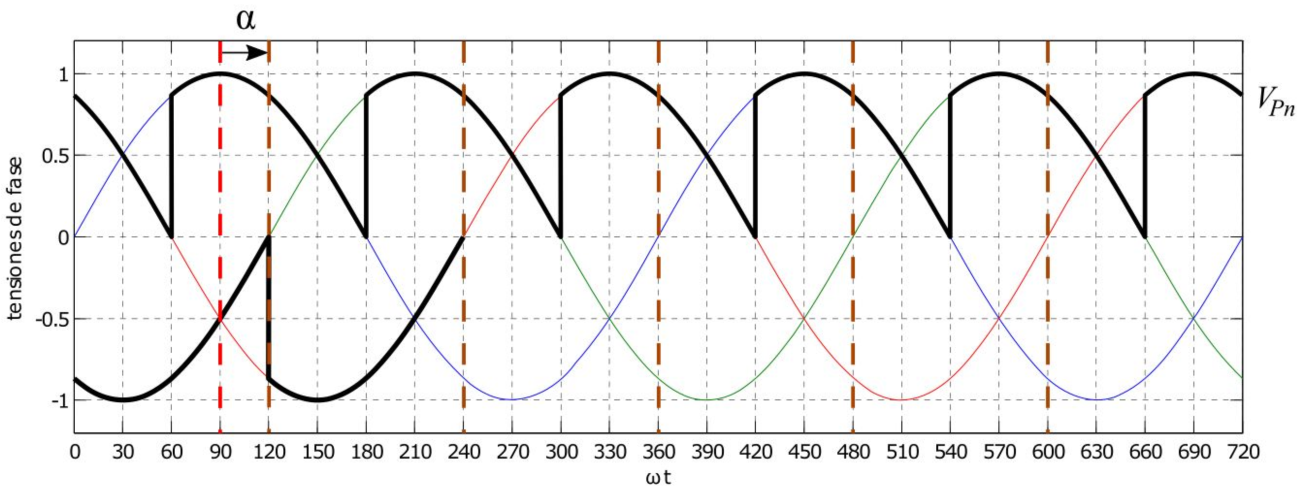


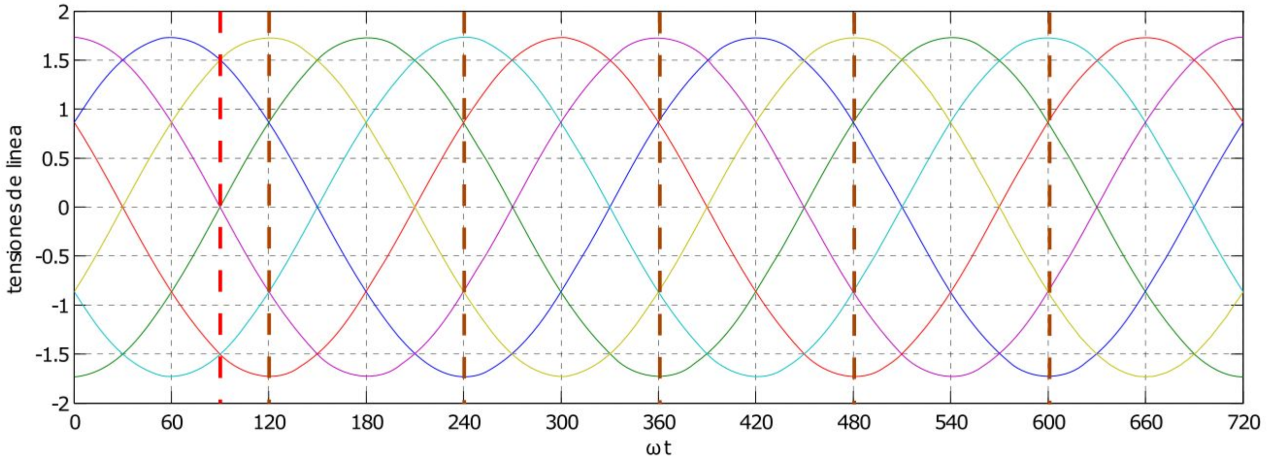
V_P
v_{T_1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
T_4
T_6
T_2
Carga
v_o
n
V_N


α
V_{Pn}
tensiones de fase
ωt
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720


tensiones de linea
ωt
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720


V_P
+
v_{T_1}
-
+
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
T_4
T_6
T_2
Carga
v_o
-
n
V_N


α
V_Pn
tensiones de fase
ωt
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720


tensiones de linea
ωt
2
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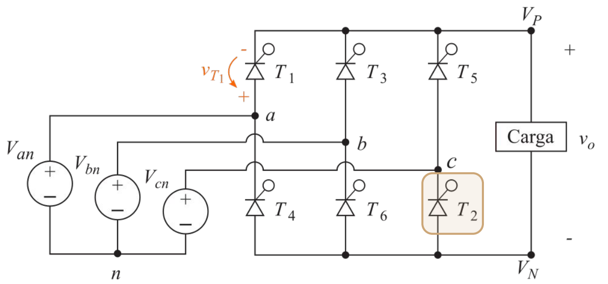
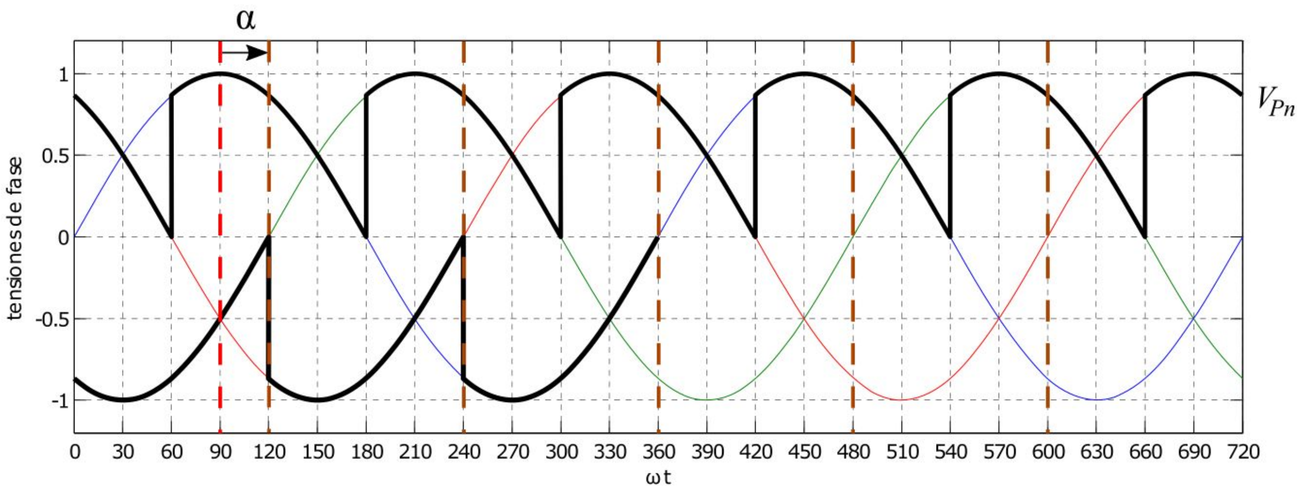
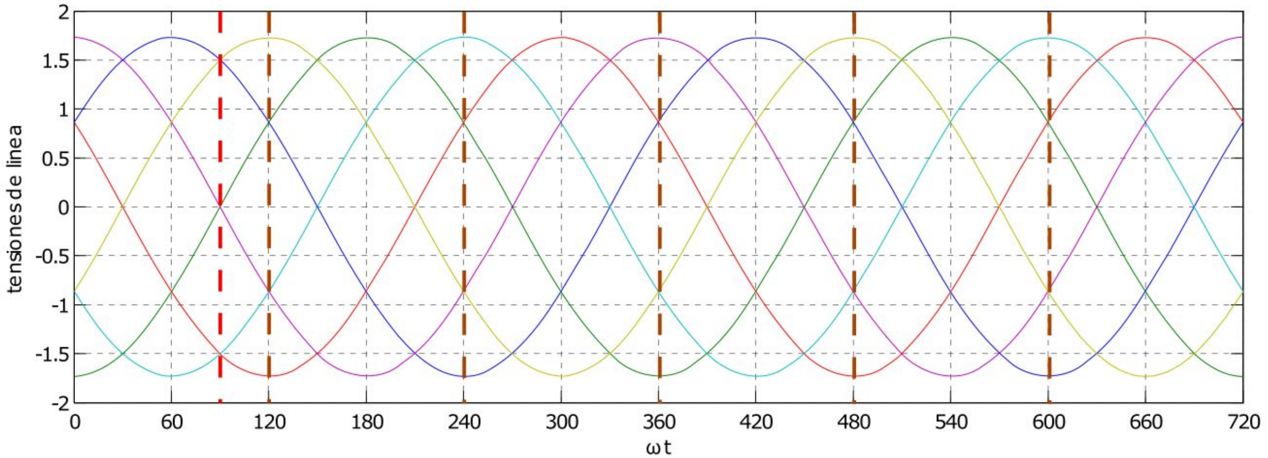
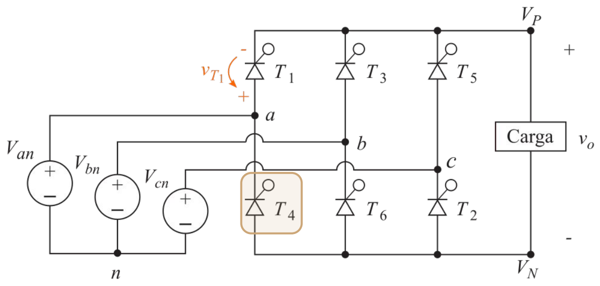
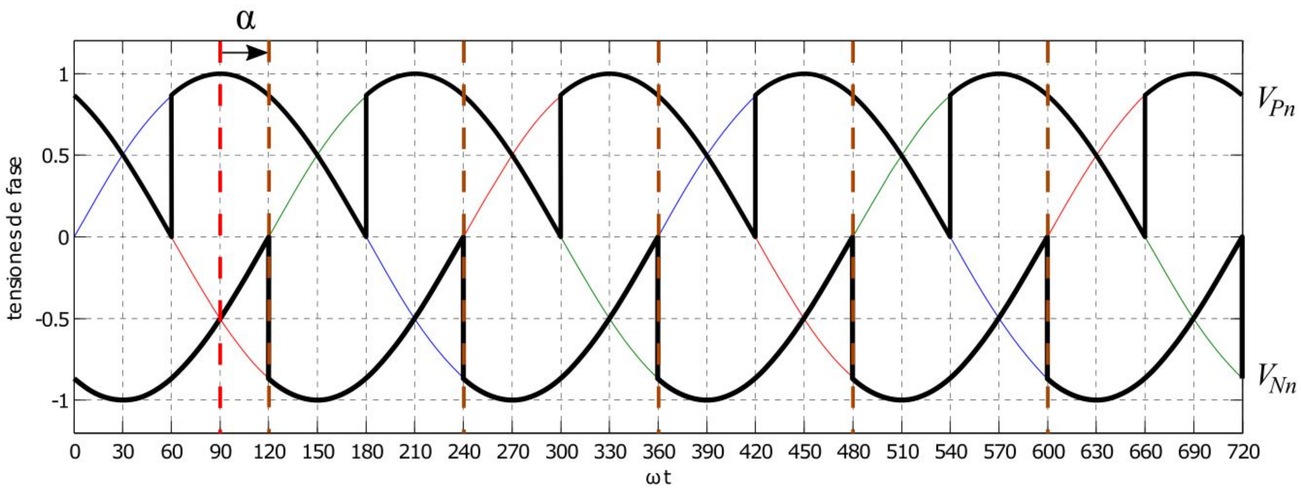
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720

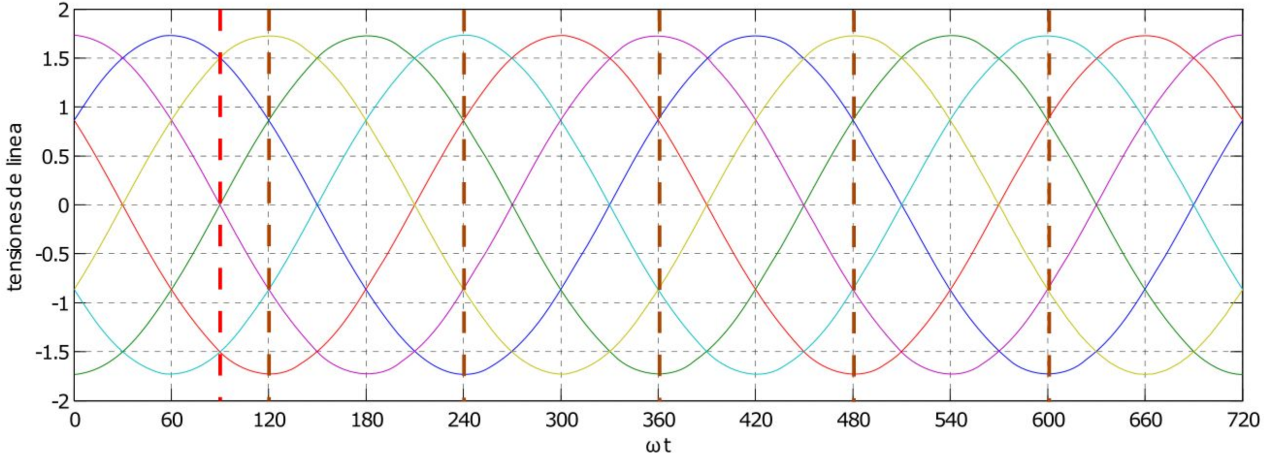

V_P
V_N
v_T1
T1
T3
T5
T4
T6
T2
a
b
c
n
V_an
V_bn
V_cn
Carga
v_o
+
-

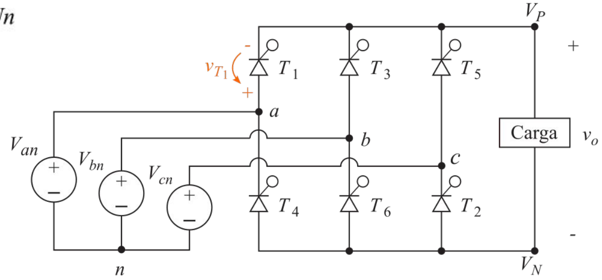

α
V_{Pn}
tensiones de fase
ωt

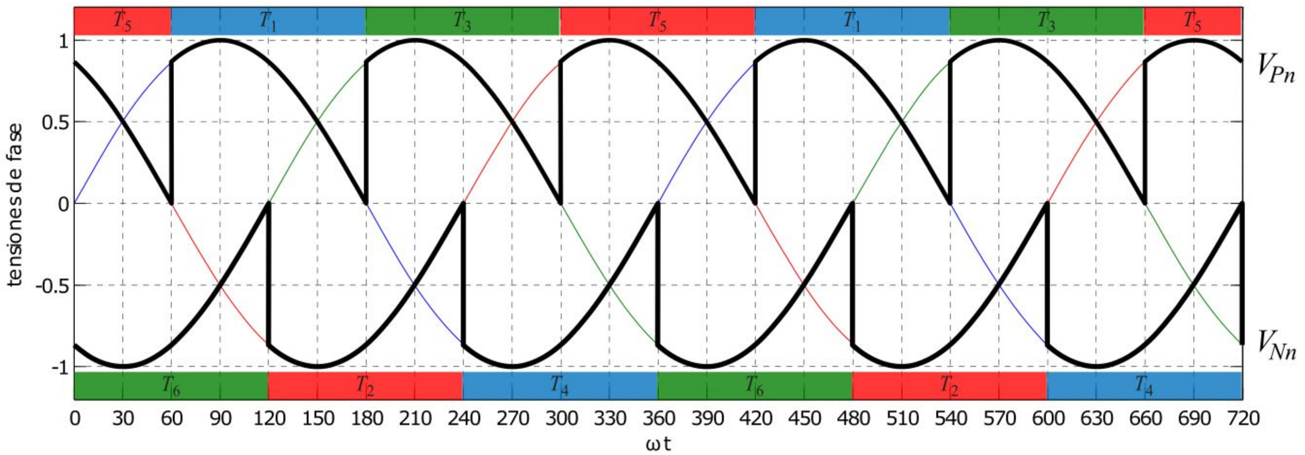

tensiones de linea
ωt


V_P
V_N
v_{T_1}
T_1
T_3
T_5
T_4
T_6
T_2
a
b
c
n
V_{an}
V_{bn}
V_{cn}
Carga
v_o

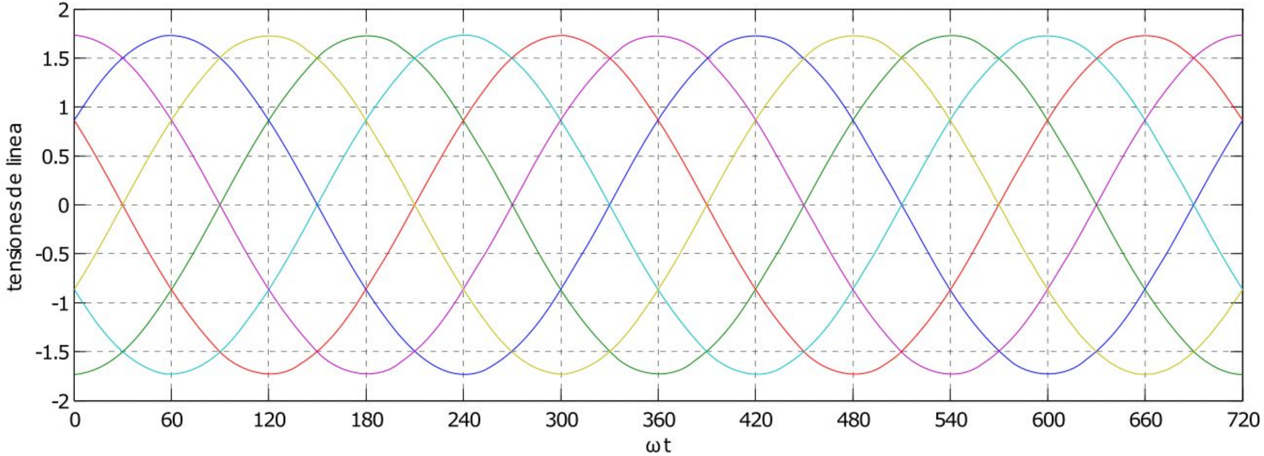

α
tensiones de fase
ωt
V_Pn
V_Nn


tensiones de linea
ωt

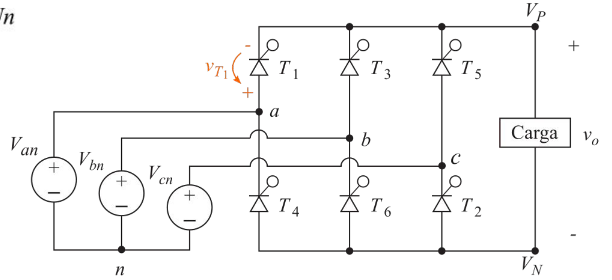

V_P
V_N
T_1
T_3
T_5
T_4
T_6
T_2
v_T1
a
b
c
n
V_an
V_bn
V_cn
Carga
v_o
+
-



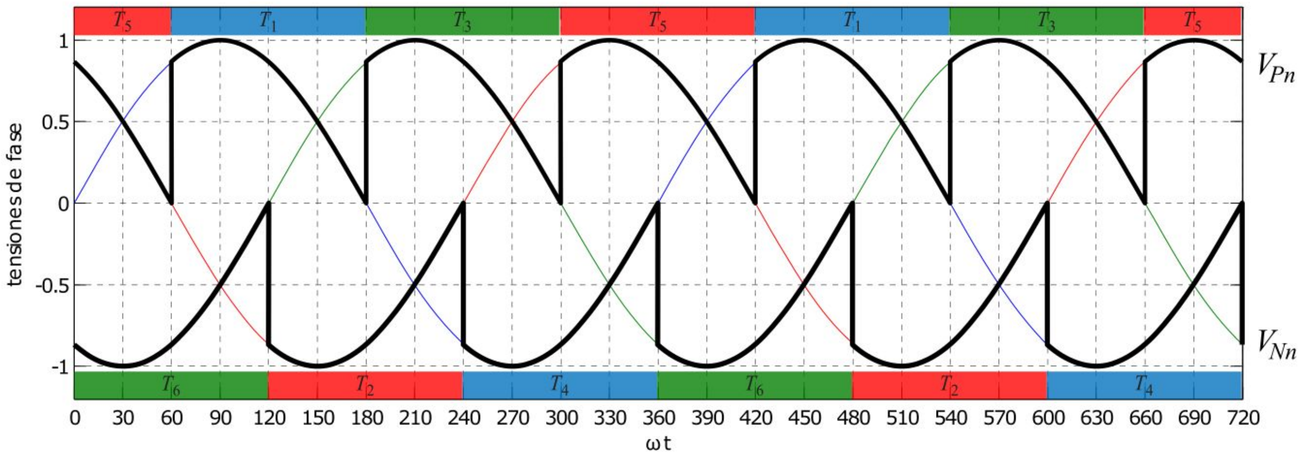
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt



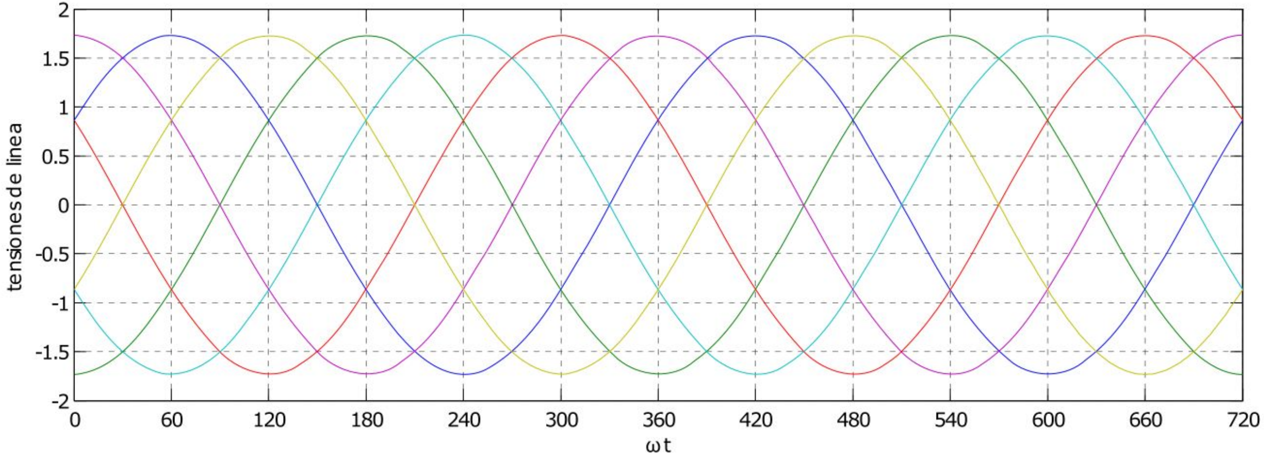
tensiones de linea
ωt



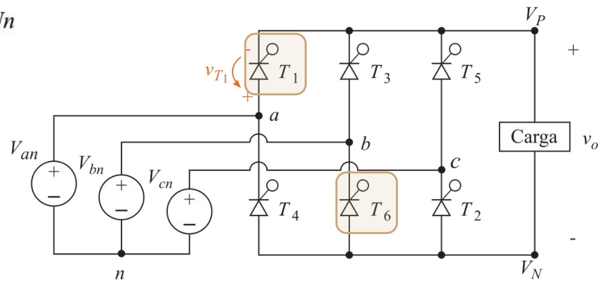
VP
VN
Van
Vbn
Vcn
vT1
T1 T3 T5
T4 T6 T2
a
b
c
n
Carga
vo



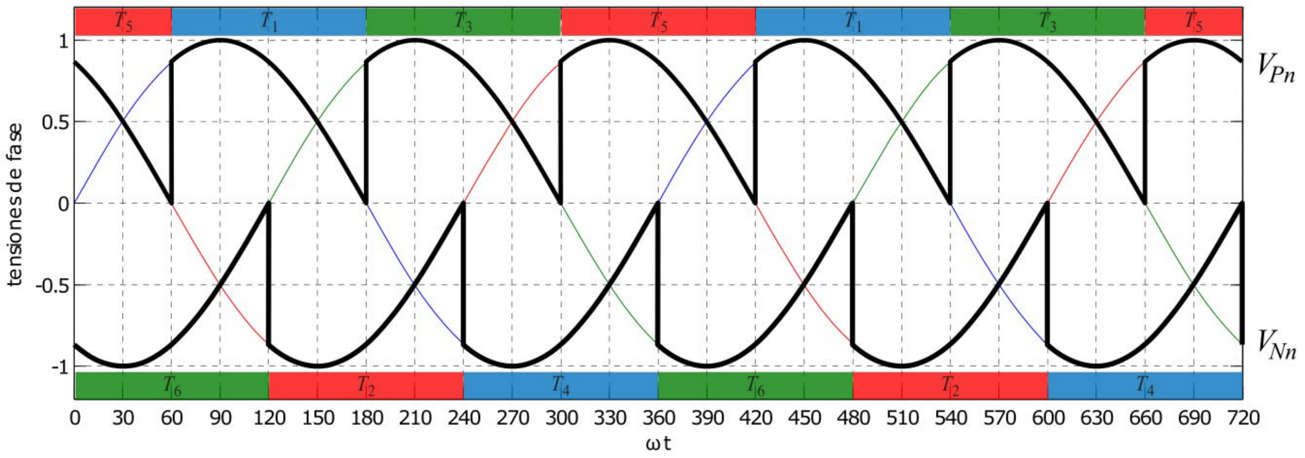
T5
T1
T3
T5
T1
T3
T5
V_Pn
V_Nn
T6
T2
T4
T6
T2
T4
tensiones de fase
ωt



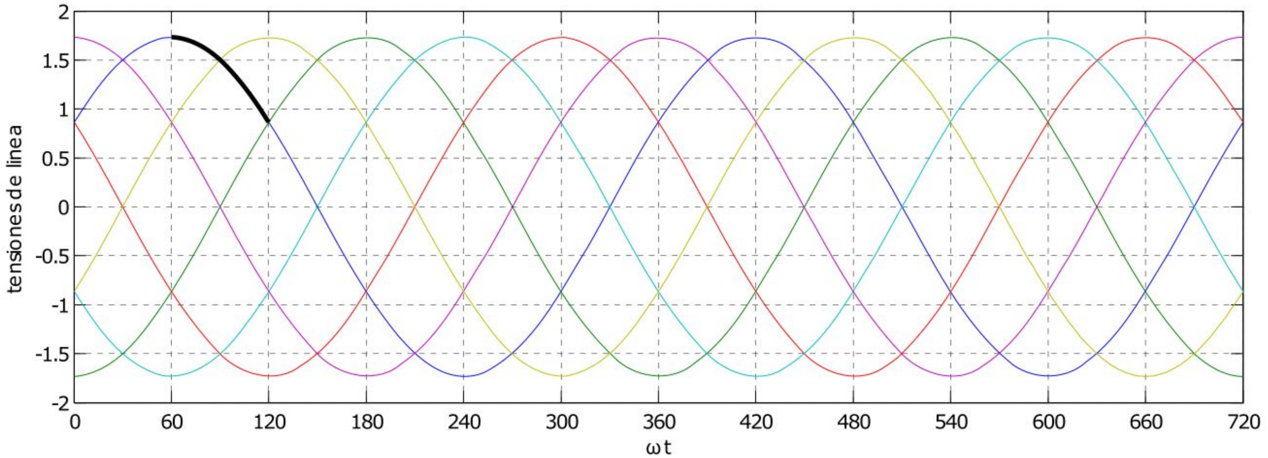
tensiones de linea
ωt



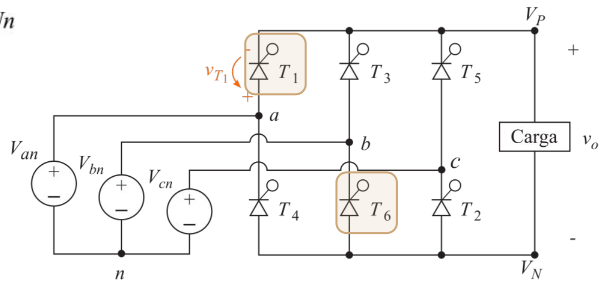
V_P
+
v_T1
T1
T3
T5
a
b
c
Carga
v_o
V_an
V_bn
V_cn
T4
T6
T2
-
n
V_N



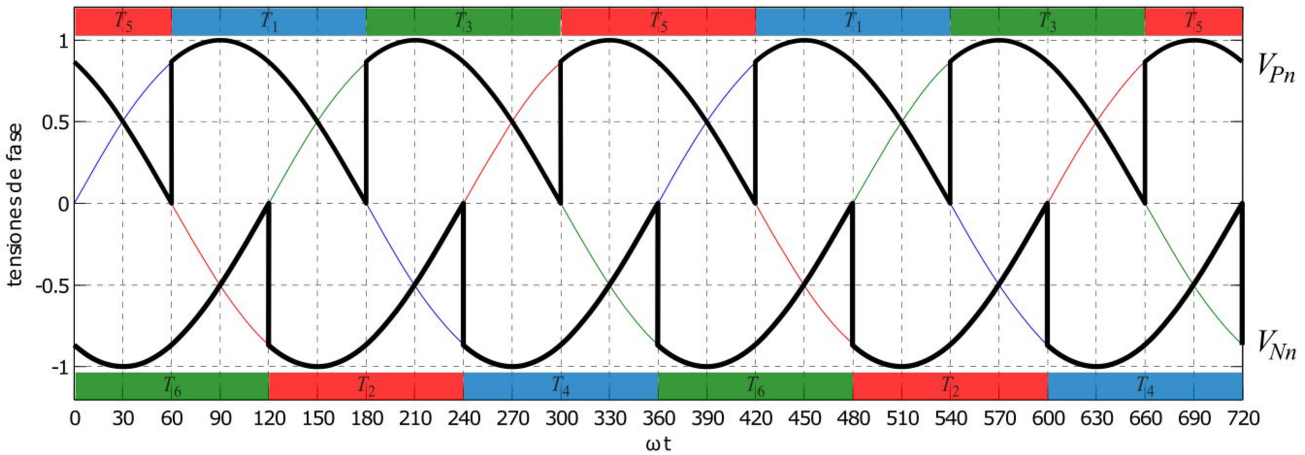
T_5
T_1
T_3
T_5
T_1
T_3
T_5
V_{Pn}
V_{Nn}
T_6
T_2
T_4
T_6
T_2
T_4
tensiones de fase
ωt



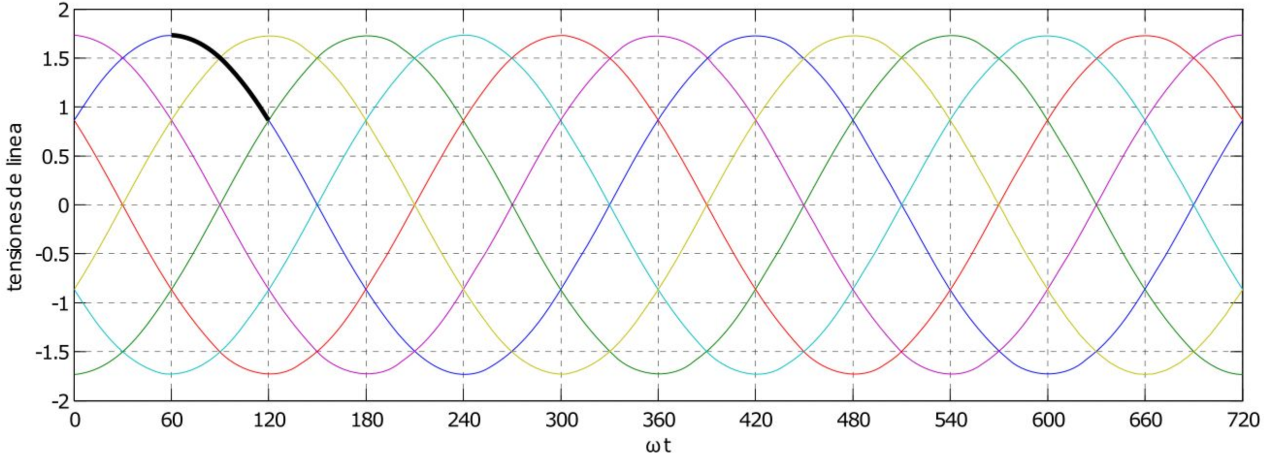
tensiones de linea
ωt



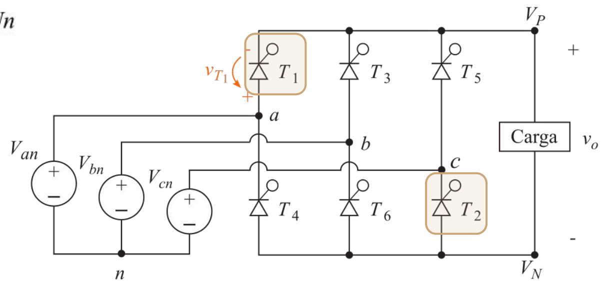
V_P
v_{T_1}
T_1
T_3
T_5
+
a
b
c
V_{an}
V_{bn}
V_{cn}
Carga
v_o
T_4
T_6
T_2
-
n
V_N



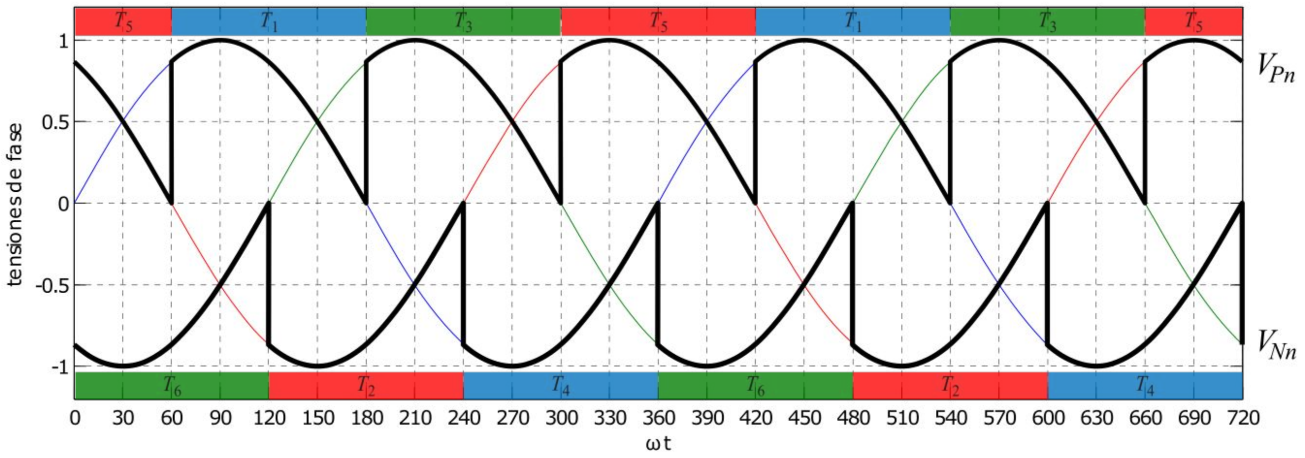
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt

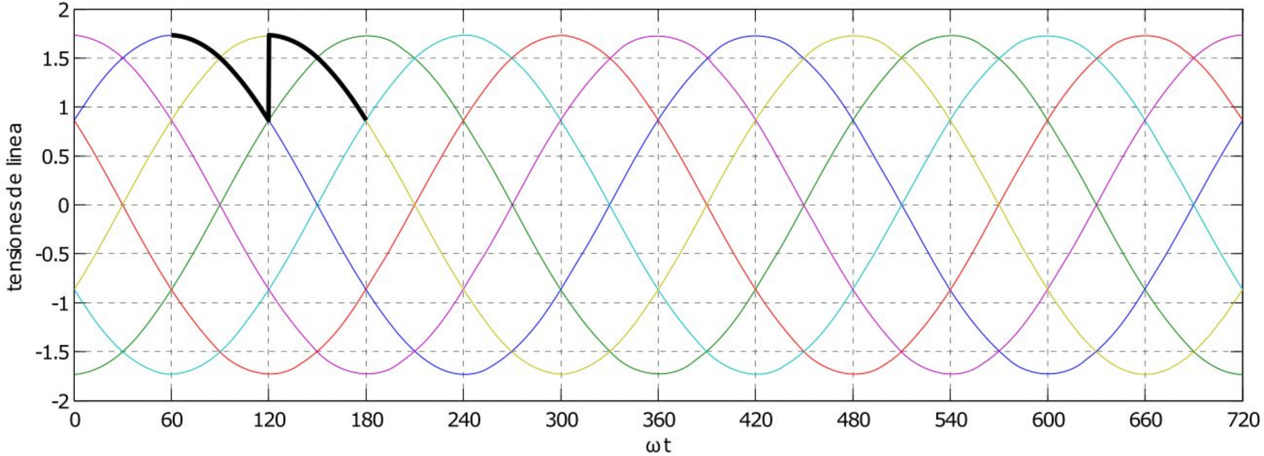


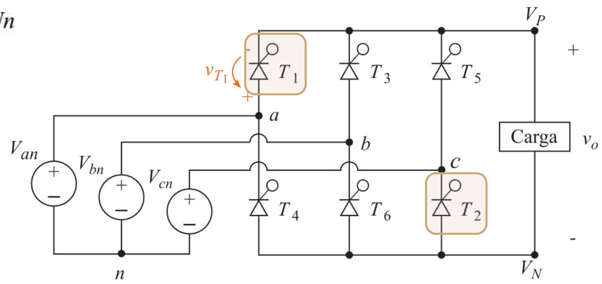
tensiones de linea
ωt



VP
VN
vT1
T1
T3
T5
T4
T6
T2
a
b
c
n
Van
Vbn
Vcn
Carga
vo
+
-


T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt


tensiones de linea
ωt


VP
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T1
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T6
T2
a
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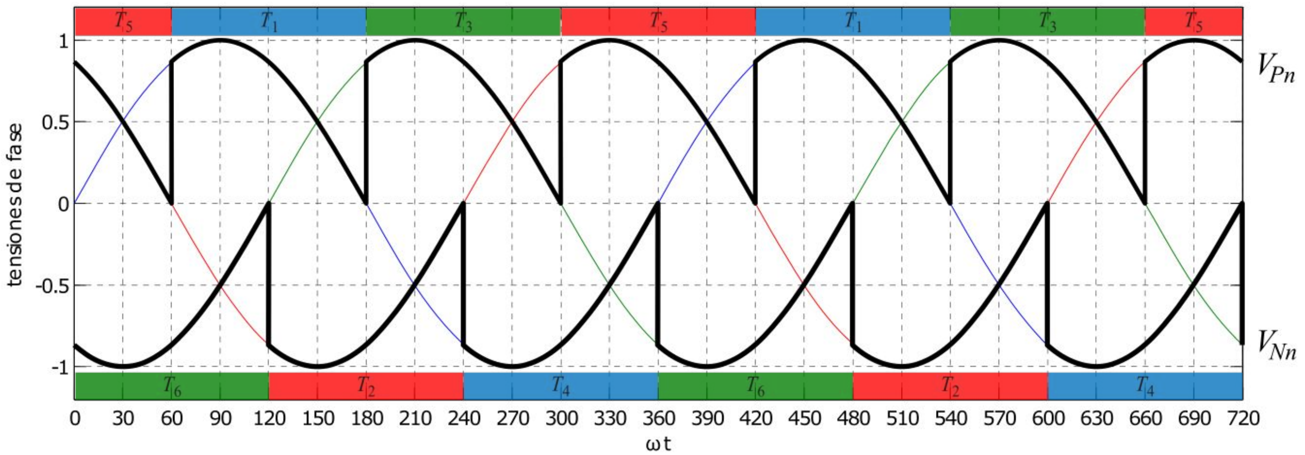
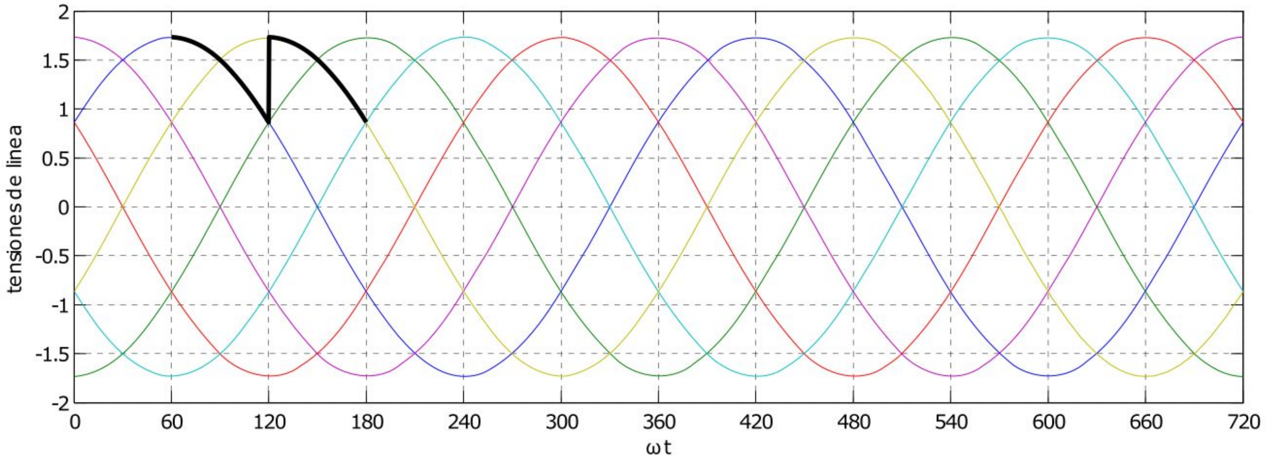
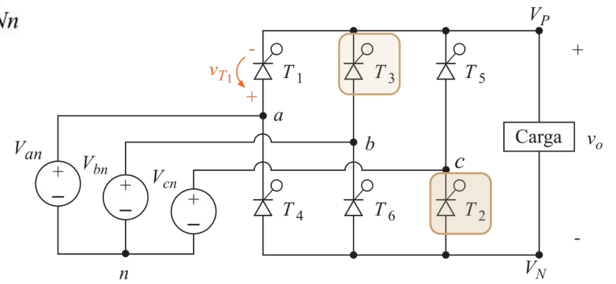
Van
Vbn
Vcn
Carga
vo



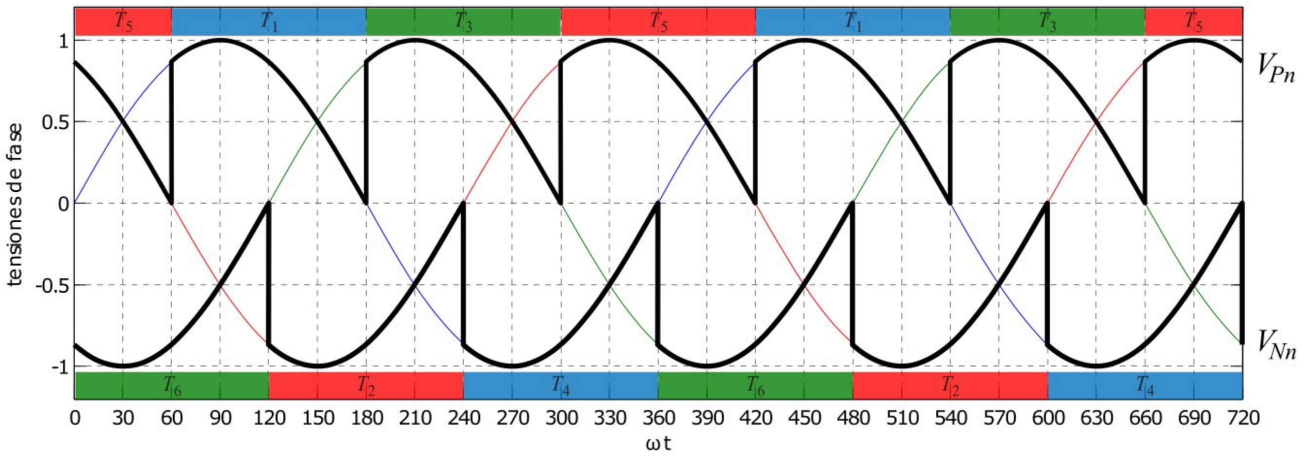
T5
T1
T3
T5
T1
T3
T5
V_Pn
V_Nn
T6
T2
T4
T6
T2
T4
tensiones de fase
ωt

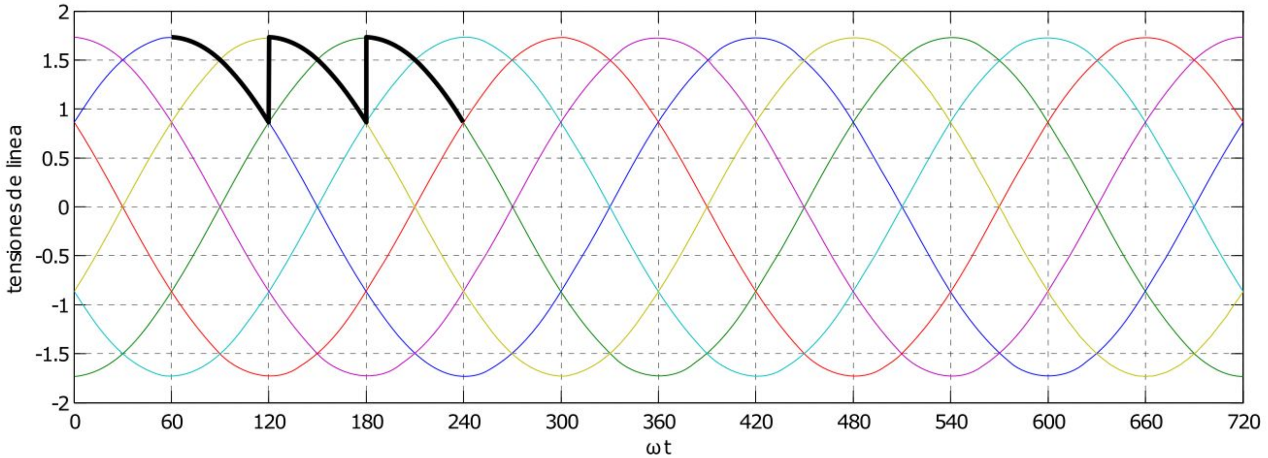


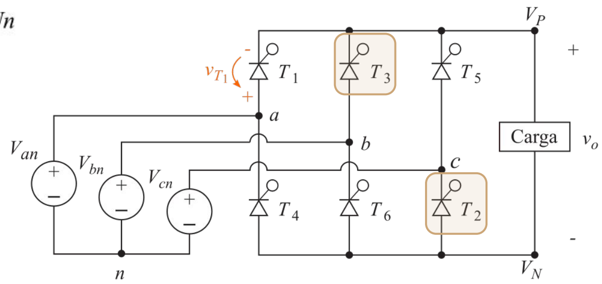
tensiones de linea
ωt



V_P
v_T1
T1
T3
T5
a
b
c
Carga
v_o
V_an
V_bn
V_cn
T4
T6
T2
n
V_N


T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
tensiones de fase
ωt
VPn
VNn


tensiones de linea
ωt


VP
VN
vT1
T1
T3
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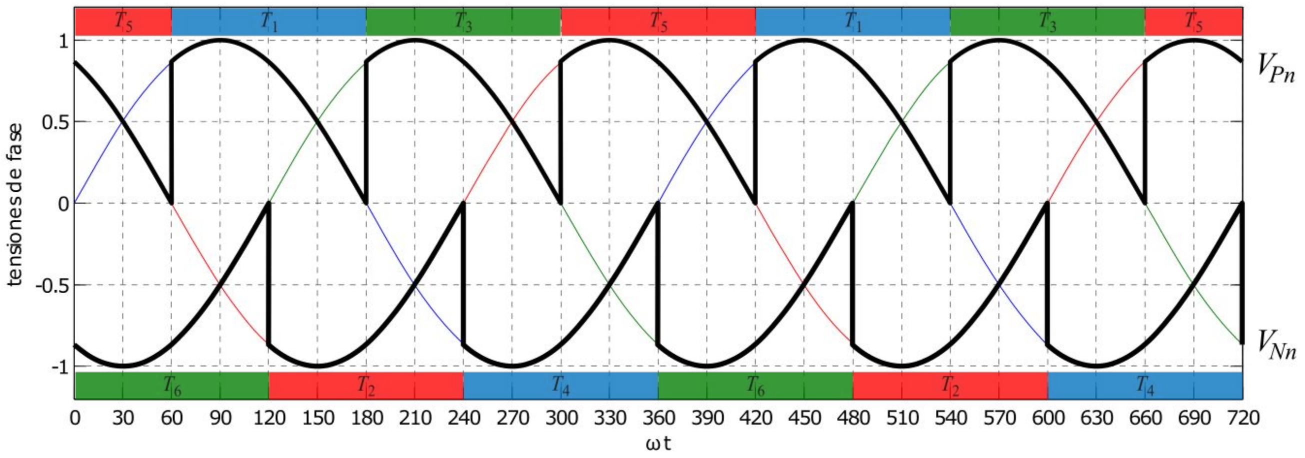
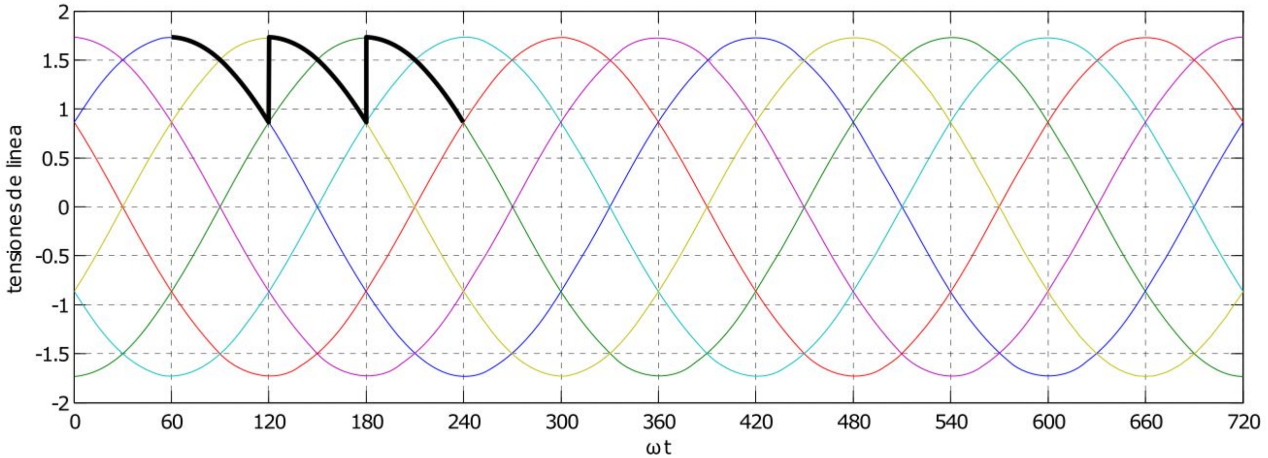
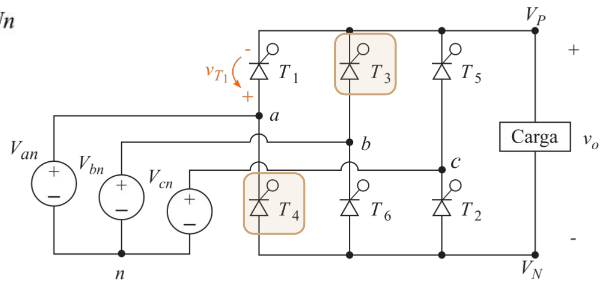
Vbn
Vcn
Carga
vo


T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt


tensiones de linea
ωt


VP
VN
T1 T3 T5
T4 T6 T2
vT1
a
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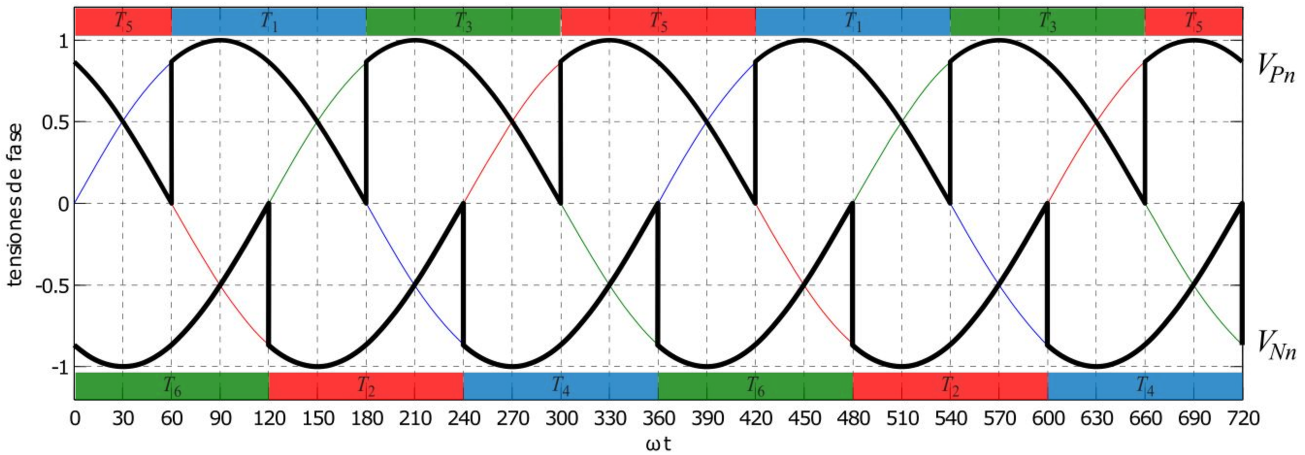
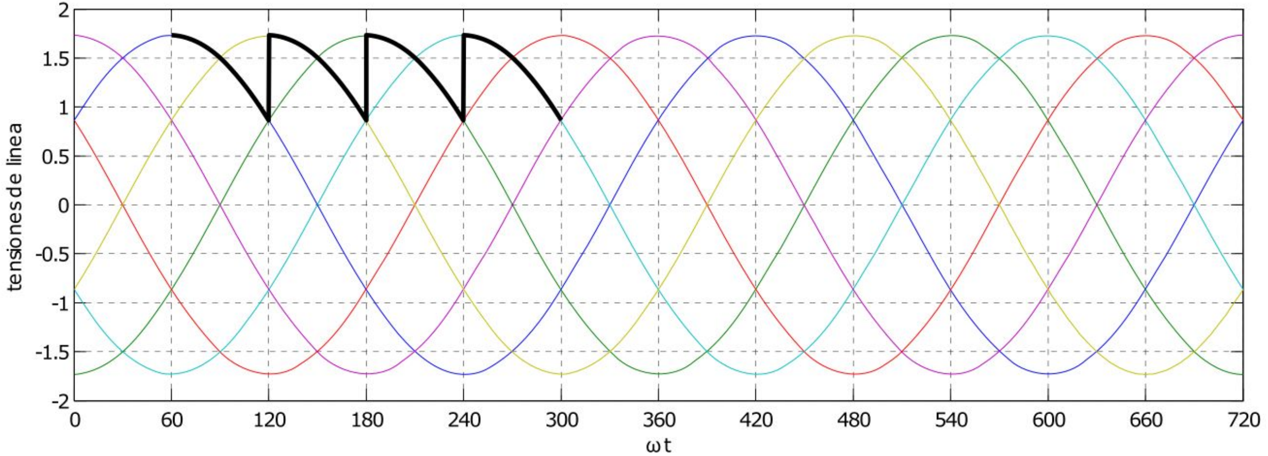
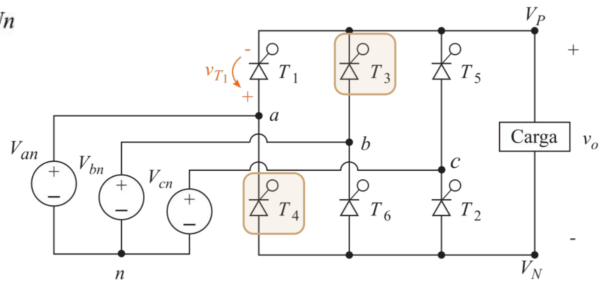
Vcn
Carga
vo



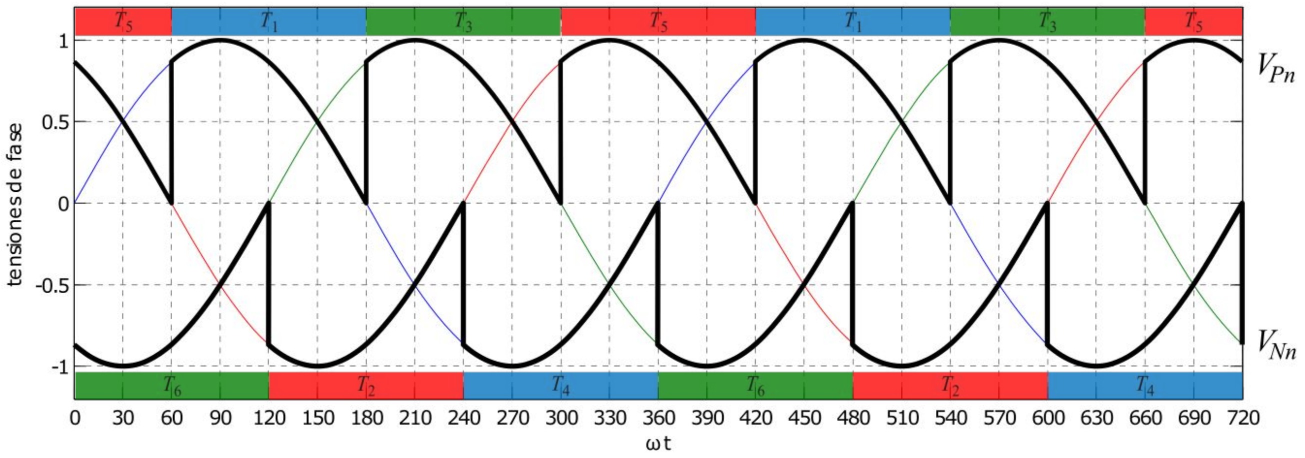
T5 T1 T3 T5 T1 T3 T5
V_Pn
V_Nn
T6 T2 T4 T6 T2 T4
tensiones de fase
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt



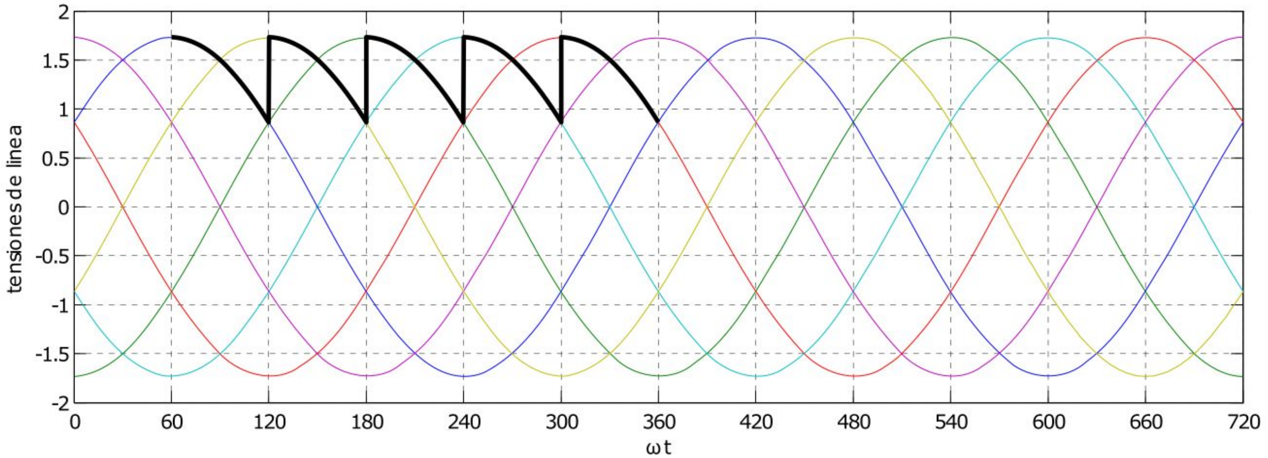
tensiones de linea
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt



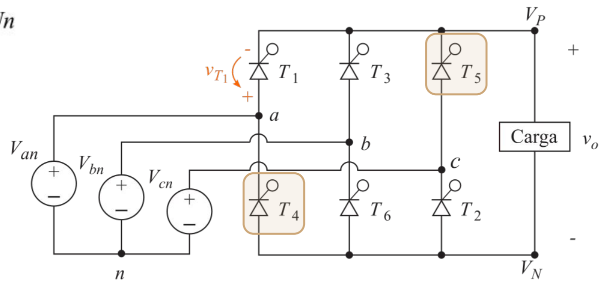
V_P
V_N
v_T1
T1 T3 T5 T4 T6 T2
a b c n
V_an V_bn V_cn
Carga
v_o



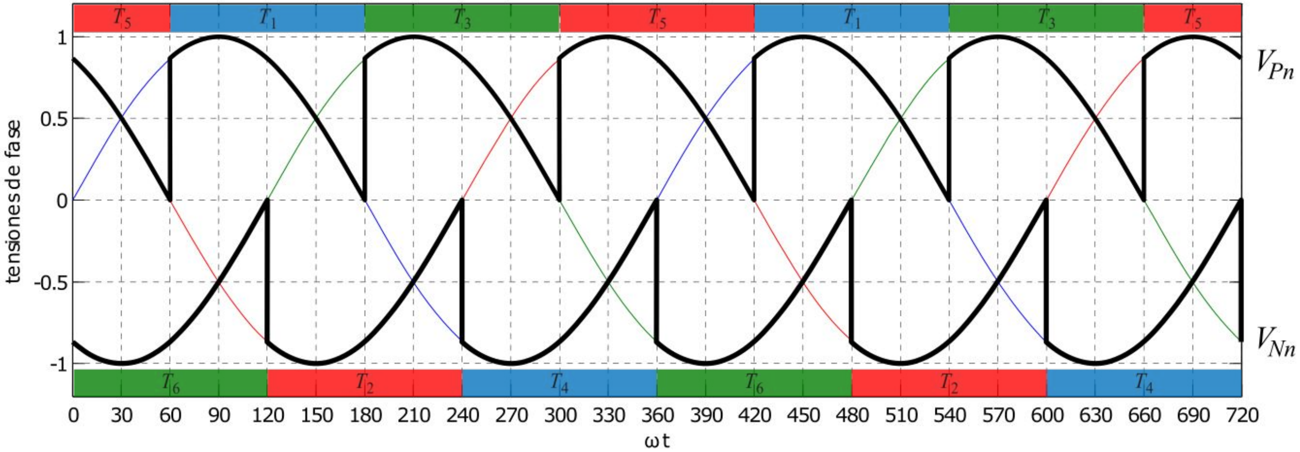
T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
VPn
VNn
tensiones de fase
ωt



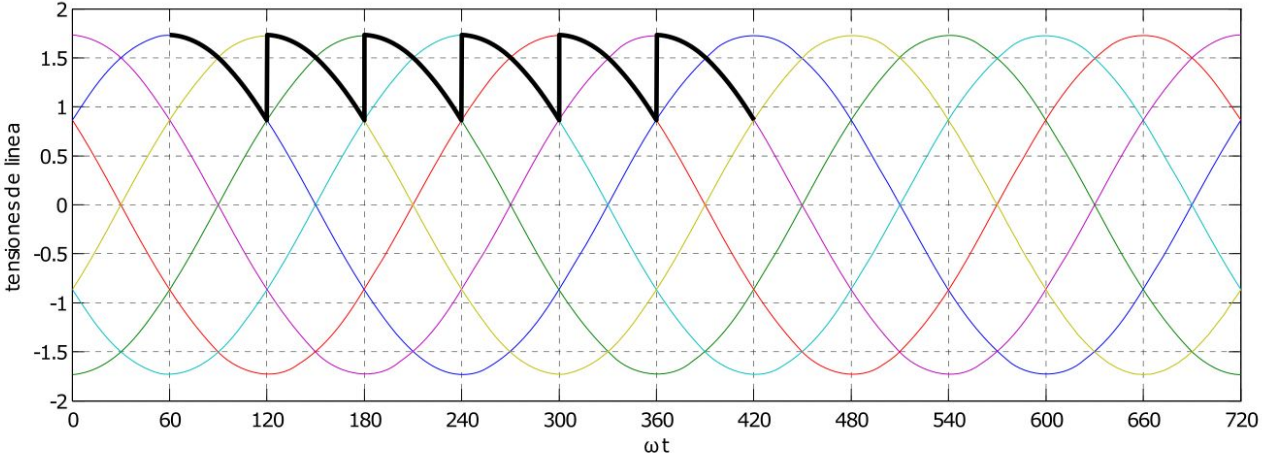
tensiones de linea
ωt



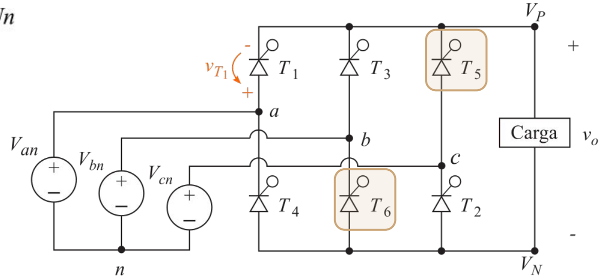
VP
VN
vT1
T1
T3
T5
T4
T6
T2
a
b
c
n
Van
Vbn
Vcn
Carga
vo



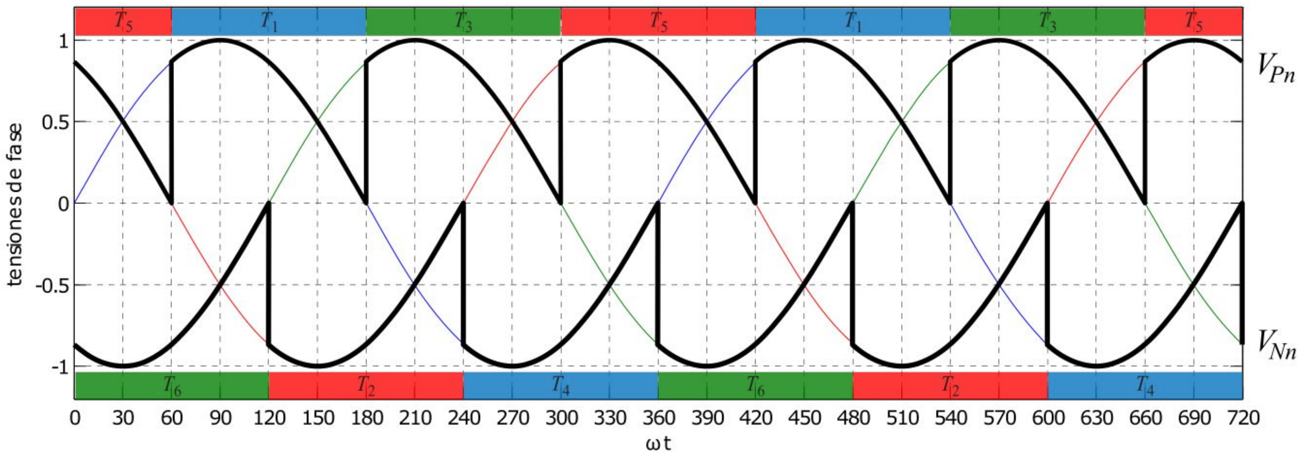
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
1 0.5 0 -0.5 -1

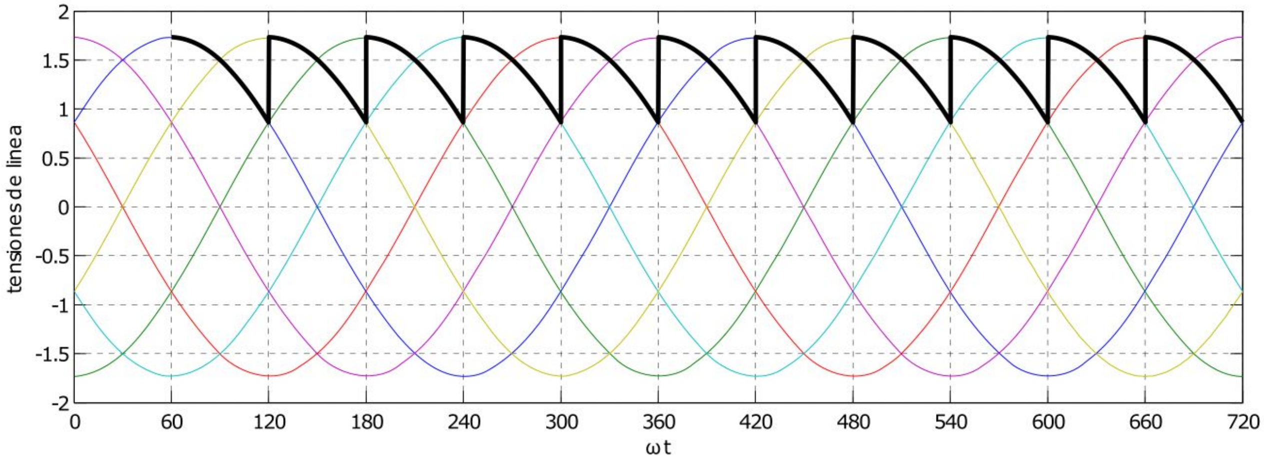


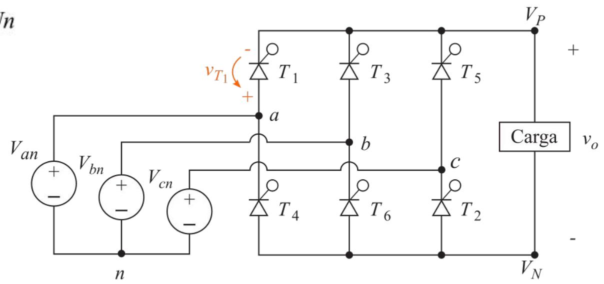
tensiones de linea
ωt
0 60 120 180 240 300 360 420 480 540 600 660 720
2 1.5 1 0.5 0 -0.5 -1 -1.5 -2

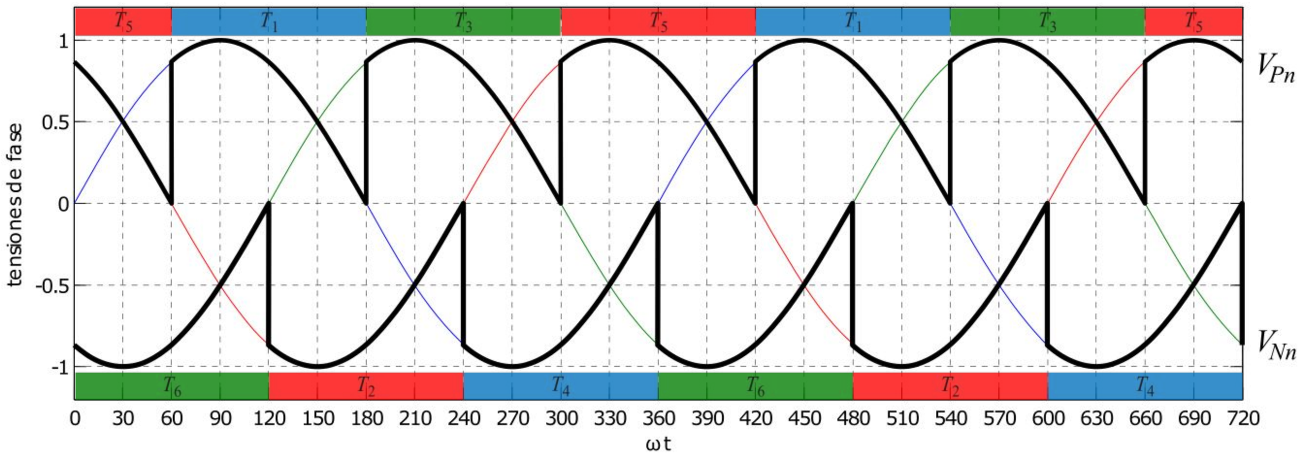


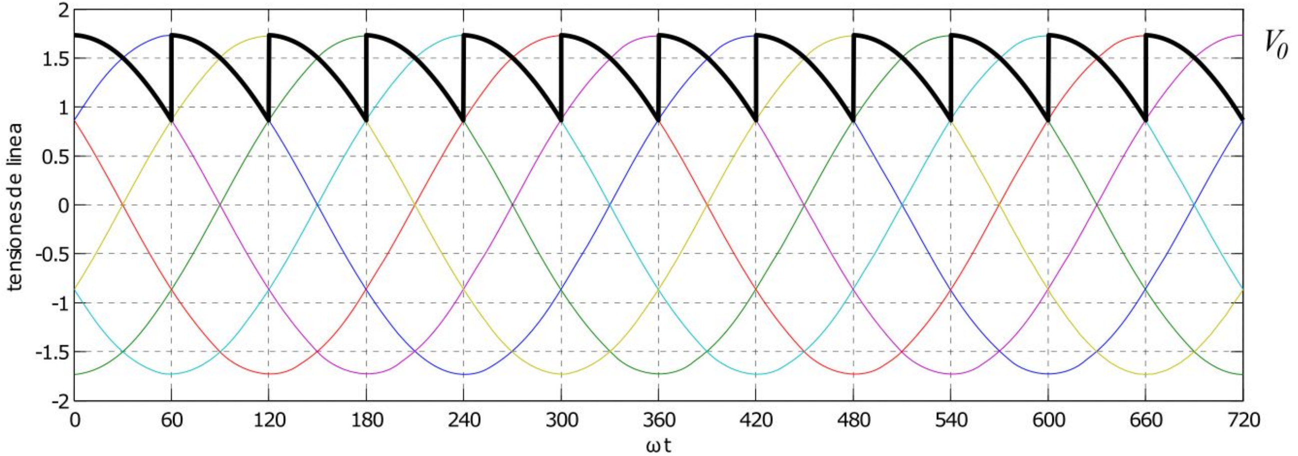
VP
VN
vT1
T1 T3 T5
T4 T6 T2
a
b
c
n
Van
Vbn
Vcn
Carga
vo
+
-

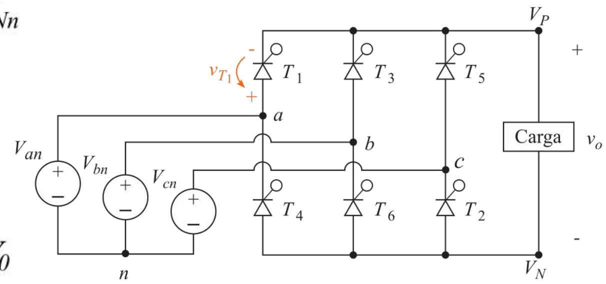

T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
VPn
VNn
tensiones de fase
ωt

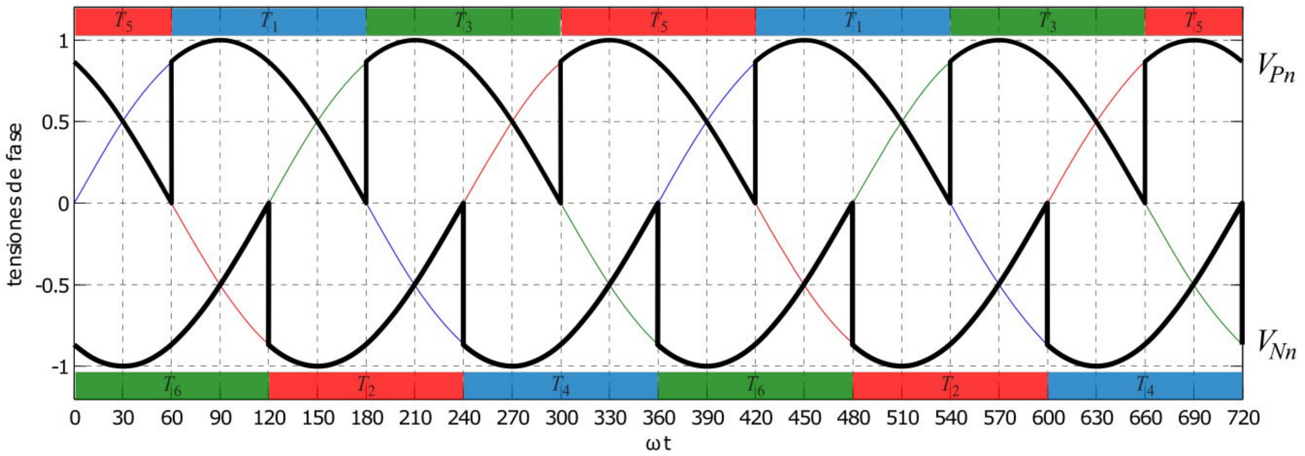

tensiones de linea
ωt

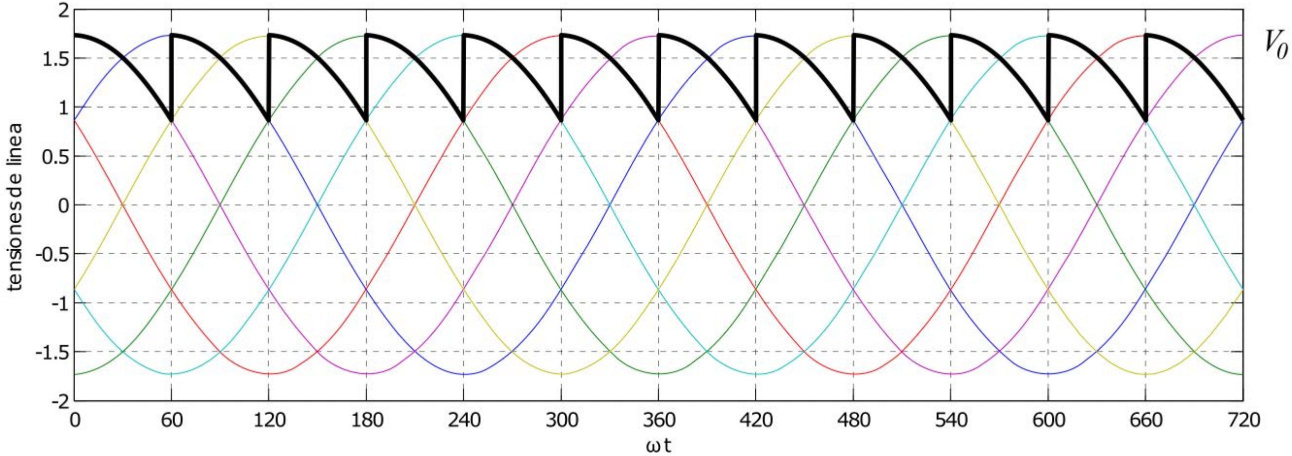

VP
VN
T1
T3
T5
T4
T6
T2
vT1
a
b
c
Van
Vbn
Vcn
n
Carga
vo
+
-

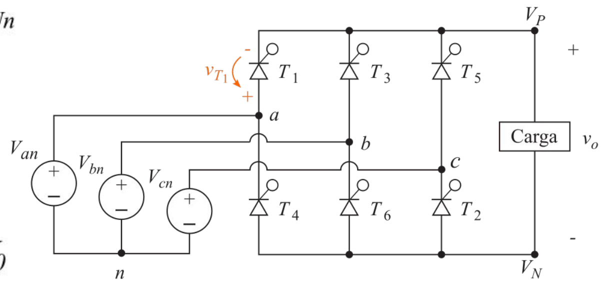

T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
VPn
VNn
tensiones de fase
ωt


V0
tensiones de linea
ωt

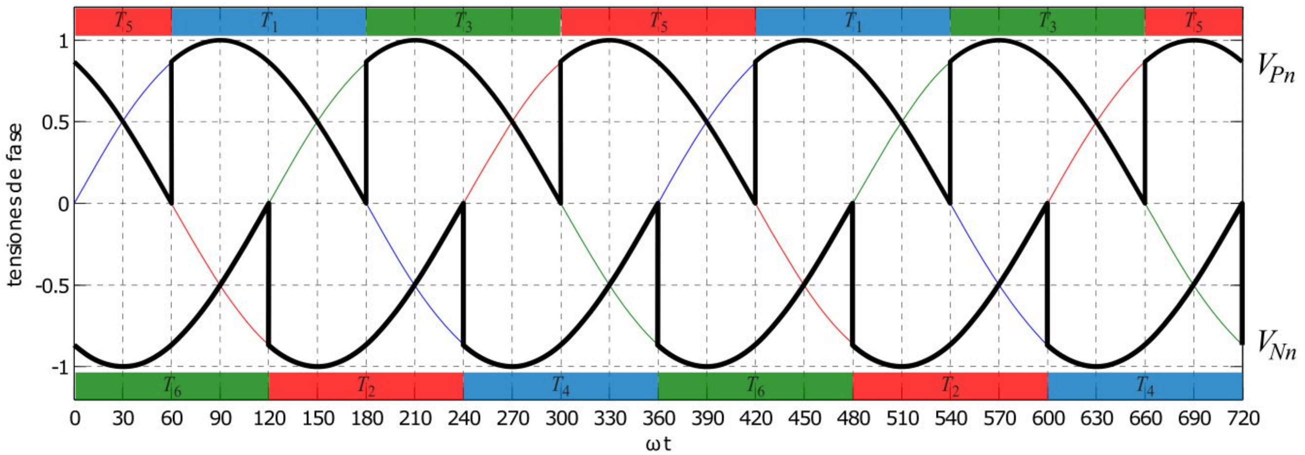

VP
VN
vT1
T1
T3
T5
T4
T6
T2
a
b
c
n
Van
Vbn
Vcn
Carga
vo

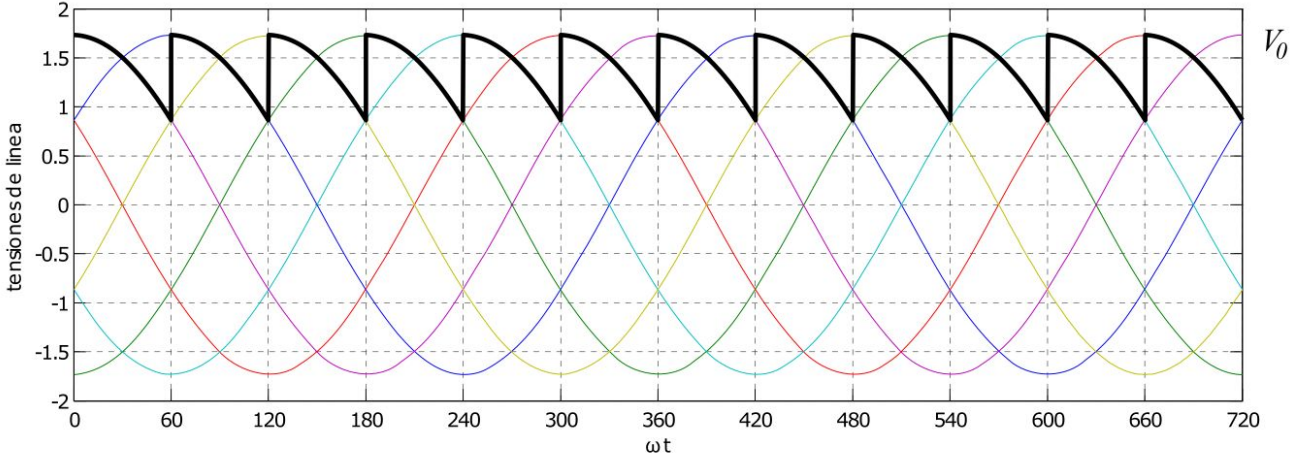


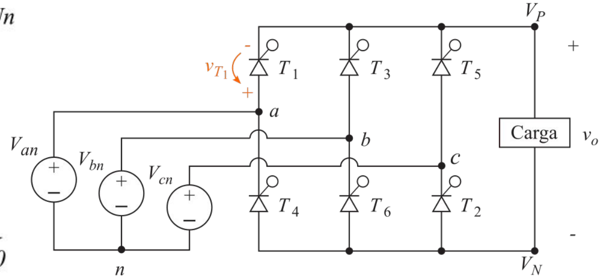




$$V_0 = \frac{1}{\pi/3} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,L-L} \sin(\omega t)\, d(\omega t)$$

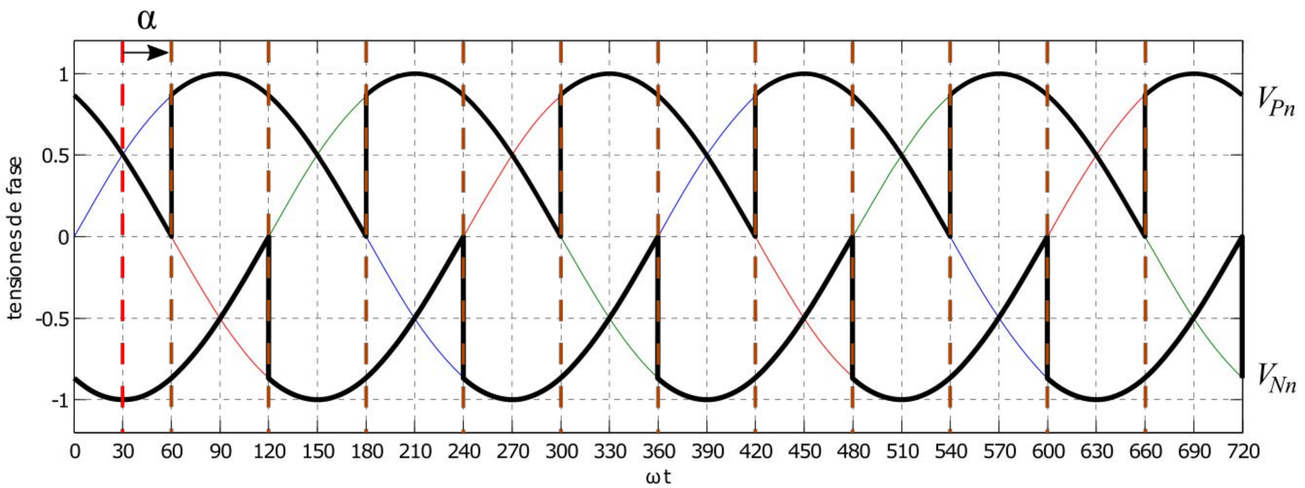


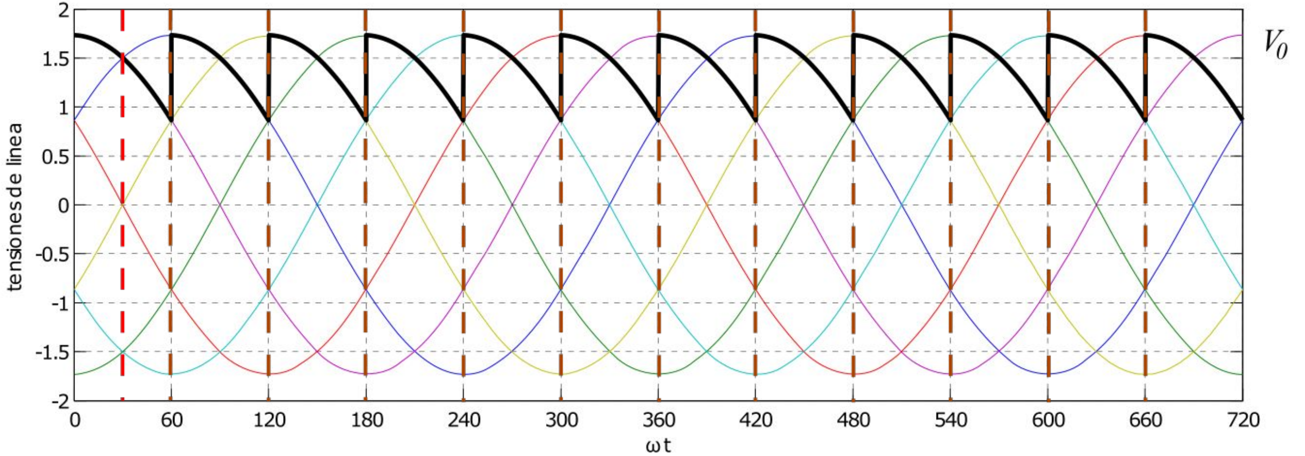


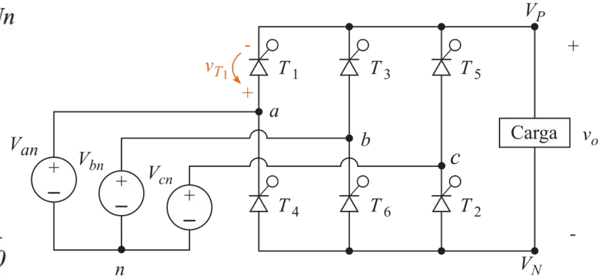


$$V_0 = \frac{1}{\pi/3} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,L-L} \sin(\omega t)\, d(\omega t) =$$

$$= \frac{3V_{m,L-L}}{\pi} \cos\alpha$$

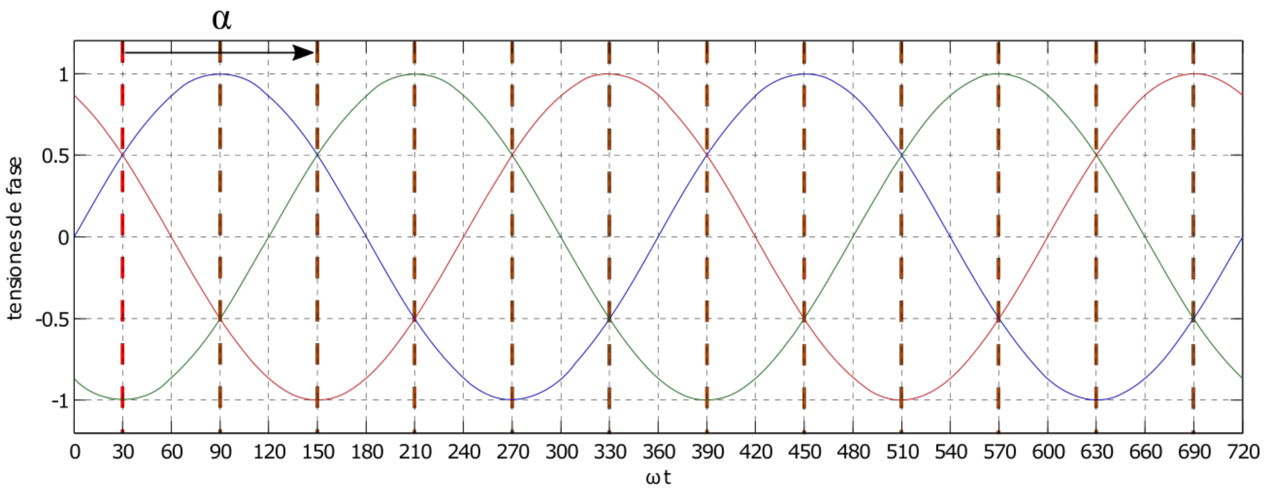


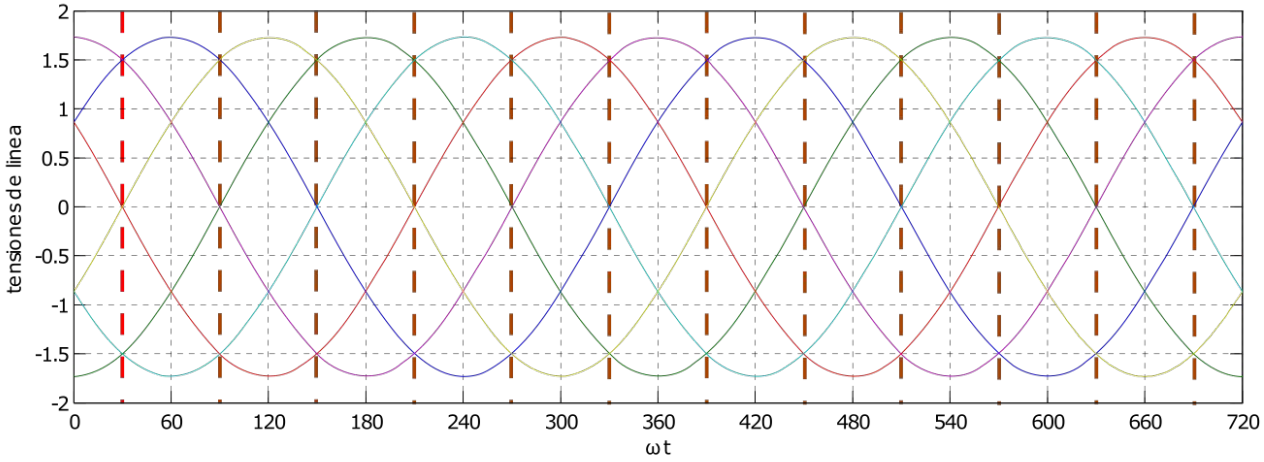


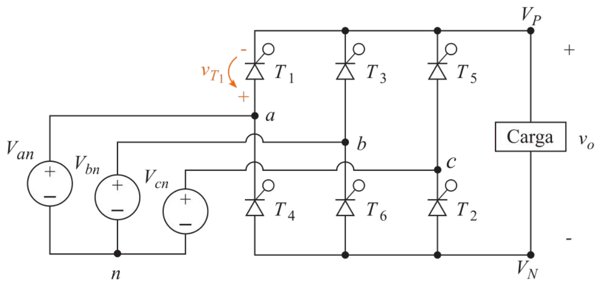


$$V_0 = \frac{1}{\pi/_3}\int_{\pi/_3+\alpha}^{2\pi/_3+\alpha} V_{m,L-L}\sin(\omega t)\, d(\omega t) =$$

$$= \frac{3V_{m,L-L}}{\pi}\cos\alpha$$

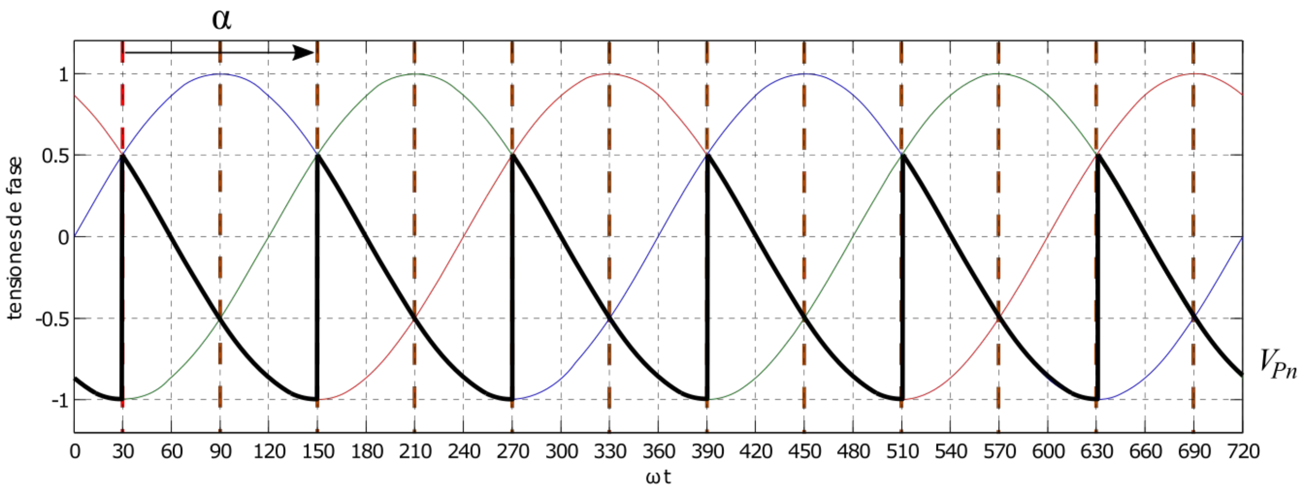


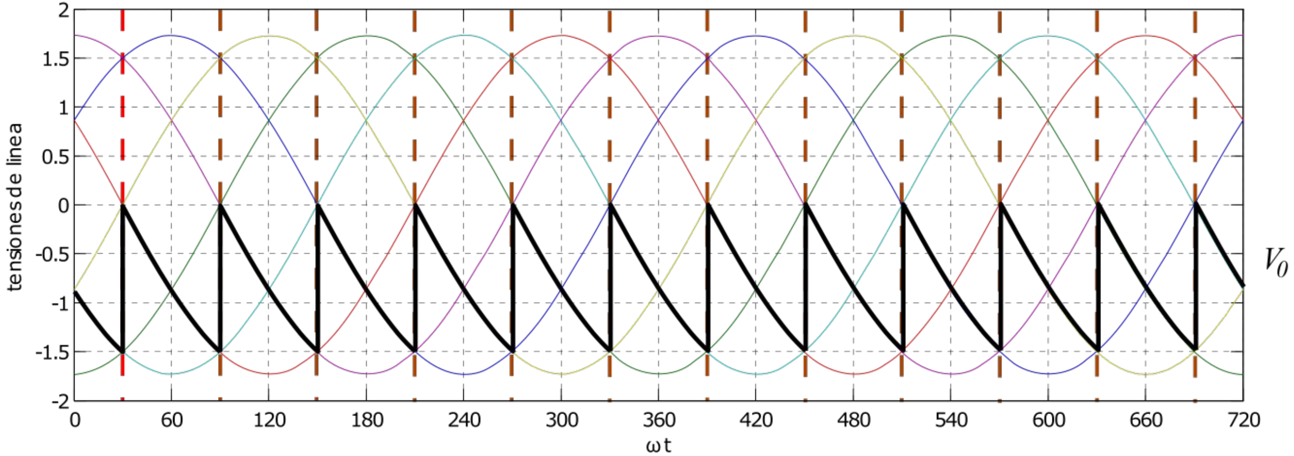


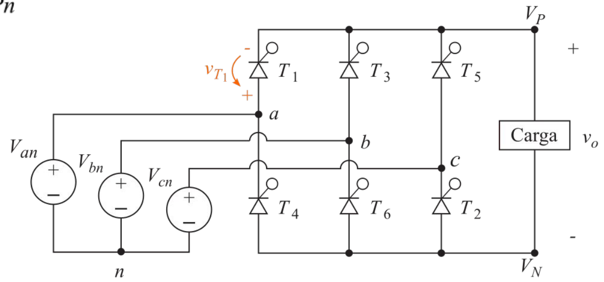


$$V_0 = \frac{1}{\pi/3}\int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,L-L}\sin(\omega t)\,d(\omega t) =$$

$$= \frac{3V_{m,L-L}}{\pi}\cos\alpha$$

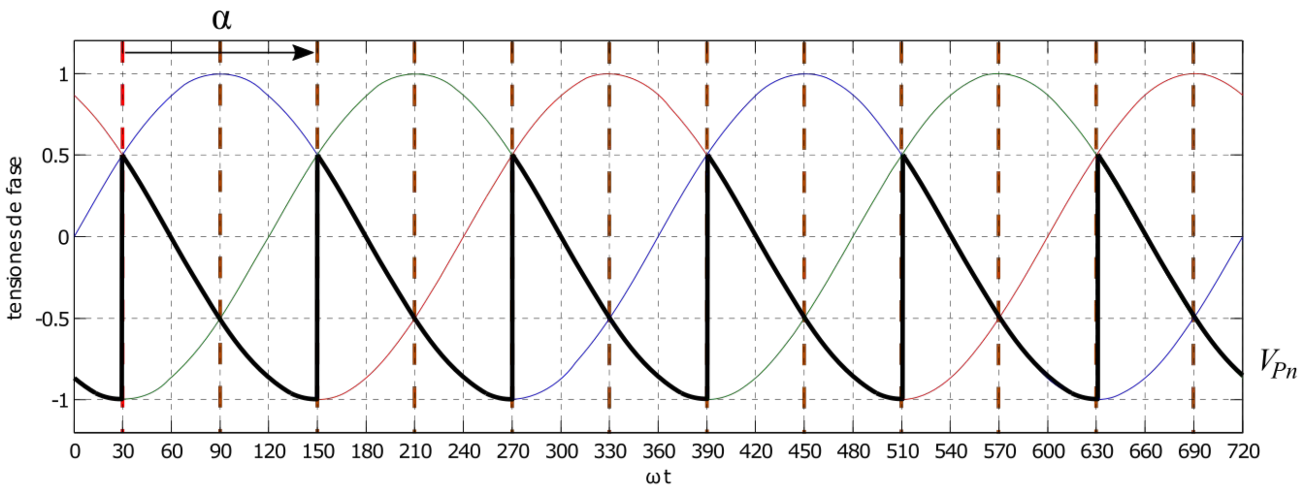


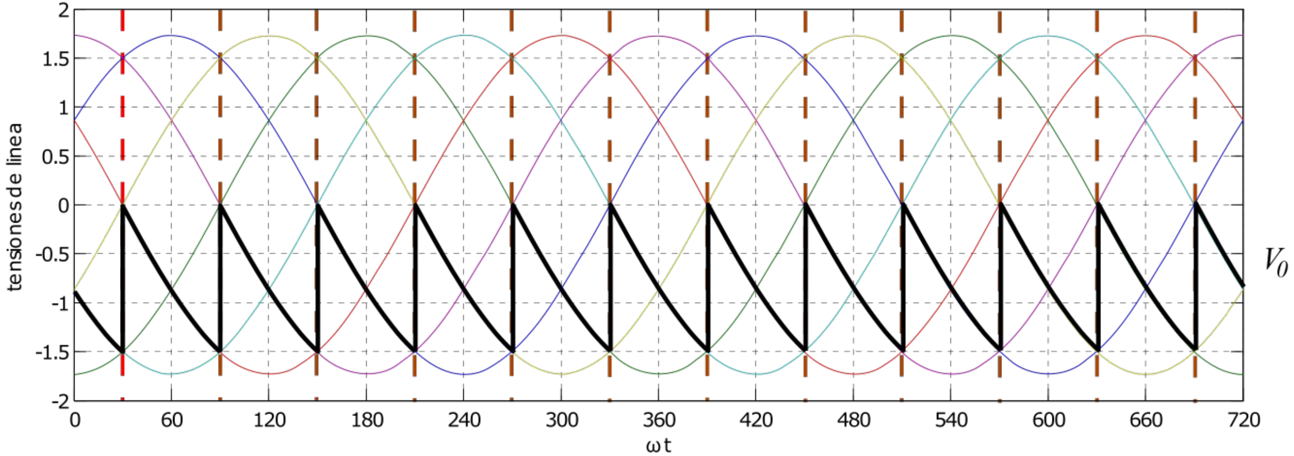


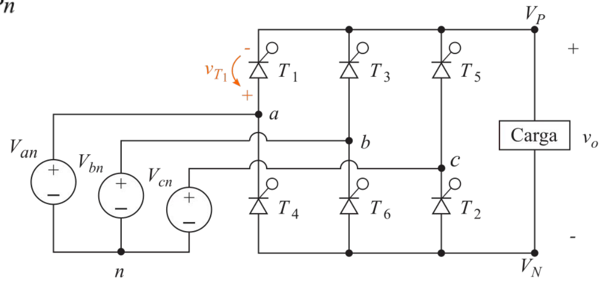


$$V_0 = \frac{1}{\pi/3}\int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,L-L}\sin(\omega t)\, d(\omega t) =$$

$$= \frac{3V_{m,L-L}}{\pi}\cos\alpha$$

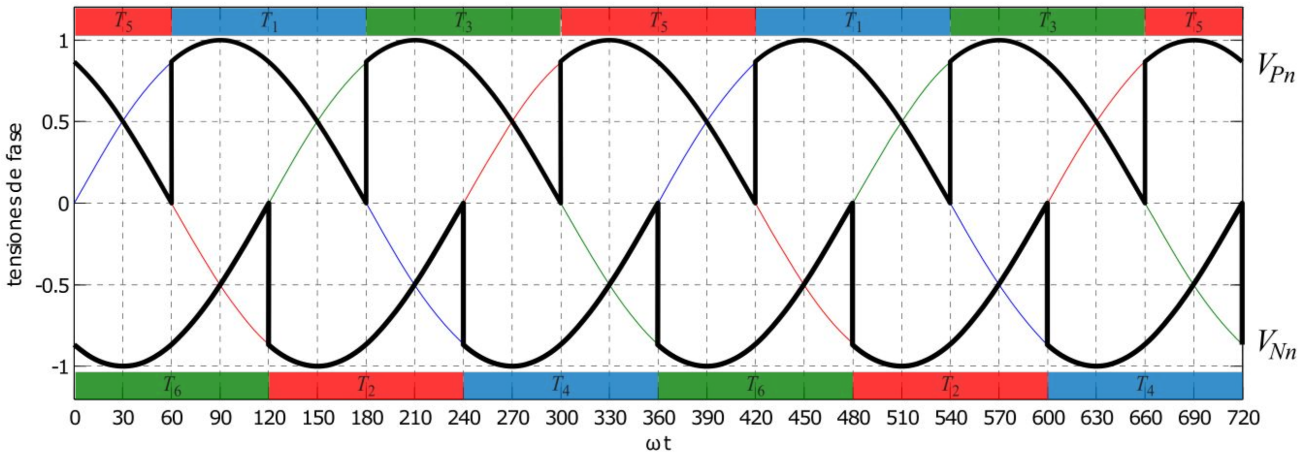




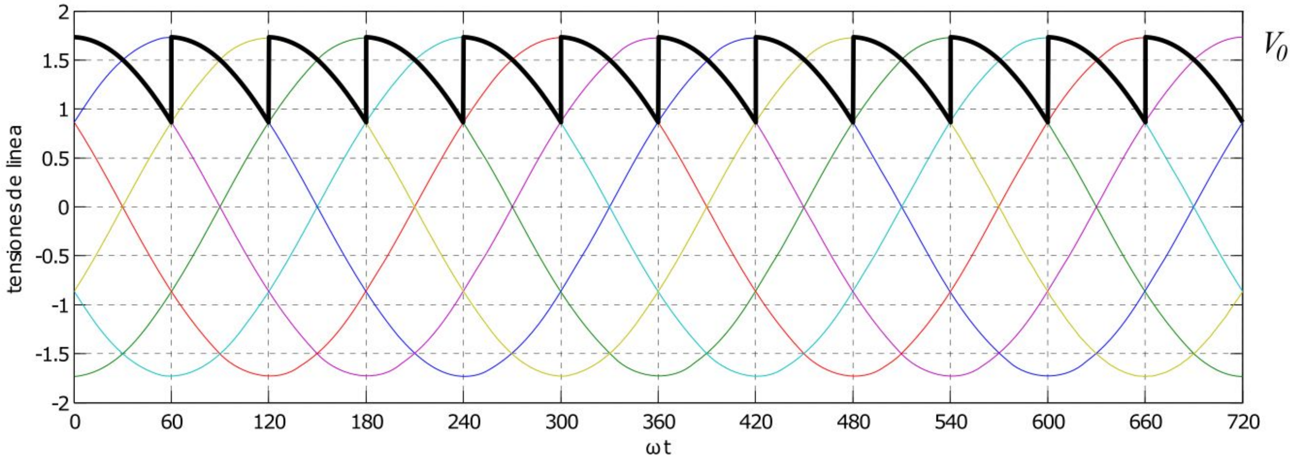


$0^\circ < \alpha < 90^\circ \longrightarrow V_0 > 0,$ **rectificador**;

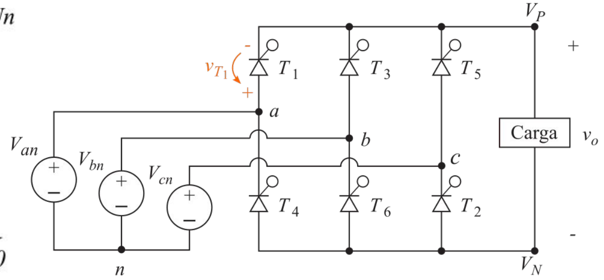
$90^\circ < \alpha < 180^\circ \longrightarrow V_0 < 0,$ **inversor**;



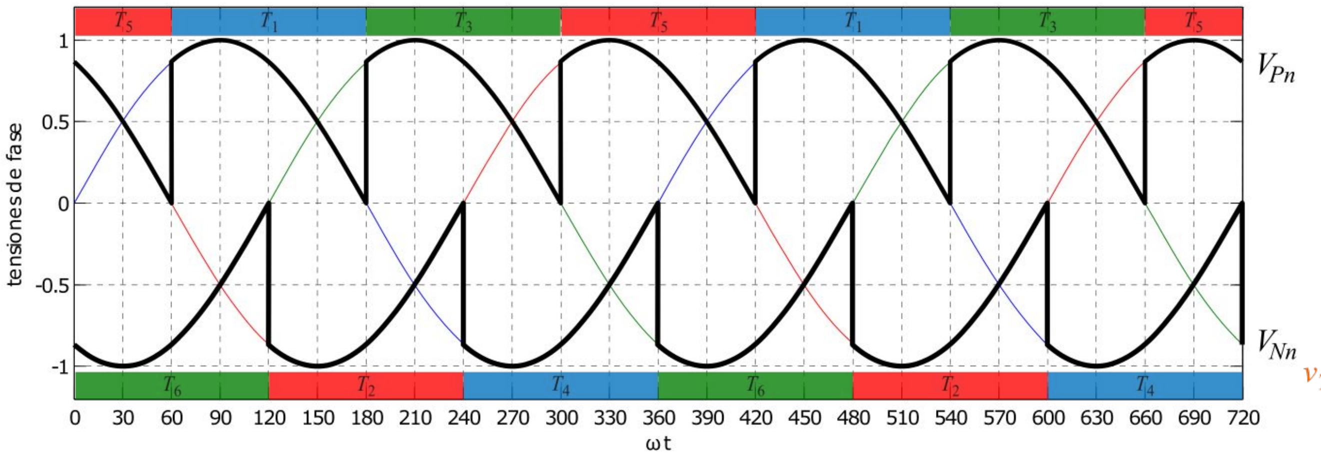
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
tensiones de fase
ωt
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
1 0.5 0 -0.5 -1
V_{Pn}
V_{Nn}

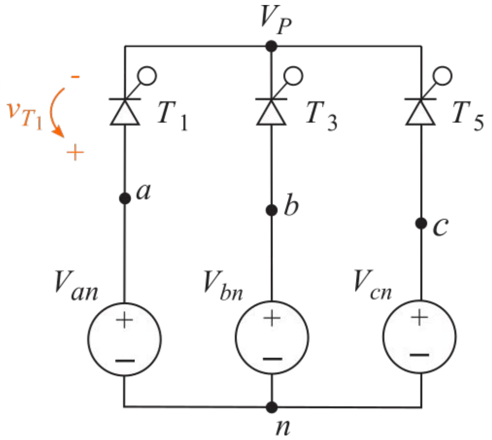


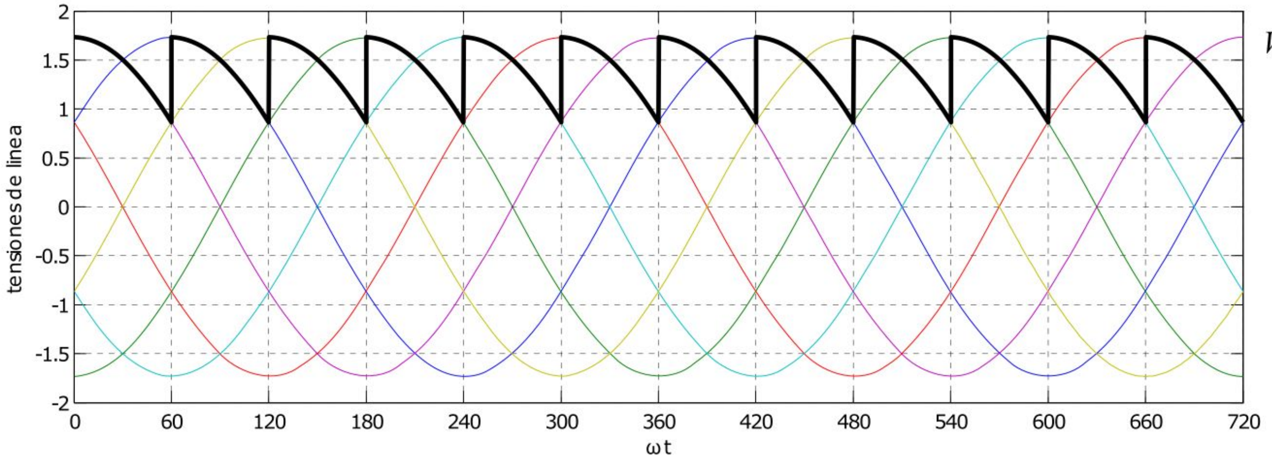
tensiones de linea
ωt
0 60 120 180 240 300 360 420 480 540 600 660 720
2 1.5 1 0.5 0 -0.5 -1 -1.5 -2
V_0

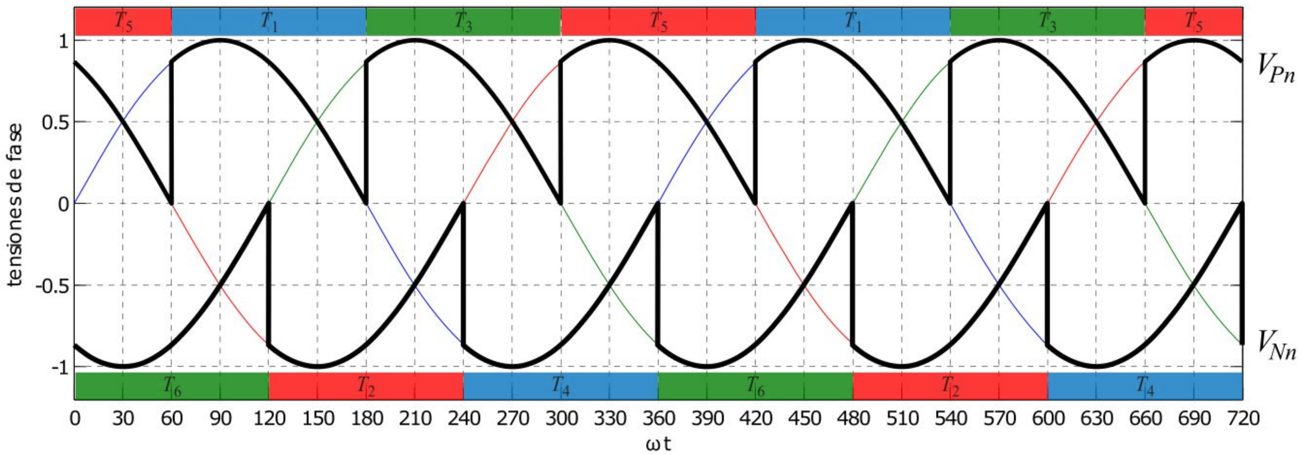


V_P
V_N
v_{T_1}
T_1 T_3 T_5
T_4 T_6 T_2
a
b
c
n
V_{an} V_{bn} V_{cn}
Carga
v_o
+
-

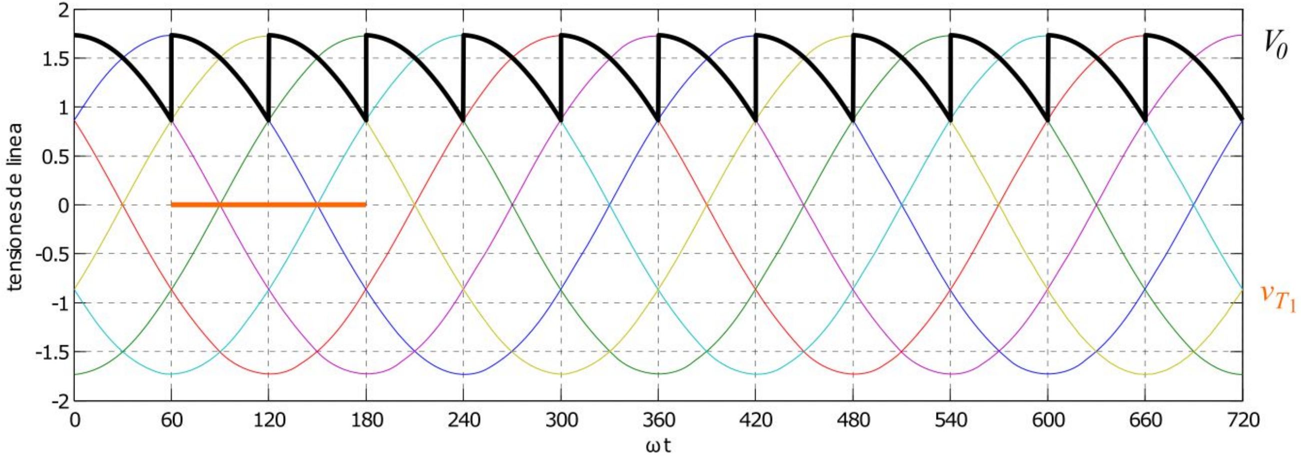

T5
T1
T3
T5
T1
T3
T5
V_Pn
V_Nn
T6
T2
T4
T6
T2
T4
tensiones de fase
ωt


V_P
T1
T3
T5
v_T1
-
+
a
b
c
V_an
V_bn
V_cn
n

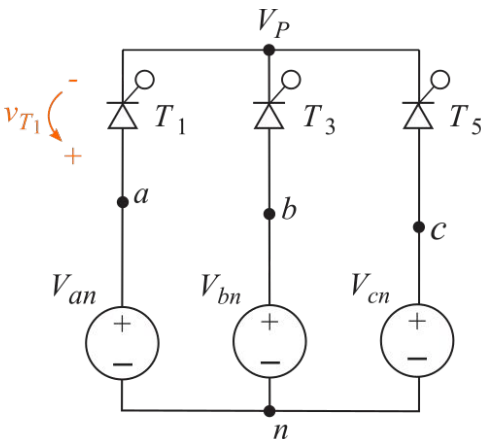

V_0
tensiones de linea
ωt



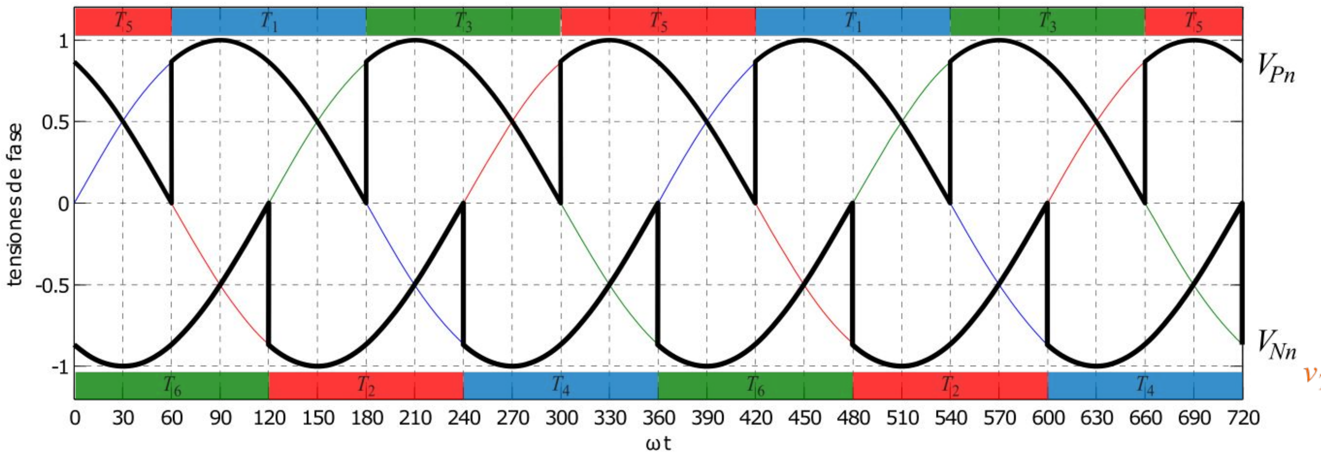
T5 T1 T3 T5 T1 T3 T5
V_Pn
V_Nn
T6 T2 T4 T6 T2 T4
tensiones de fase
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
ωt



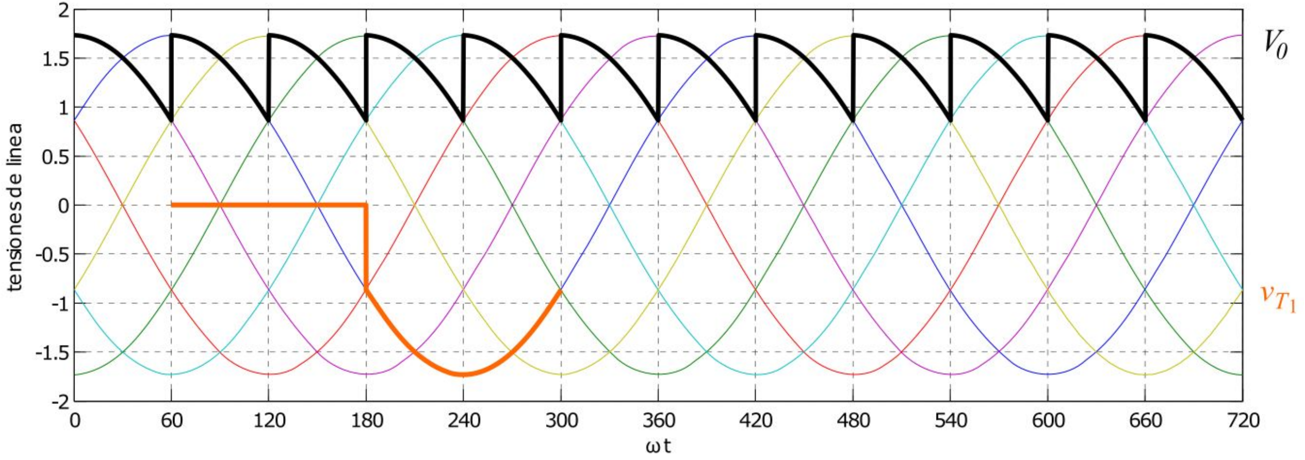
V_0
v_T1
tensiones de linea
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0 60 120 180 240 300 360 420 480 540 600 660 720
ωt



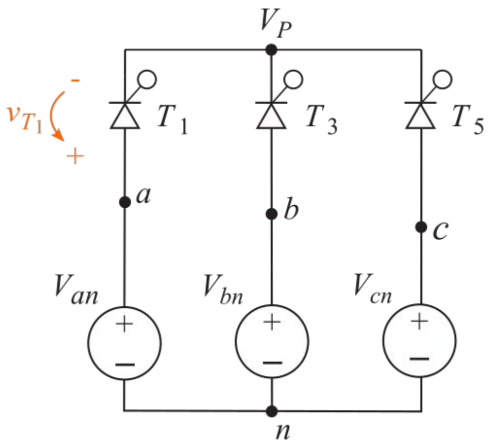
V_P
v_T1
-
+
T1
T3
T5
a
b
c
V_an
V_bn
V_cn
n



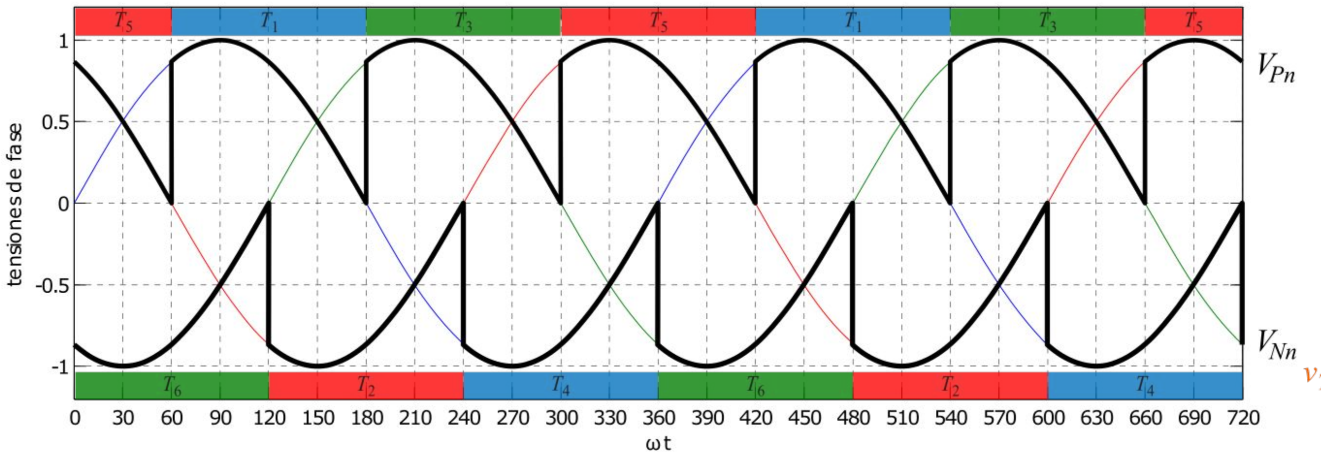
T_5
T_1
T_3
T_5
T_1
T_3
T_5
T_6
T_2
T_4
T_6
T_2
T_4
V_{Pn}
V_{Nn}
tensiones de fase
ωt

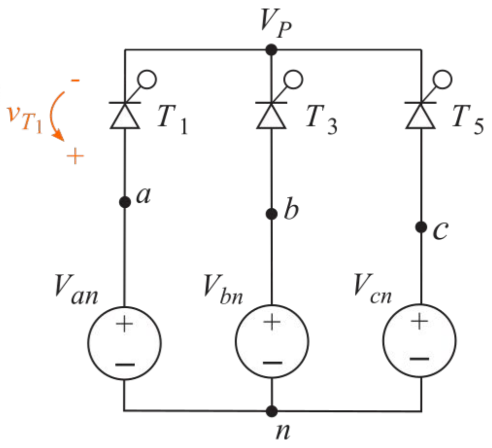


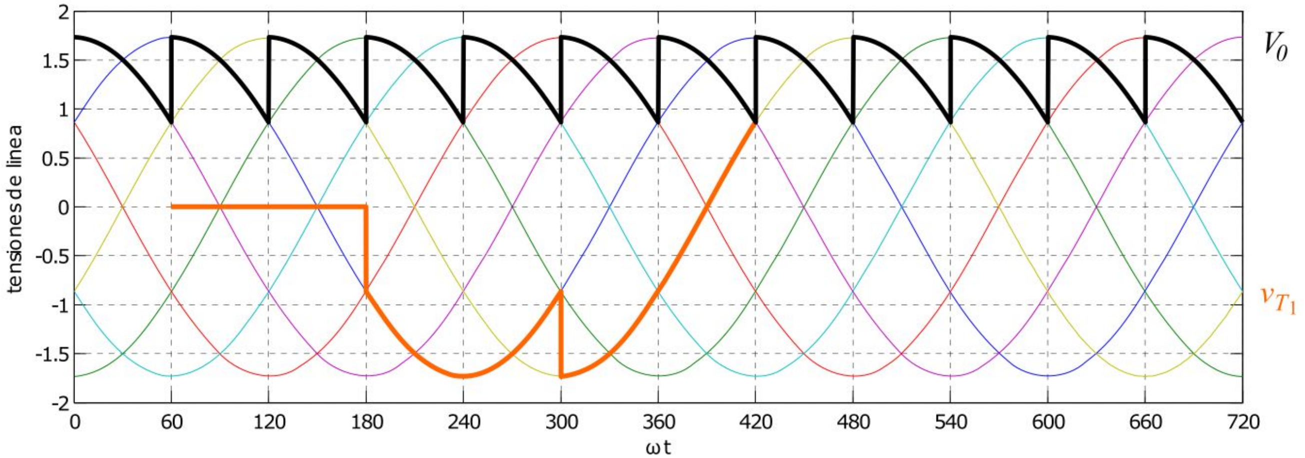
V_0
v_{T_1}
tensiones de linea
ωt

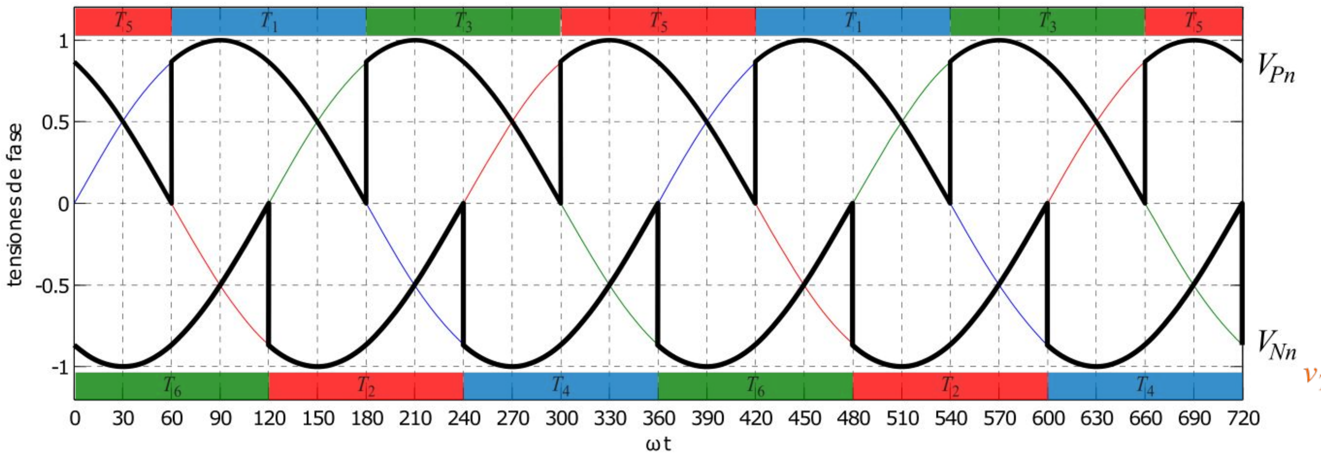


V_P
v_{T1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
n

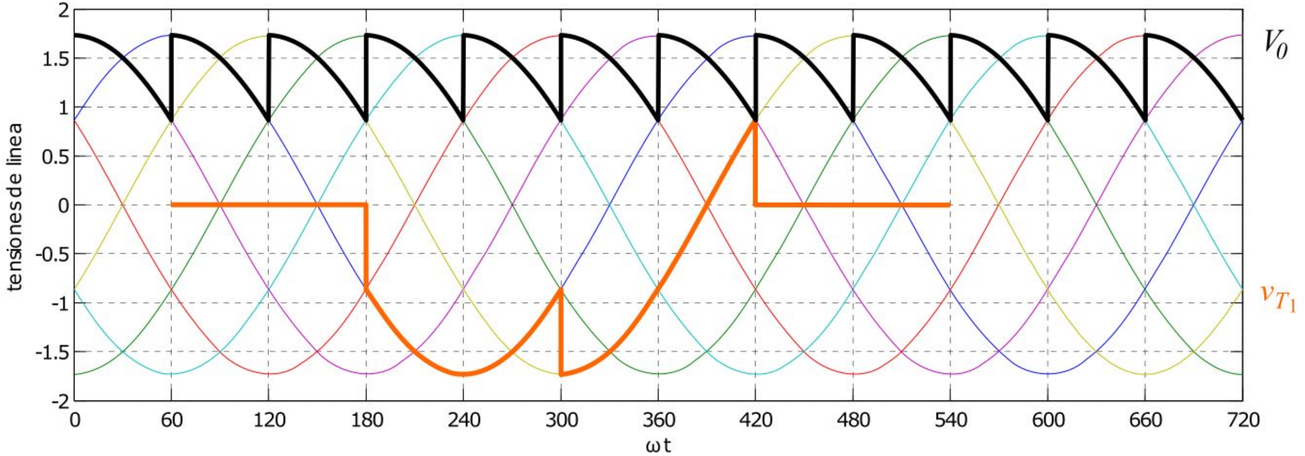

T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
V_Pn
V_Nn
tensiones de fase
ωt


V_P
T 1
T 3
T 5
v_T1
-
+
a
b
c
V_an
V_bn
V_cn
n

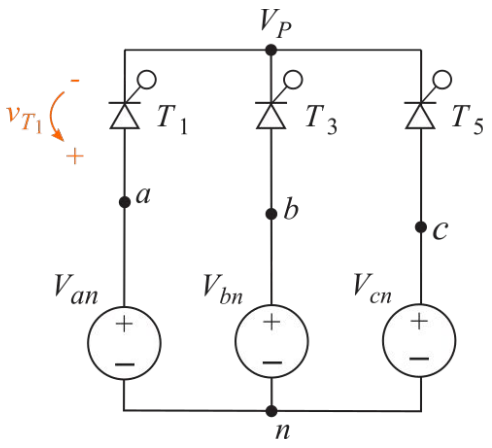

V_0
v_T1
tensiones de linea
ωt



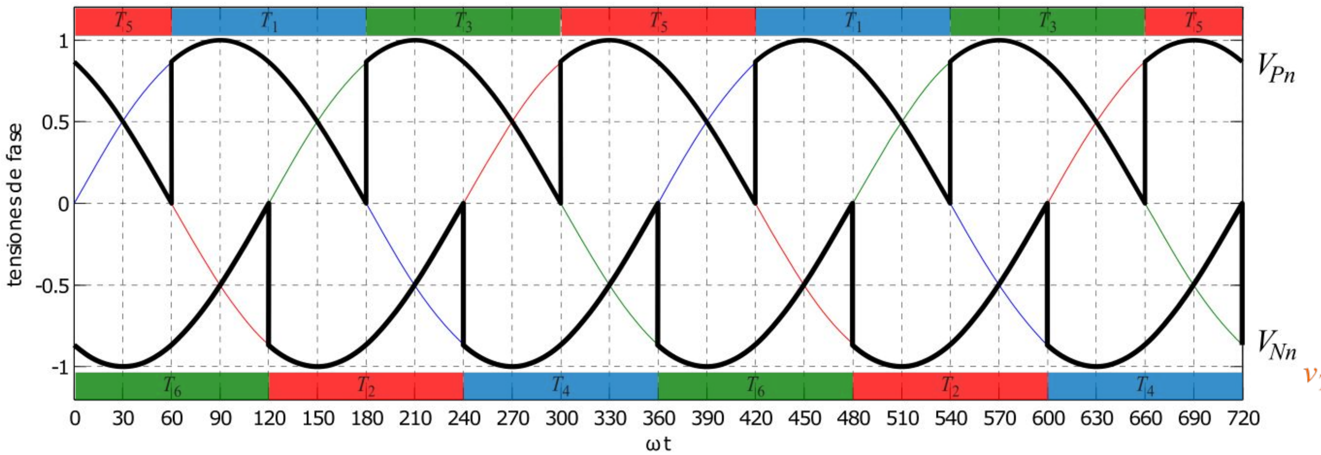
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt



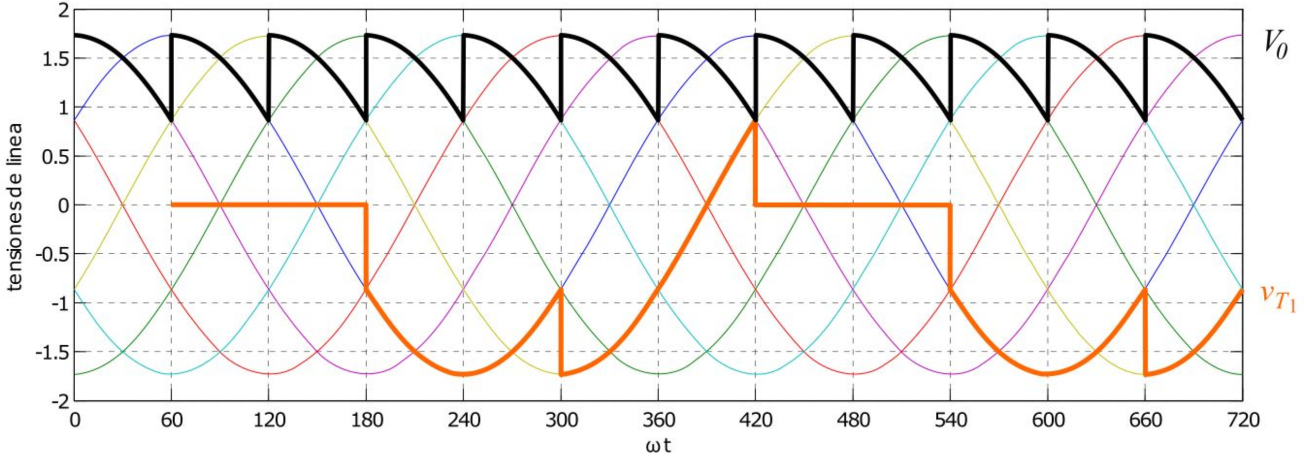
V0
vT1
tensiones de linea
ωt



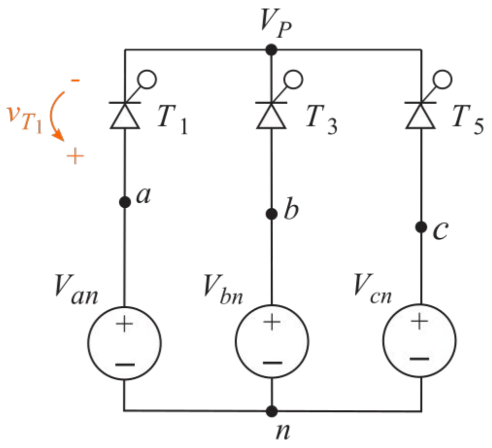
VP
vT1
T1
T3
T5
a
b
c
Van
Vbn
Vcn
n



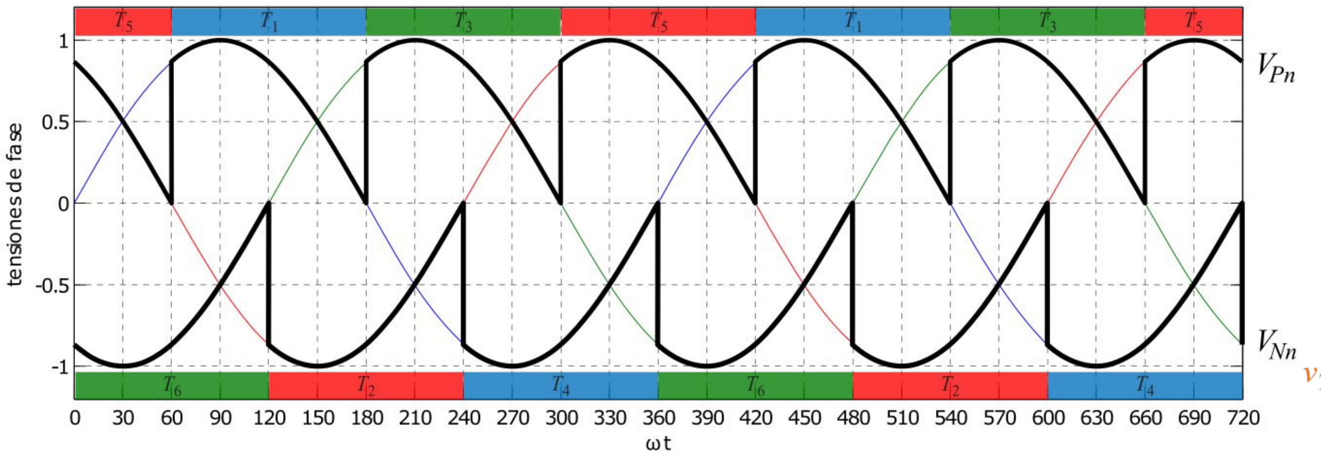
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
VPn
VNn
tensiones de fase
ωt



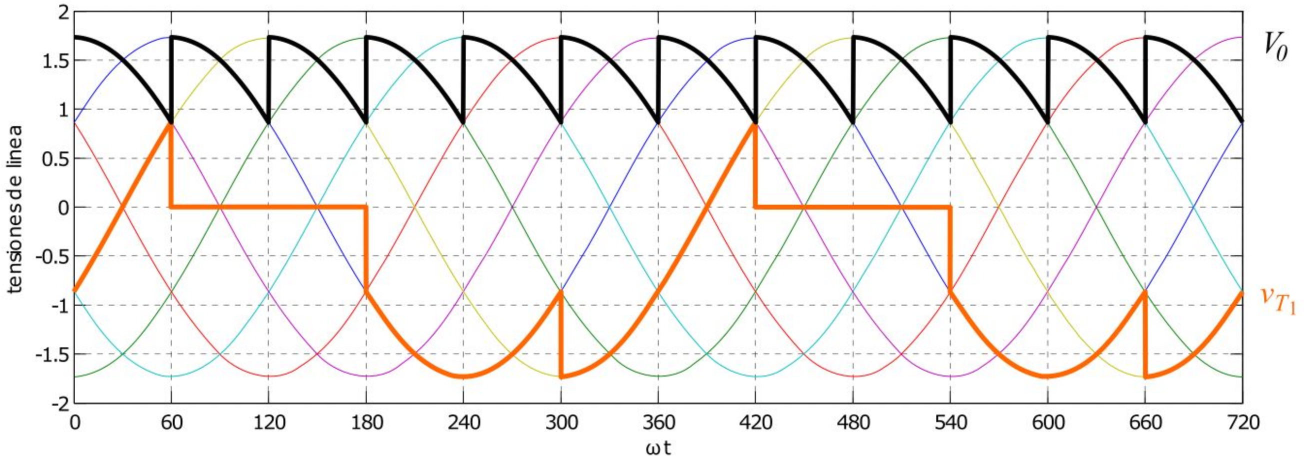
V0
vT1
tensiones de linea
ωt



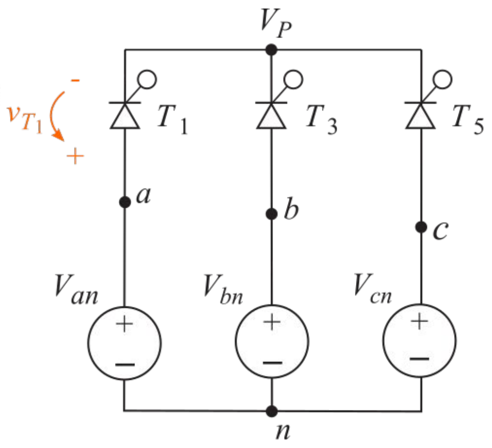
VP
vT1
T1
T3
T5
a
b
c
Van
Vbn
Vcn
n



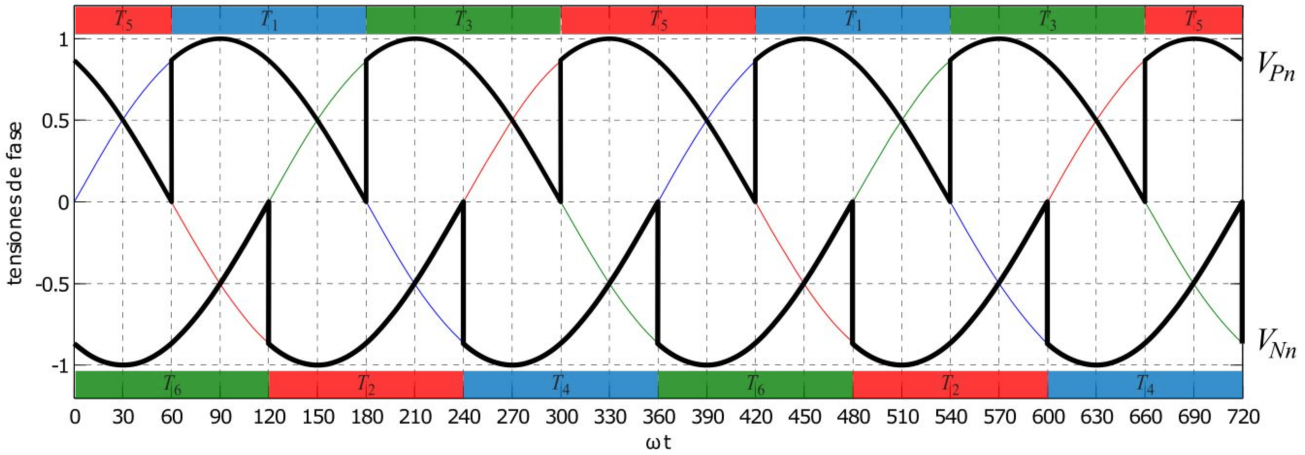
T5
T1
T3
T5
T1
T3
T5
VPn
VNn
T6
T2
T4
T6
T2
T4
tensiones de fase
ωt

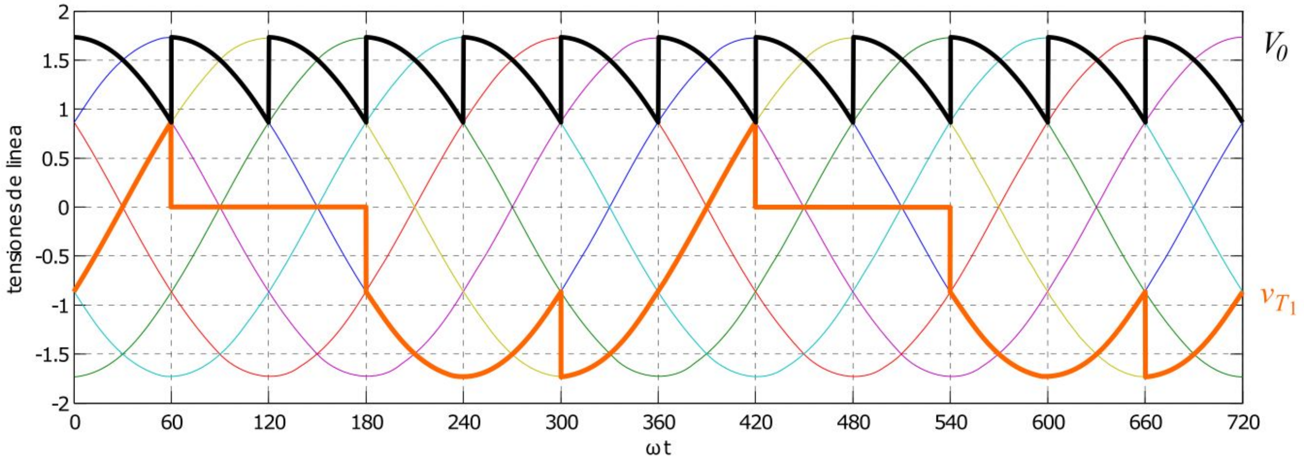


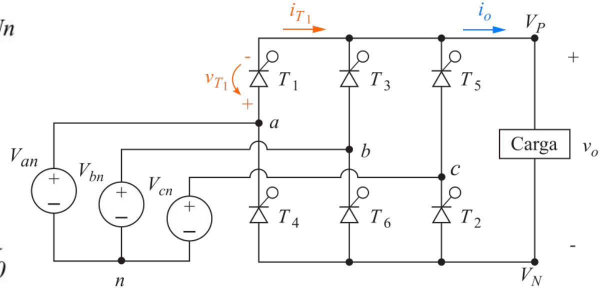
V0
vT1
tensiones de linea
ωt

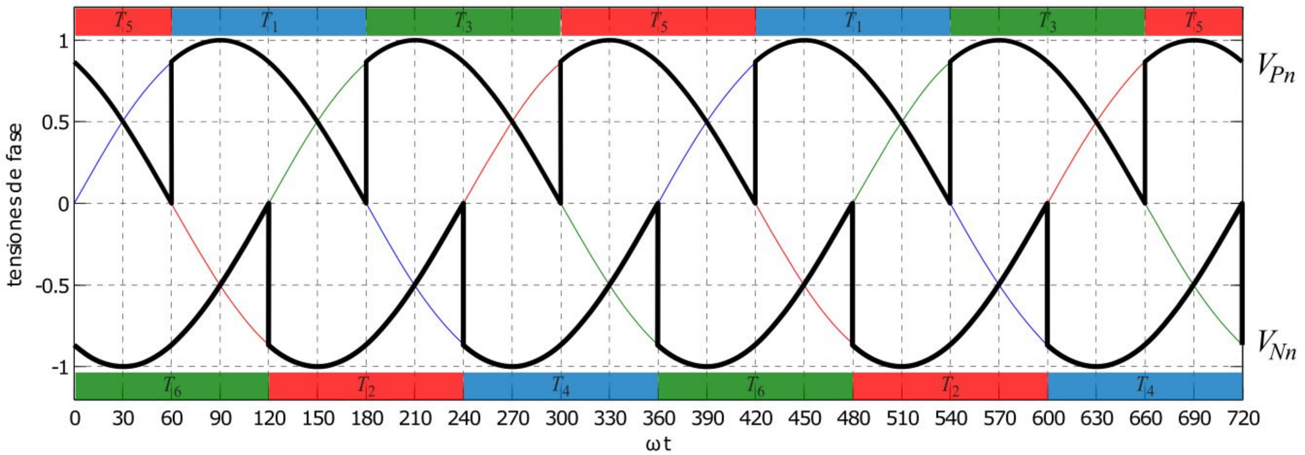


VP
vT1
T1
T3
T5
a
b
c
Van
Vbn
Vcn
n

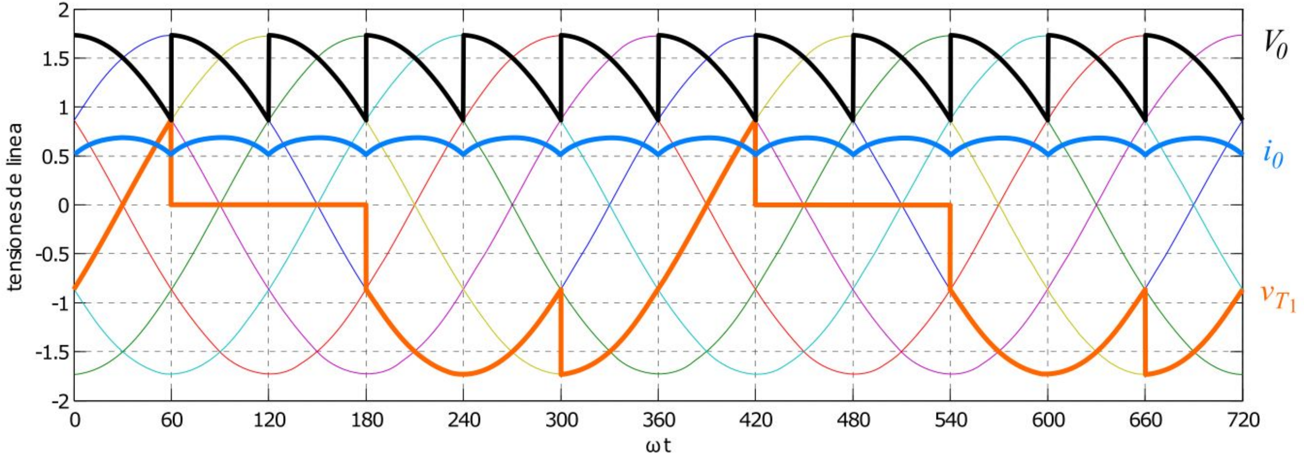

T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
V_Pn
V_Nn
tensiones de fase
ωt


V_0
v_T1
tensiones de linea
ωt

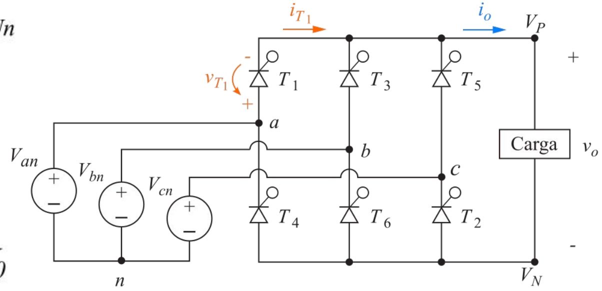

i_T1
i_o
V_P
V_N
v_T1
T1
T3
T5
T4
T6
T2
a
b
c
n
V_an
V_bn
V_cn
Carga
v_o



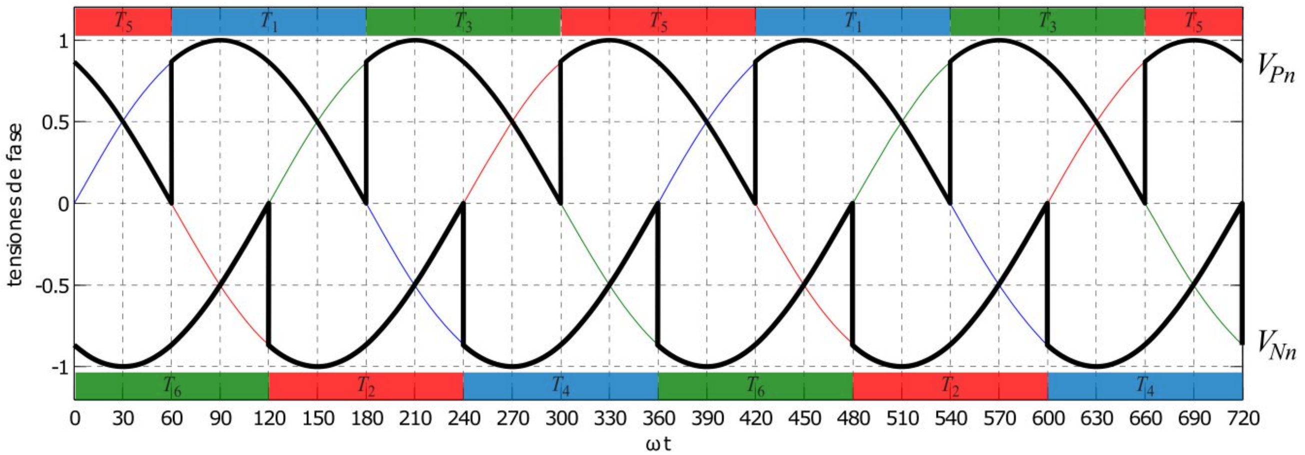
T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
tensiones de fase
ωt
V_Pn
V_Nn



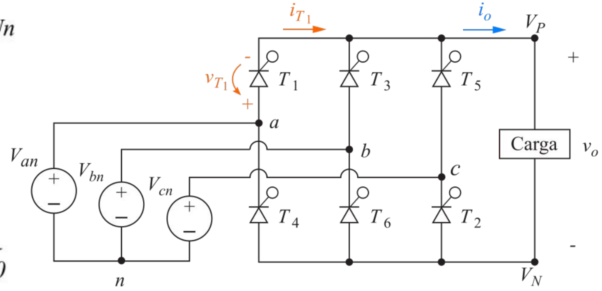
tensiones de linea
ωt
V_0
i_0
v_T1



i_T1
i_o
V_P
V_N
v_T1
T1 T3 T5
T4 T6 T2
a
b
c
Carga
v_o
V_an
V_bn
V_cn
n



T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
tensiones de fase
ωt
V_{Pn}
V_{Nn}



i_{T_1}
i_o
V_P
v_{T_1}
T1 T3 T5
T4 T6 T2
a
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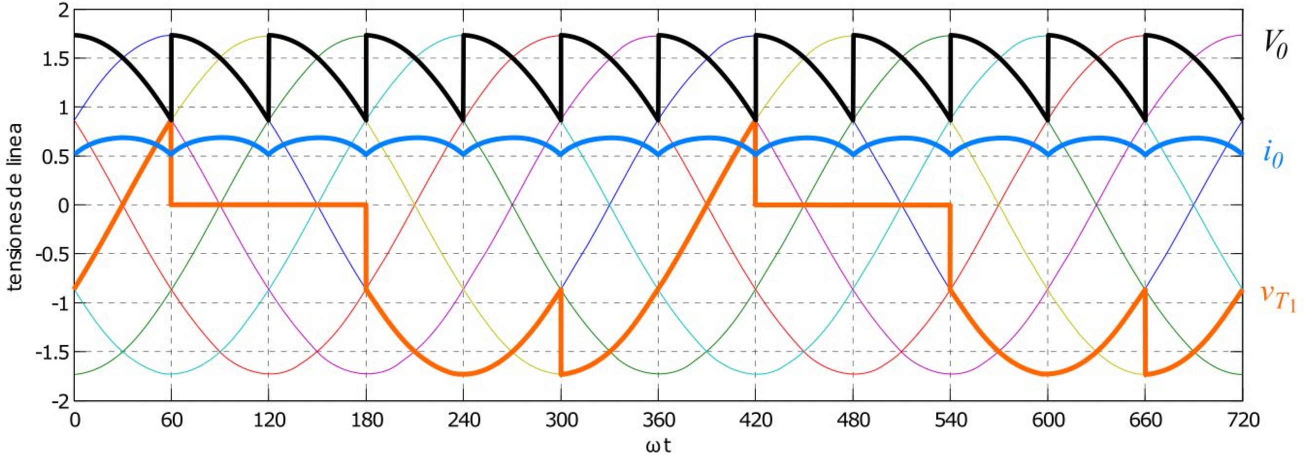
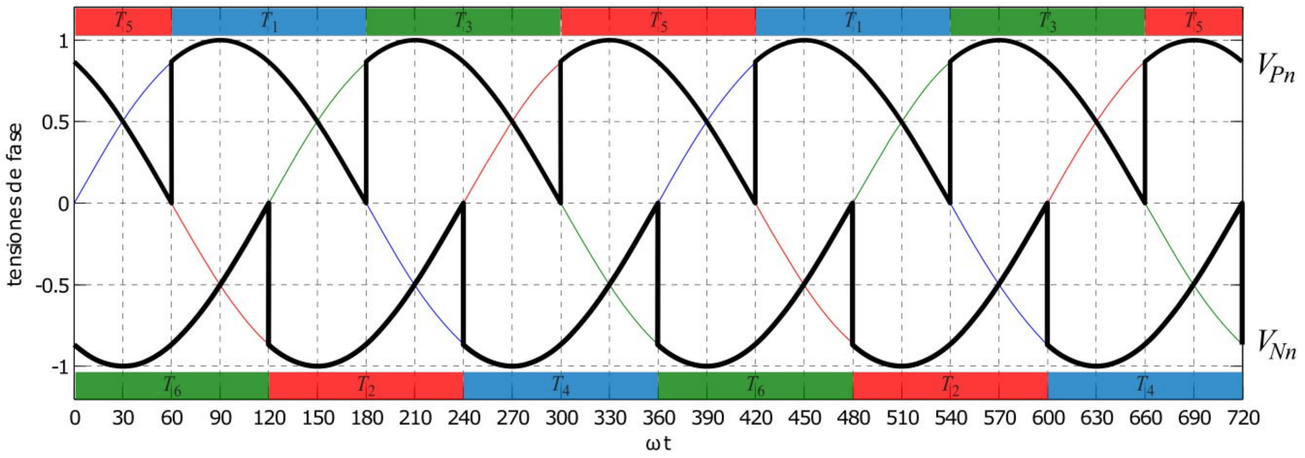
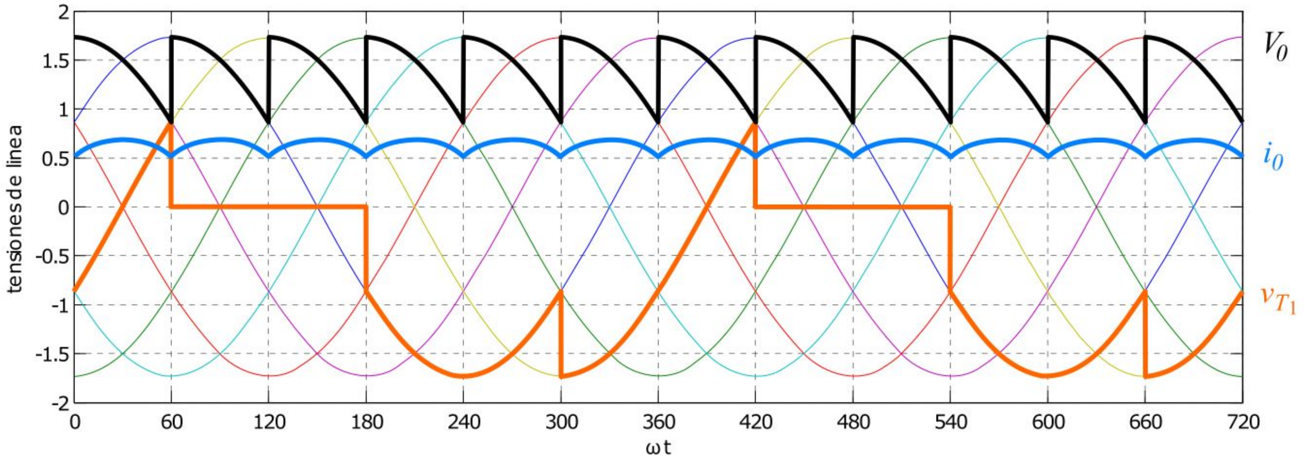
c
V_{an}
V_{bn}
V_{cn}
Carga
v_o
n
V_N

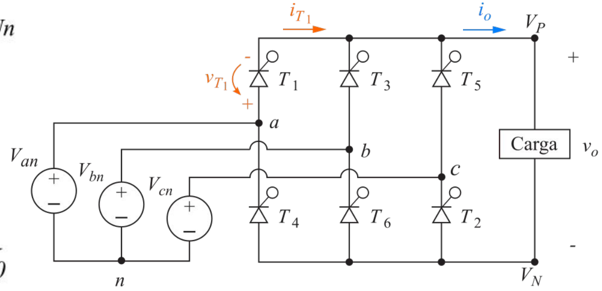


tensiones de linea
ωt
V_0
i_0
v_{T_1}

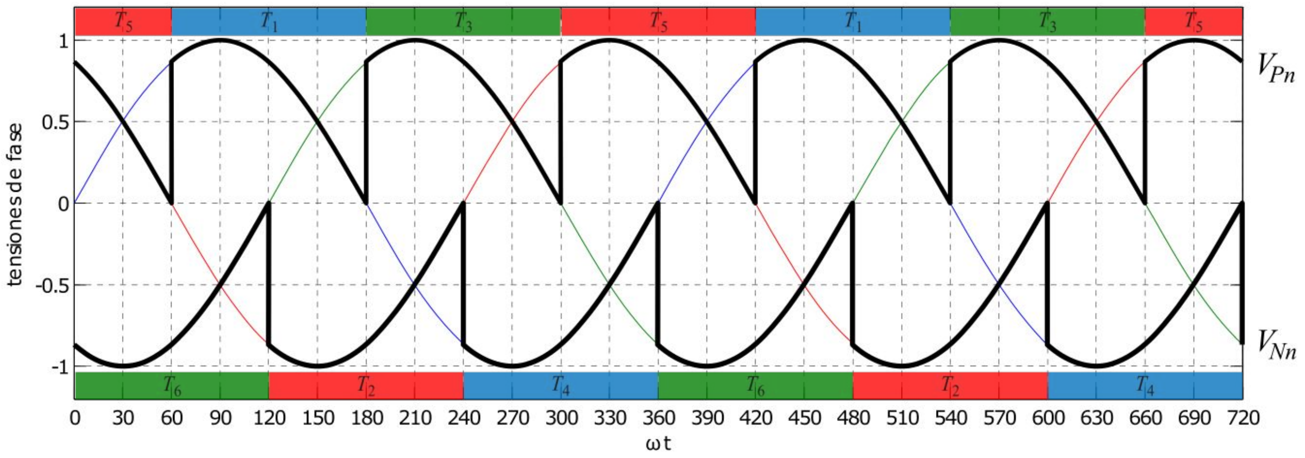
$$P_0 = V_0 \cdot I_0$$

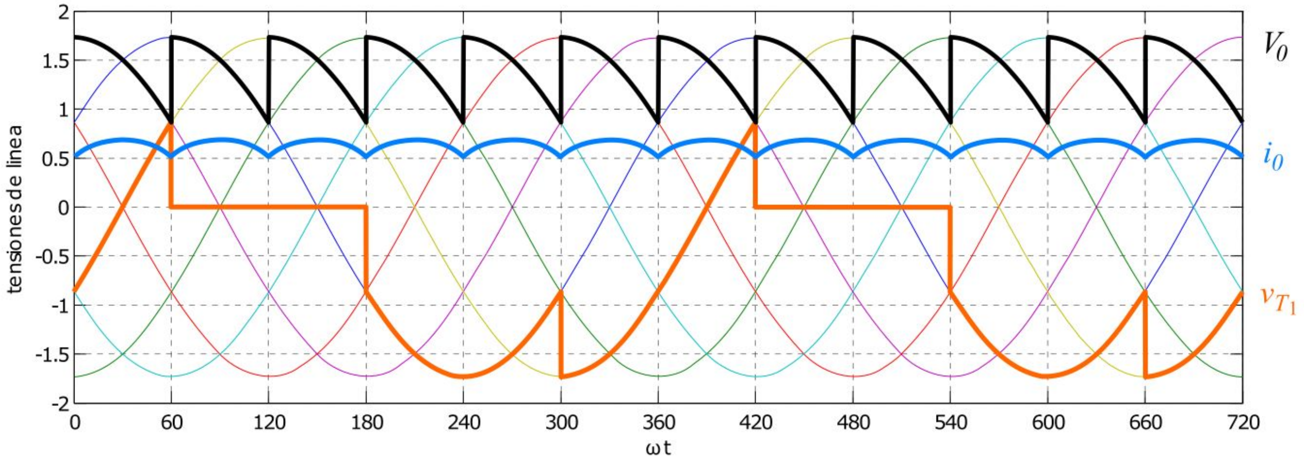


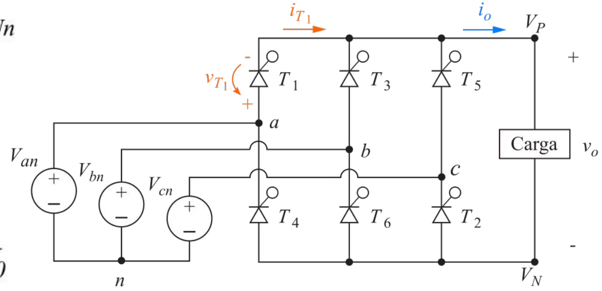




$$P_0 = V_0 \cdot I_0 = \frac{3}{\pi} V_{m,L-L} \cdot I_0 \cdot \cos\alpha$$


T5 T1 T3 T5 T1 T3 T5
T6 T2 T4 T6 T2 T4
tensiones de fase
ωt
VPn
VNn


tensiones de linea
ωt
V0
i0
vT1


iT1
io
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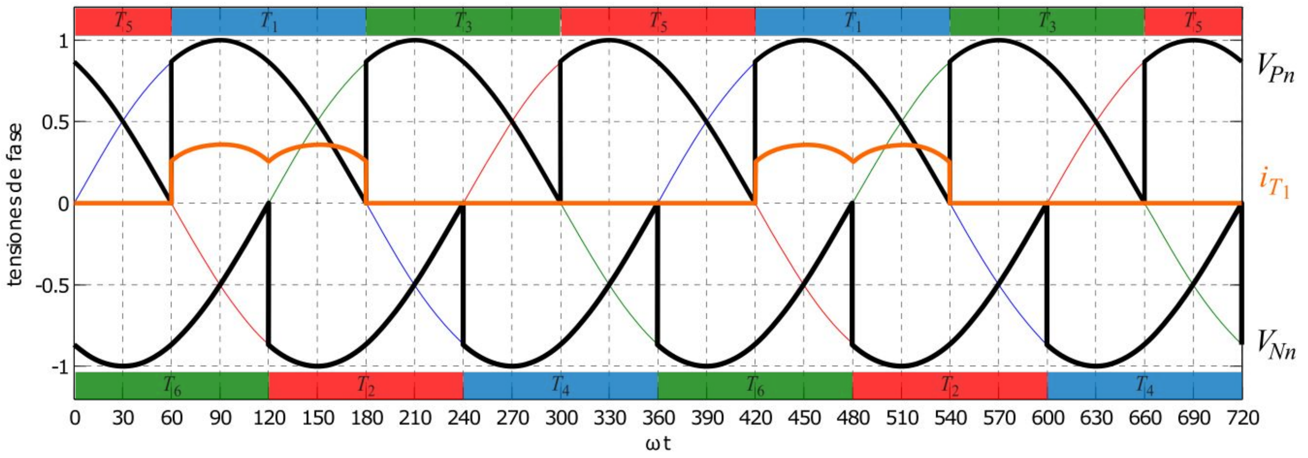
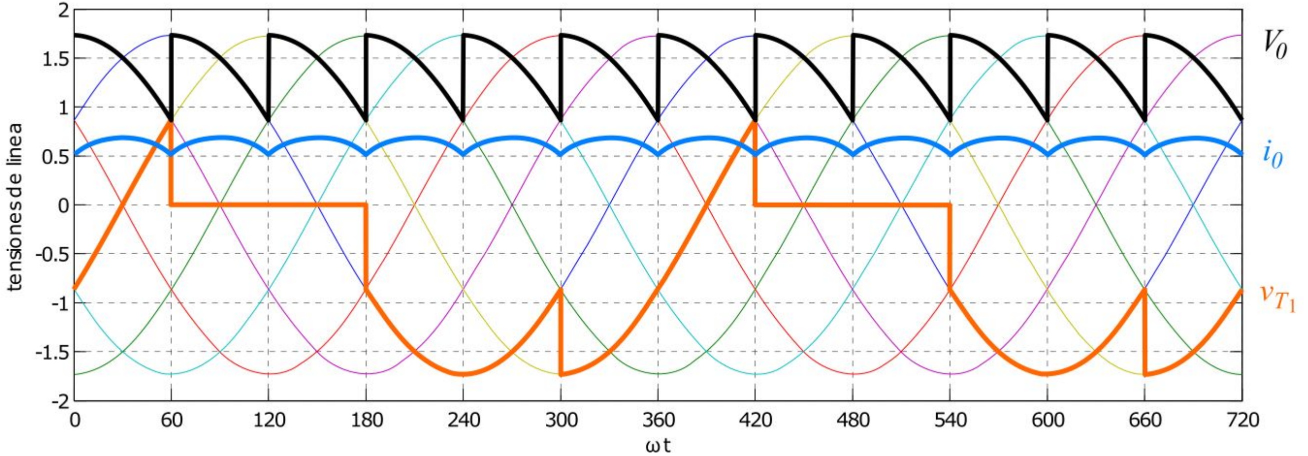
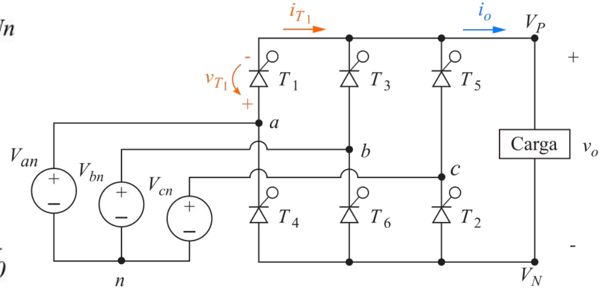
vT1
T1 T3 T5
T4 T6 T2
a
b
c
Van
Vbn
Vcn
n
Carga
vo
VN



T5
T1
T3
T5
T1
T3
T5
T6
T2
T4
T6
T2
T4
V_Pn
i_T1
V_Nn
tensiones de fase
ωt



V_0
i_0
v_T1
tensiones de linea
ωt



i_T1
i_o
V_P
V_N
v_T1
T1
T3
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T4
T6
T2
a
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V_an
V_bn
V_cn
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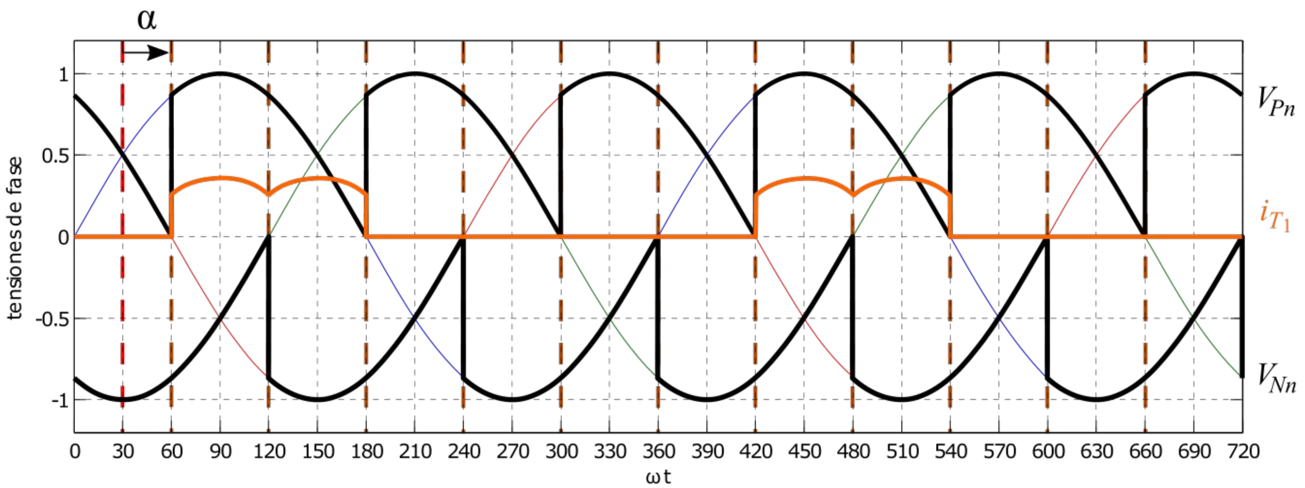
v_o


α
tensiones de fase
ωt
0
30
60
90
120
150
180
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240
270
300
330
360
390
420
450
480
510
540
570
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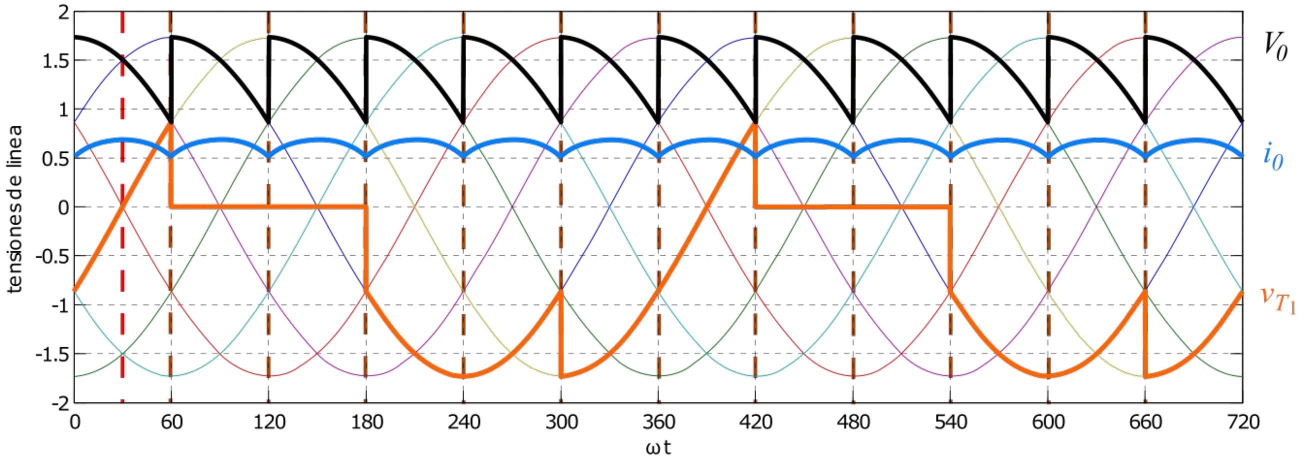
630
660
690
720
1
0.5
0
-0.5
-1
V_{Pn}
i_{T_1}
V_{Nn}


tensiones de linea
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360
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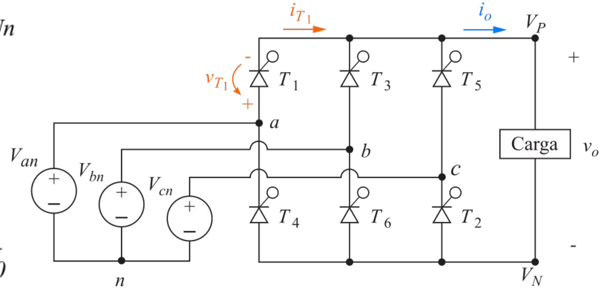
-1.5
-2
V_0
i_0
v_{T_1}


i_{T_1}
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v_{T_1}
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T_3
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V_{bn}
V_{cn}
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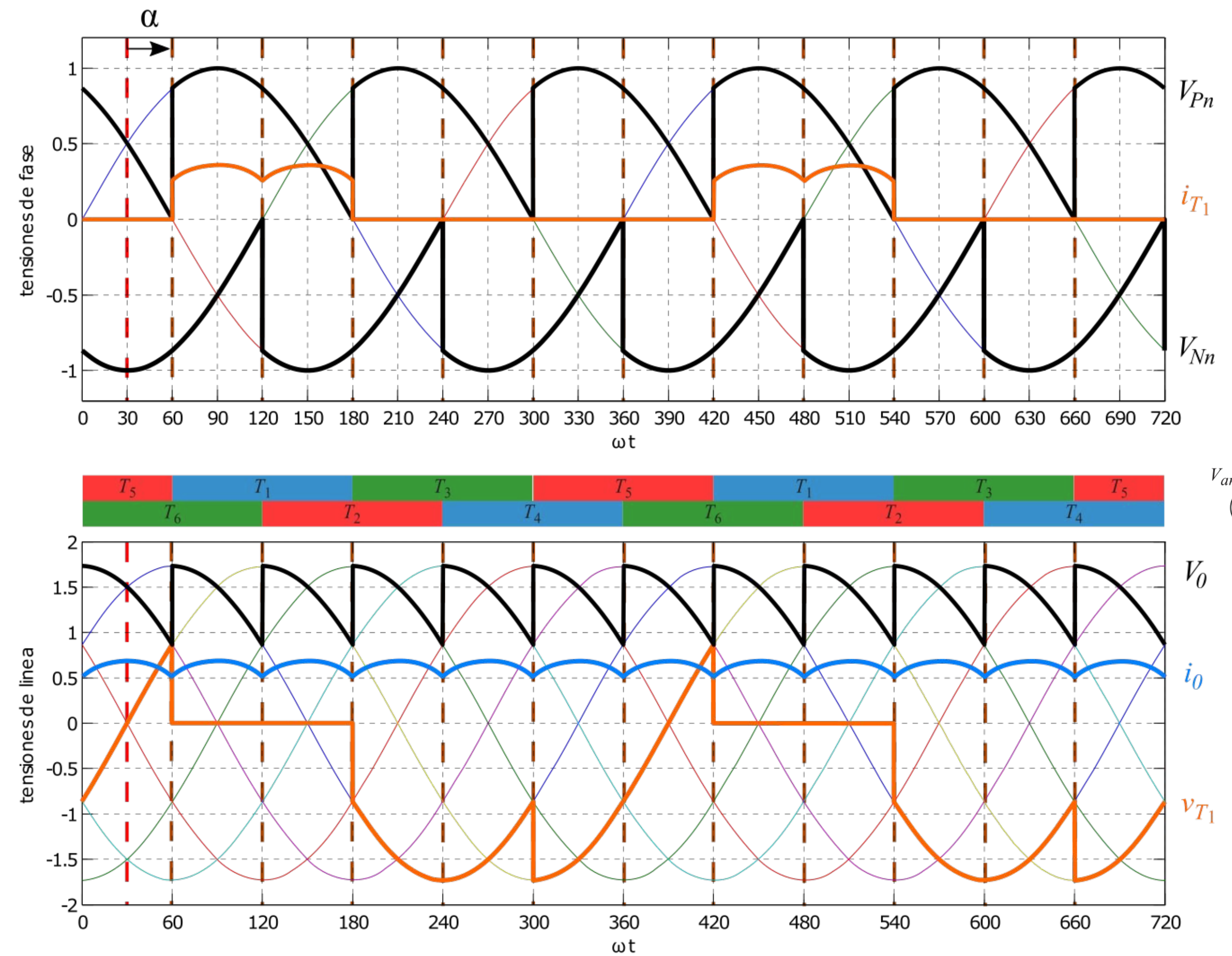
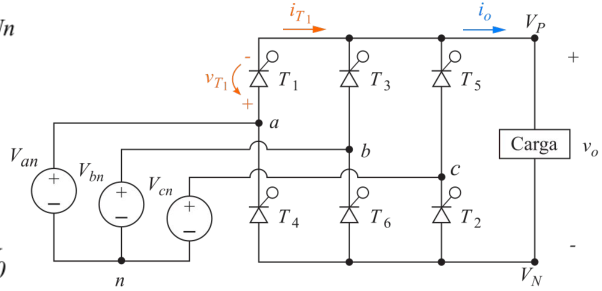
T_2
Carga
v_o
n
V_N


α
tensiones de fase
V_{Pn}
i_{T_1}
V_{Nn}
ωt
T_5 T_1 T_3 T_5 T_1 T_3 T_5
T_6 T_2 T_4 T_6 T_2 T_4
tensiones de linea
V_0
i_0
v_{T_1}
ωt


i_{T_1}
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V_P
v_{T_1}
T_1 T_3 T_5
T_4 T_6 T_2
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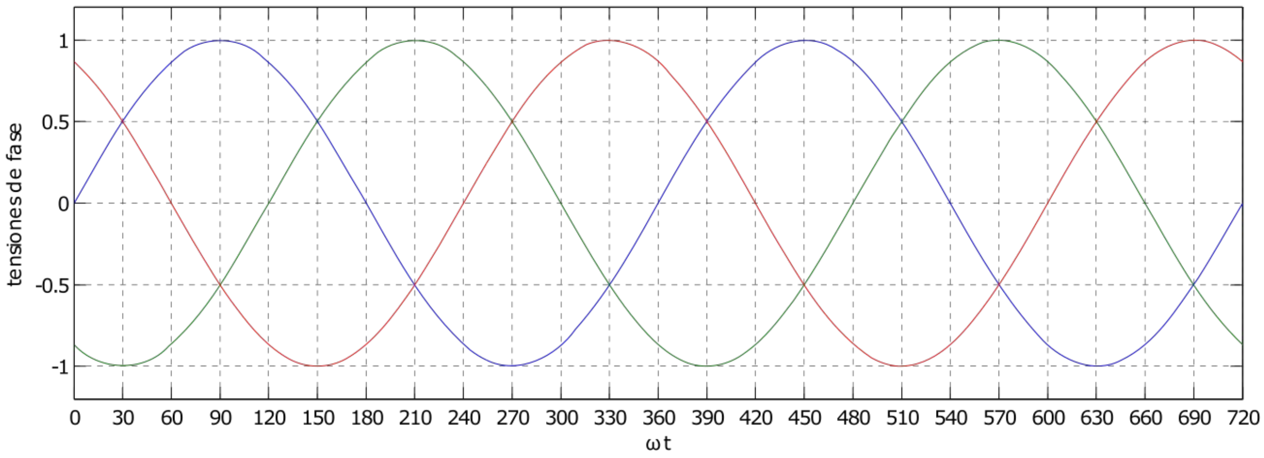
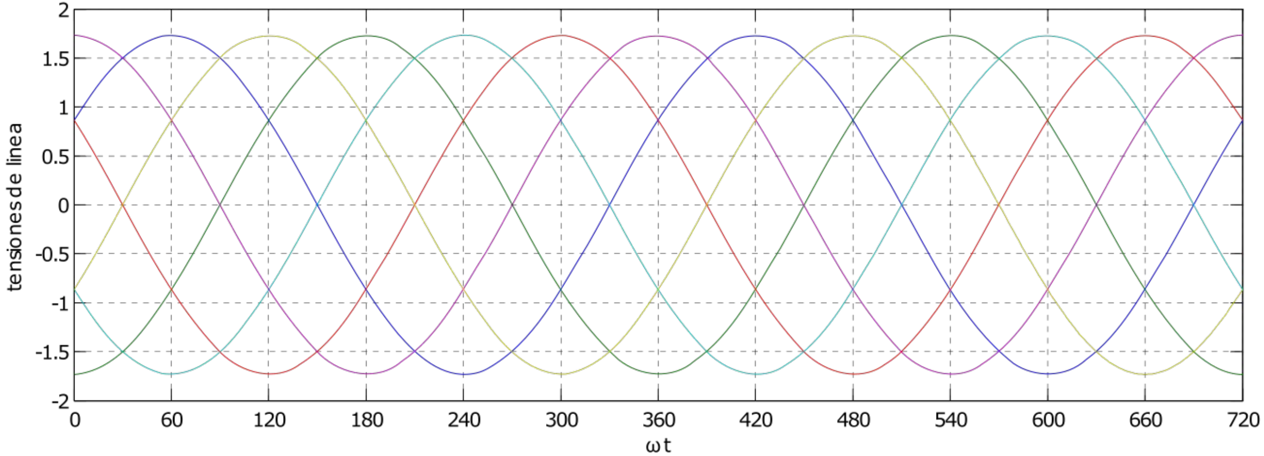
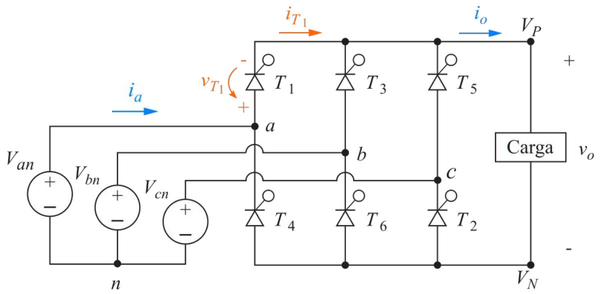
V_{bn}
V_{cn}
Carga
v_o
n
V_N


tensiones de fase
ωt


tensiones de linea
ωt


i_{T1}
i_o
V_P
v_{T1}
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T_5
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Carga
v_o
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T_6
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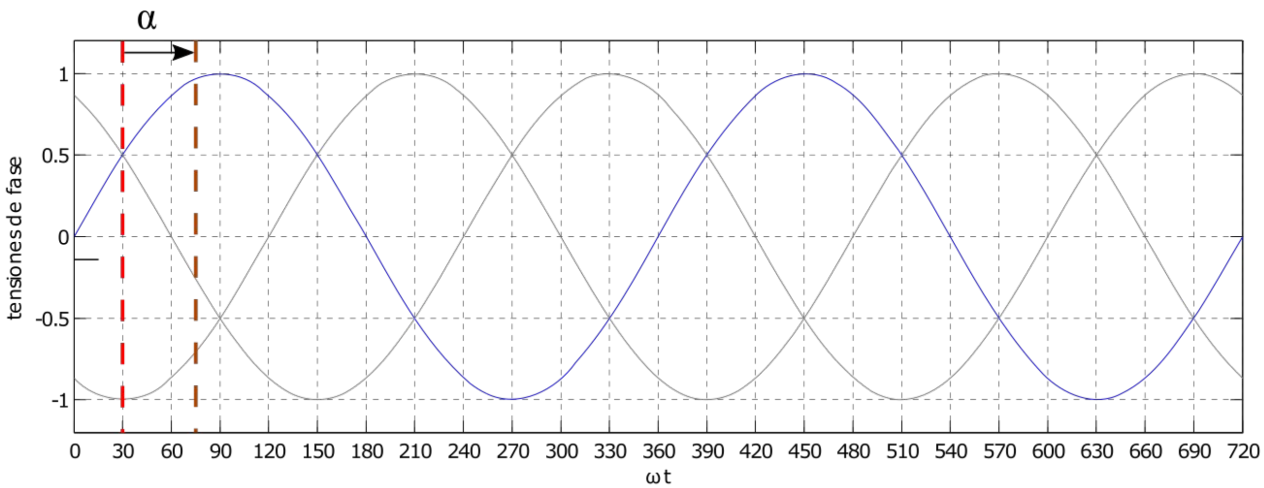
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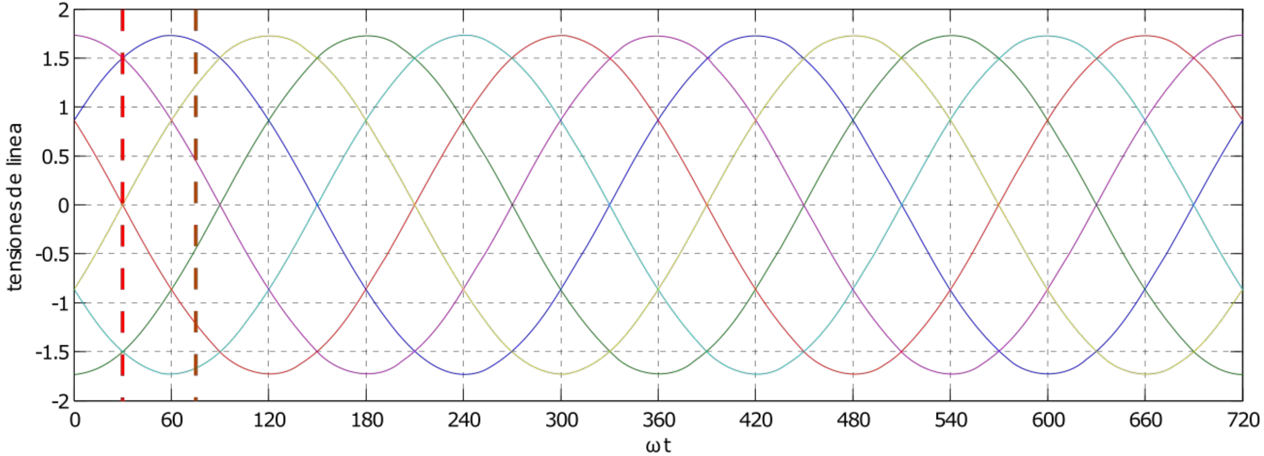
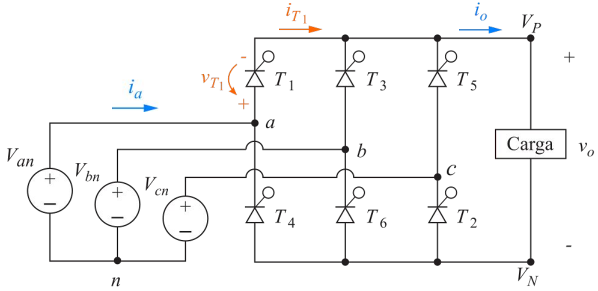
tensiones de fase
ωt


tensiones de linea
ωt


i_{T1}
i_o
V_P
i_a
v_{T1}
T_1
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a
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Carga
v_o
V_{an}
V_{bn}
V_{cn}
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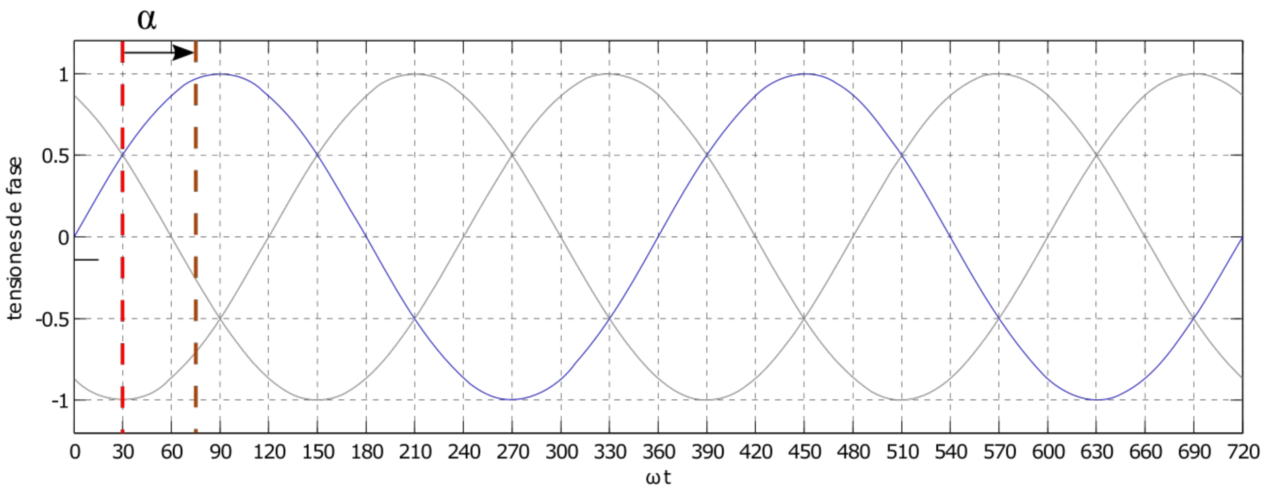
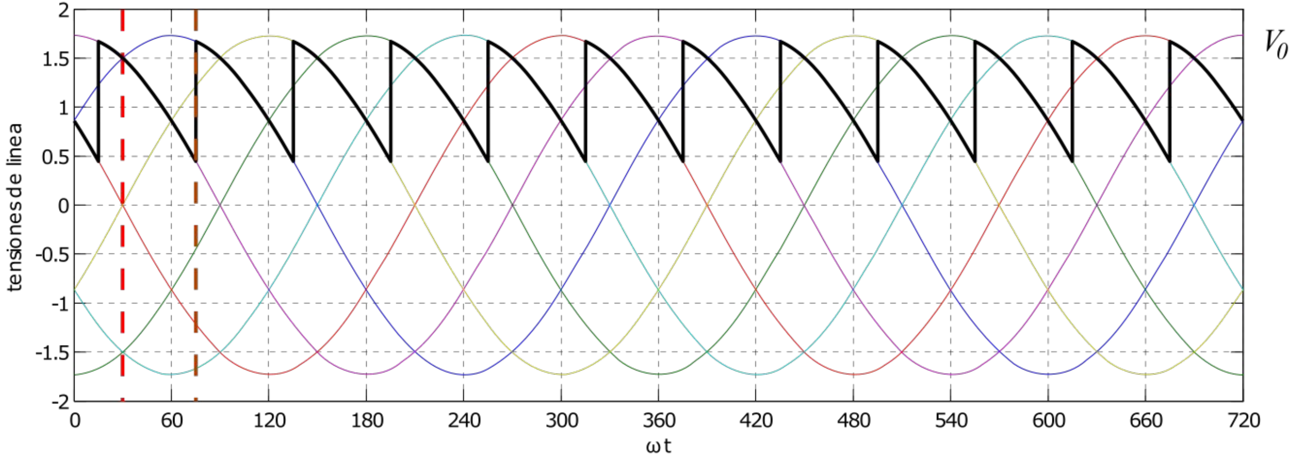
V_N


α
tensiones de fase
ωt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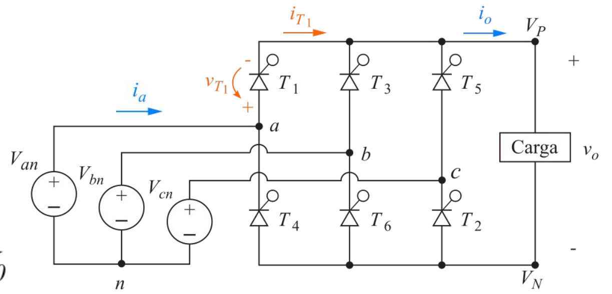
ωt
V_0


i_{T_1}
i_o
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v_{T_1}
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T_3
T_5
i_a
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Carga
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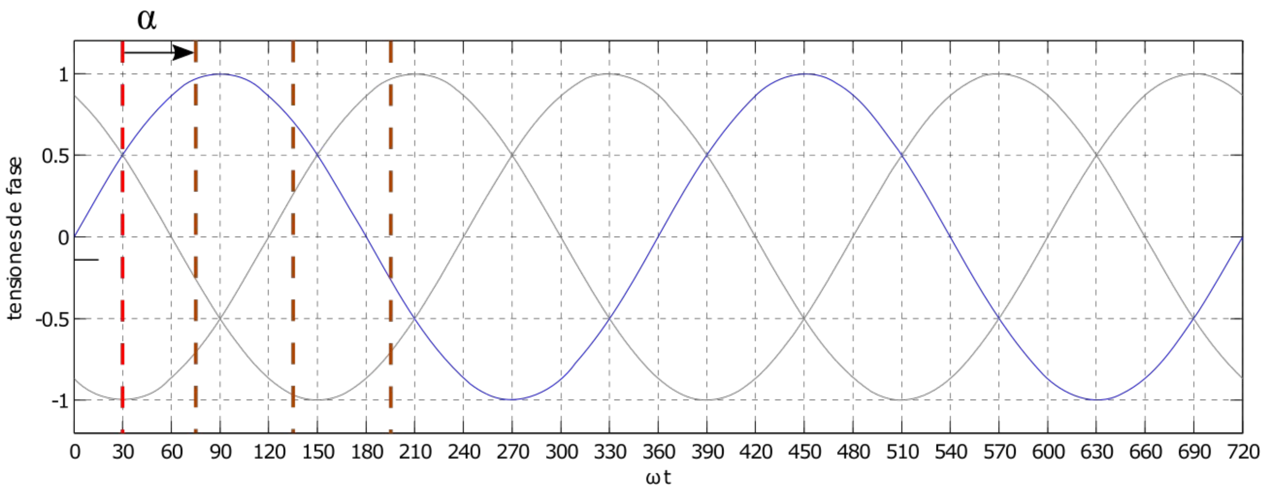
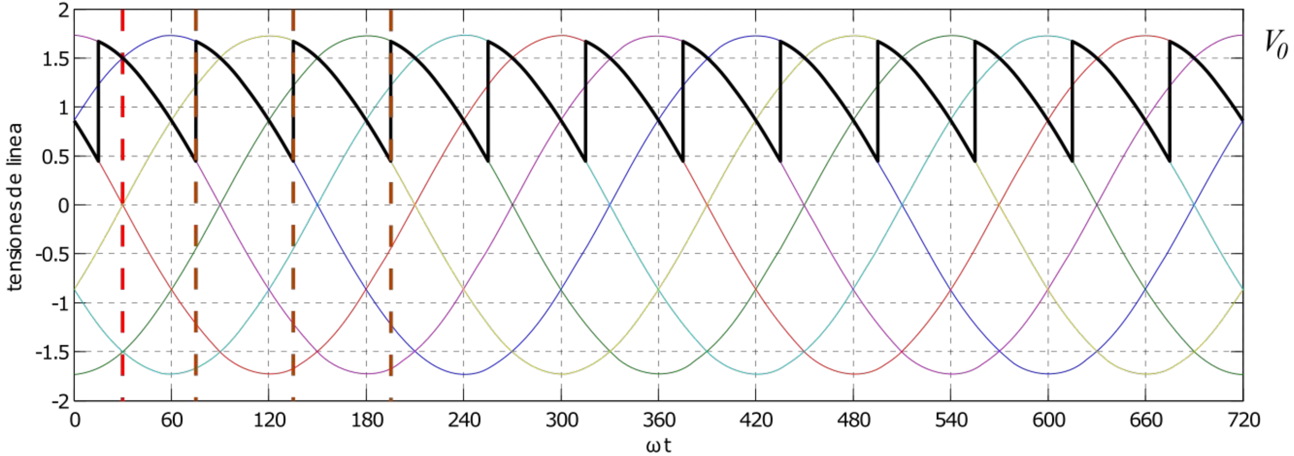
V_{cn}
T_4
T_6
T_2
n
V_N


α
tensiones de fase
ωt
1
0.5
0
-0.5
-1
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720


tensiones de linea
ωt
V_0
2
1.5
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-1
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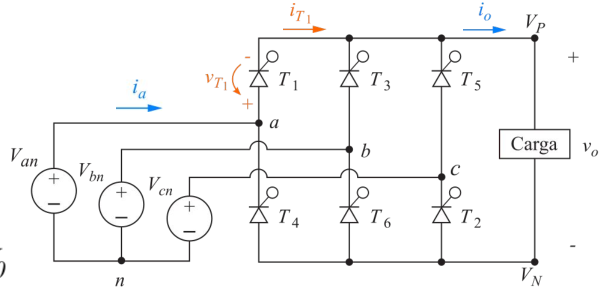
-2
0 60 120 180 240 300 360 420 480 540 600 660 720


i_{T1}
i_o
V_P
i_a
v_{T1}
T_1
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T_6
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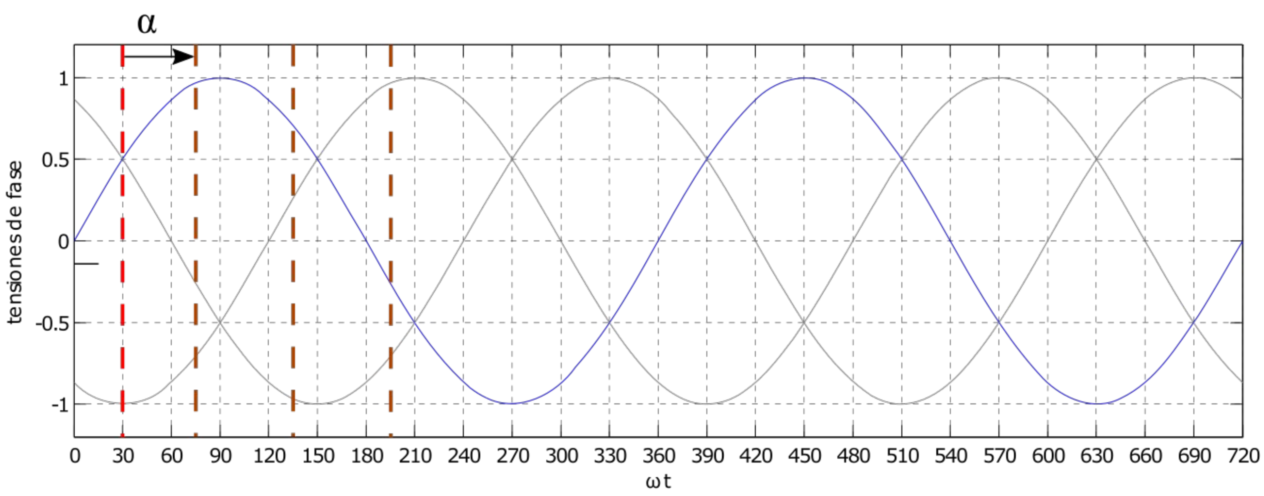
V_{cn}
Carga
v_o
V_N


α
tensiones de fase
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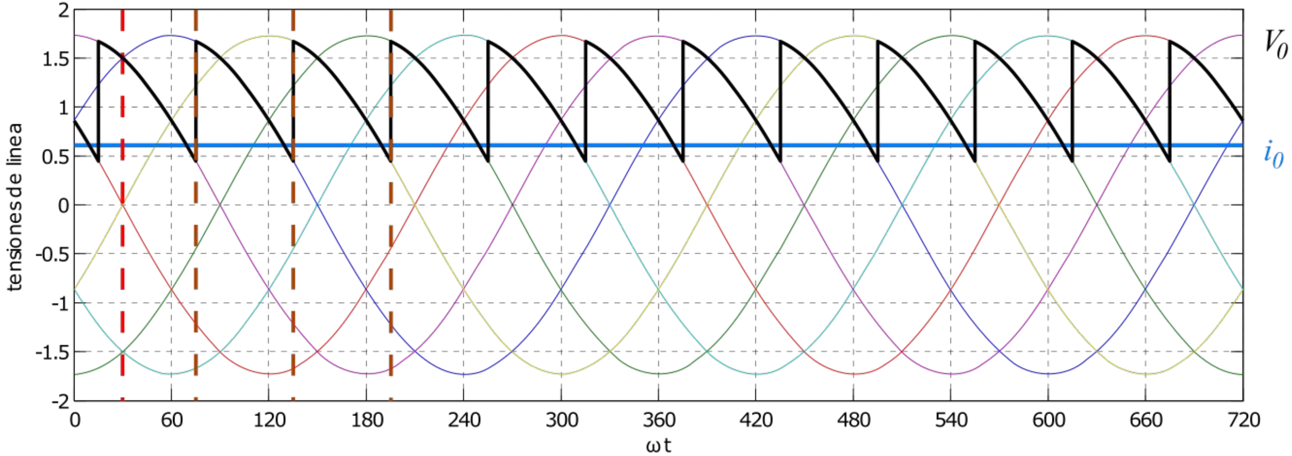
tensiones de linea
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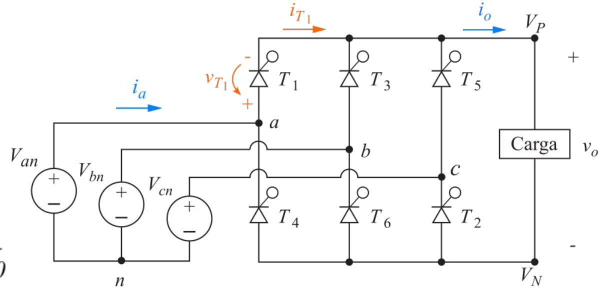
V_0
i_0


i_{T_1}
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V_P
v_{T_1}
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T_1
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Carga
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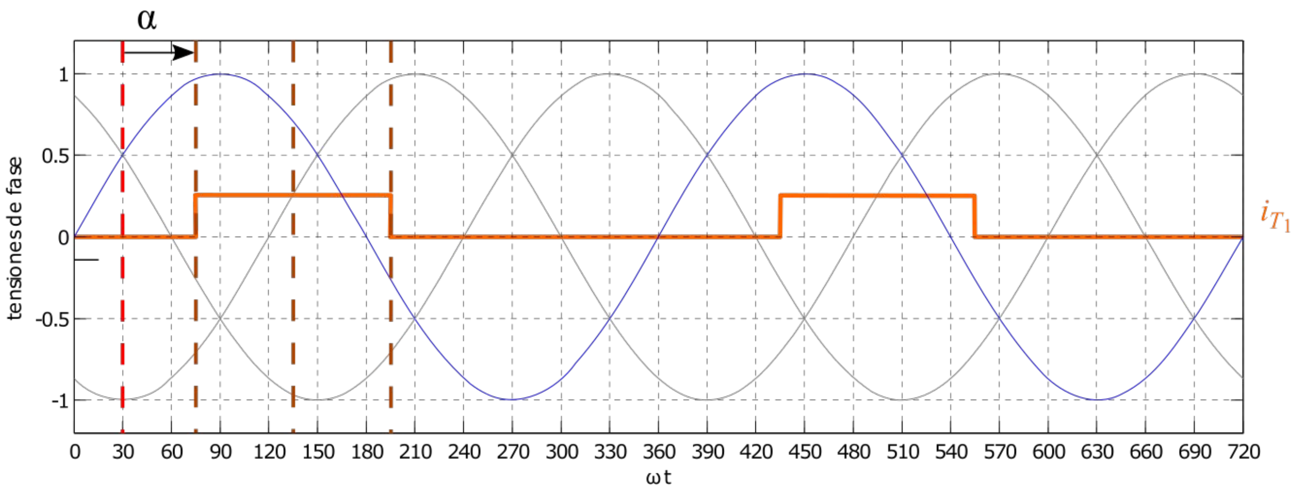
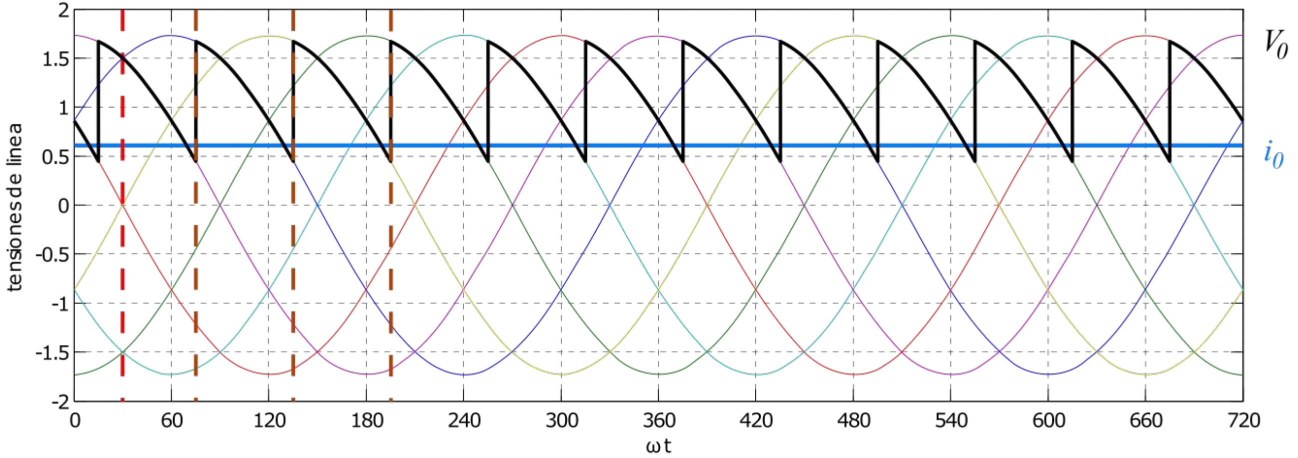
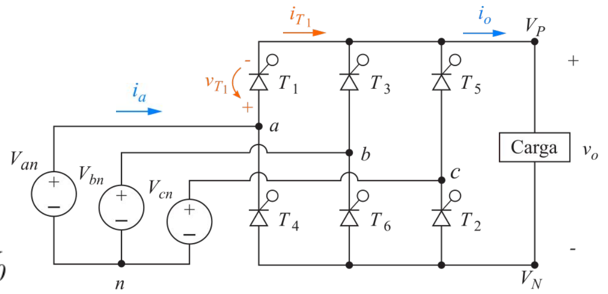
V_{an}
V_{bn}
V_{cn}
T_4
T_6
T_2
n
V_N


α
tensiones de fase
ωt
i_{T_1}


tensiones de linea
ωt
V_0
i_0


i_{T_1}
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v_o
T_4
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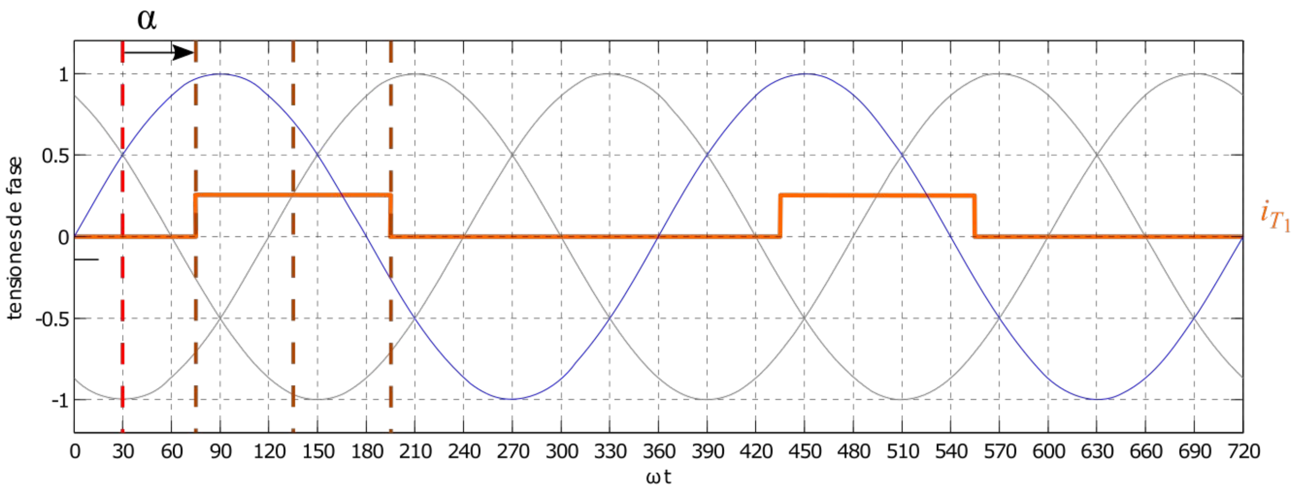
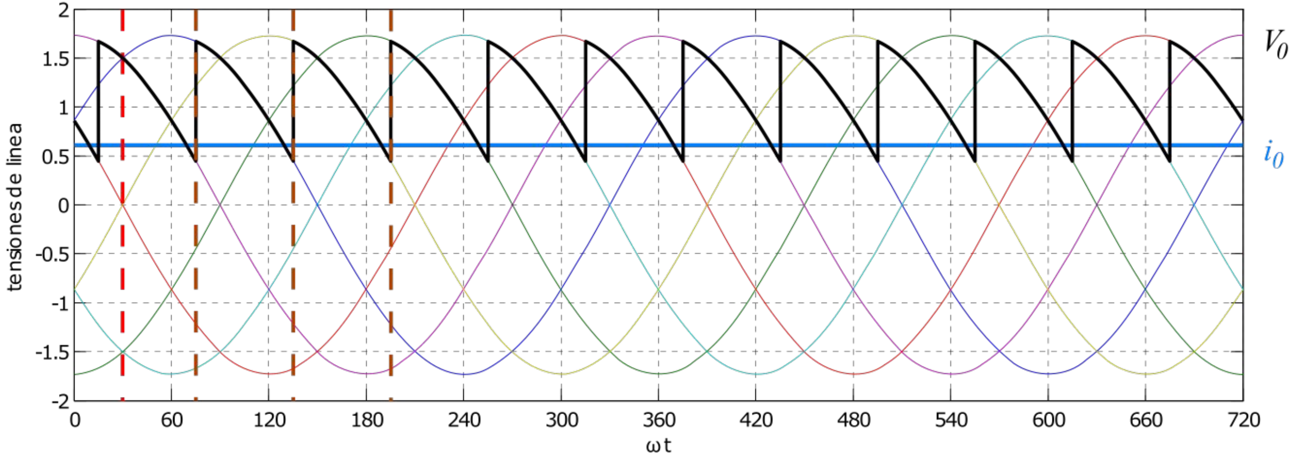
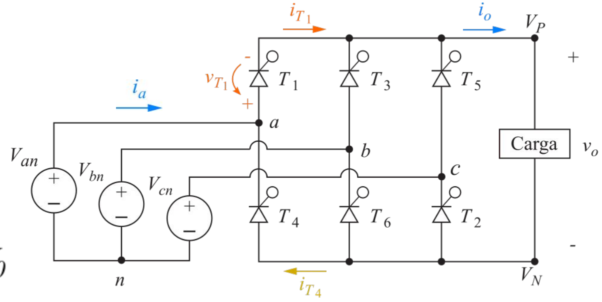


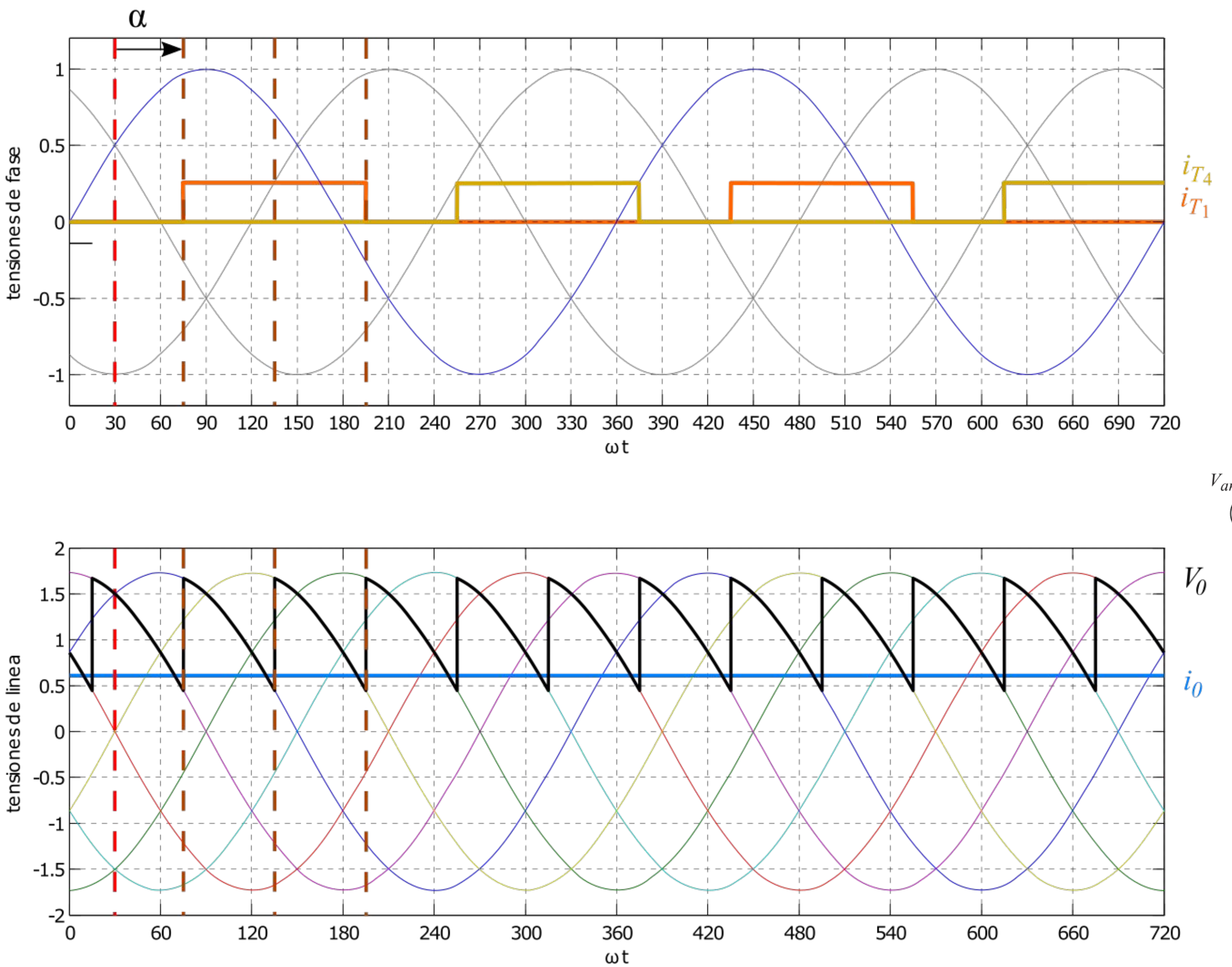
α
tensiones de fase
ωt
i_{T1}

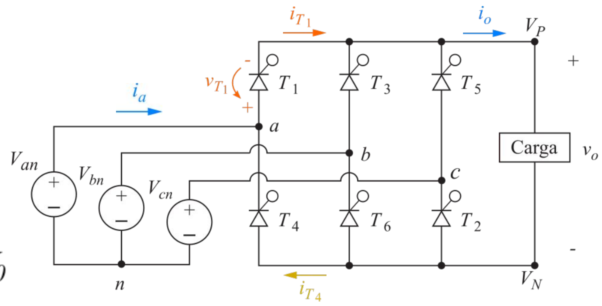


tensiones de línea
ωt
V_0
i_0



i_{T1}
i_o
V_P
i_a
v_{T1}
T_1
T_3
T_5
T_4
T_6
T_2
a
b
c
n
V_{an}
V_{bn}
V_{cn}
Carga
v_o
i_{T4}
V_N


α
tensiones de fase
ωt
i_{T4}
i_{T1}
tensiones de linea
ωt
V_0
i_0


i_{T1}
i_o
V_P
v_{T1}
i_a
T_1
T_3
T_5
a
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Carga
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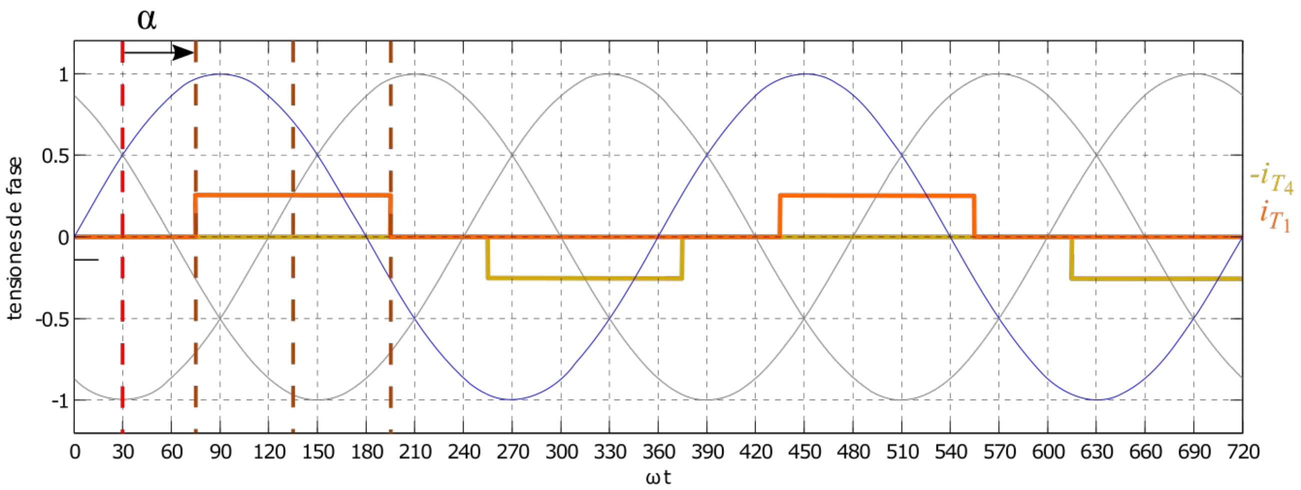
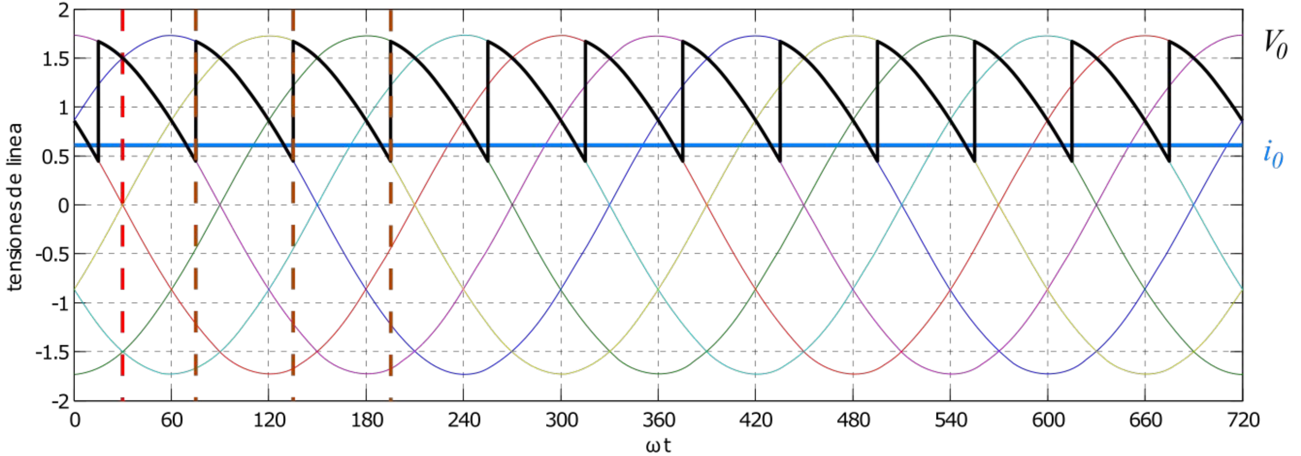
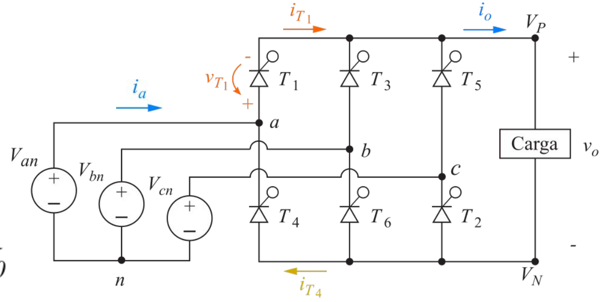
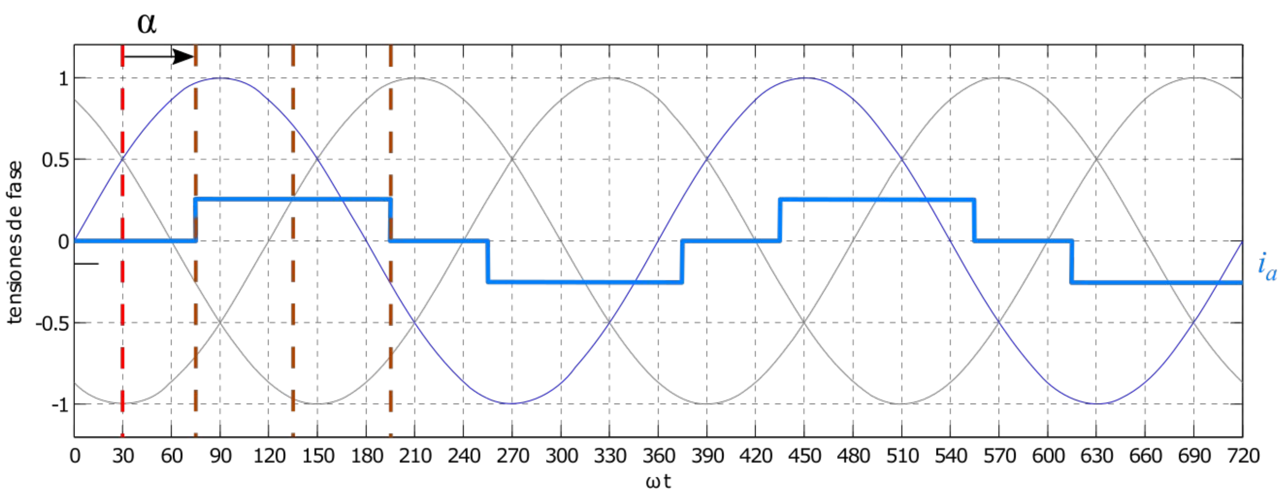
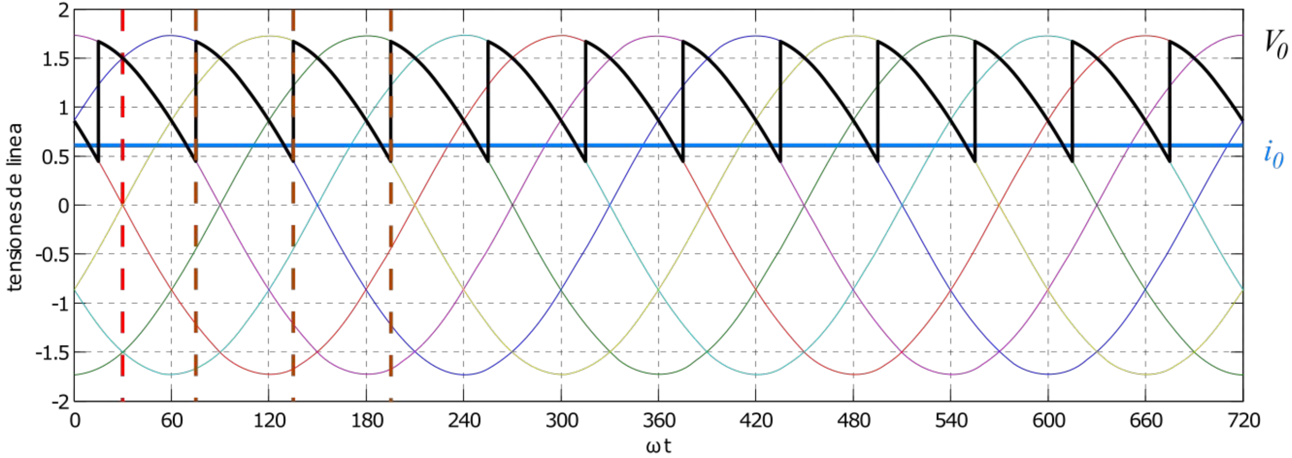
V_{an}
V_{bn}
V_{cn}
T_4
T_6
T_2
n
i_{T4}
V_N

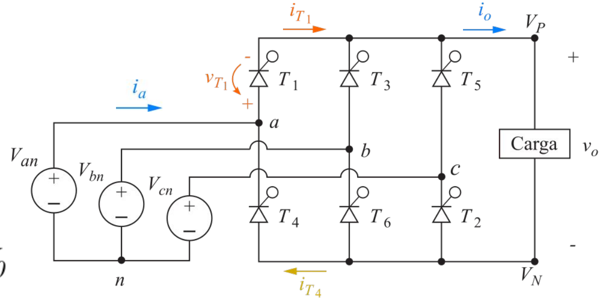

α
tensiones de fase
ωt
$-i_{T4}$
i_{T1}

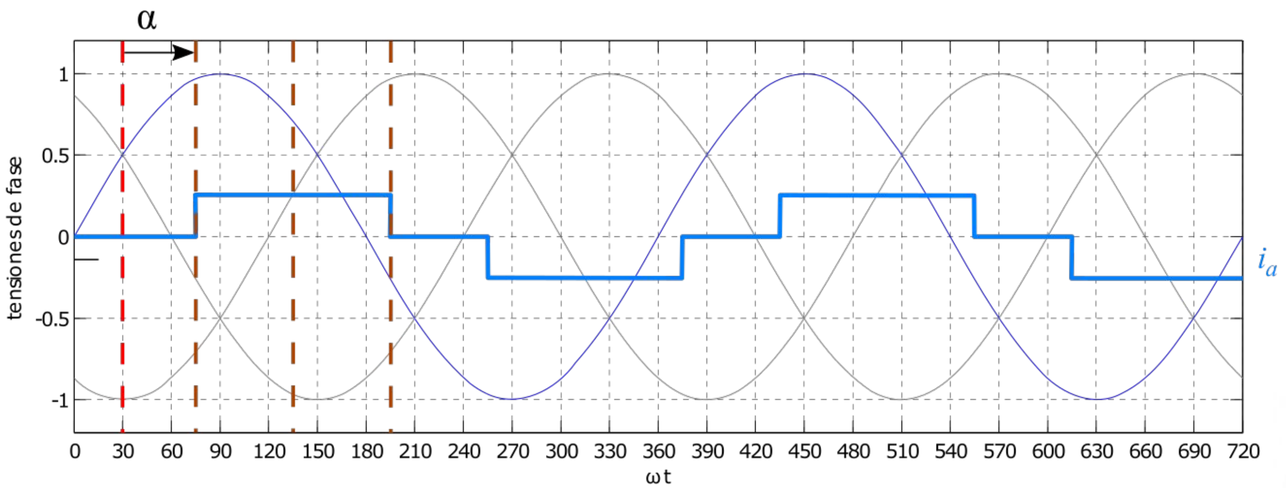

tensiones de linea
ωt
V_0
i_0

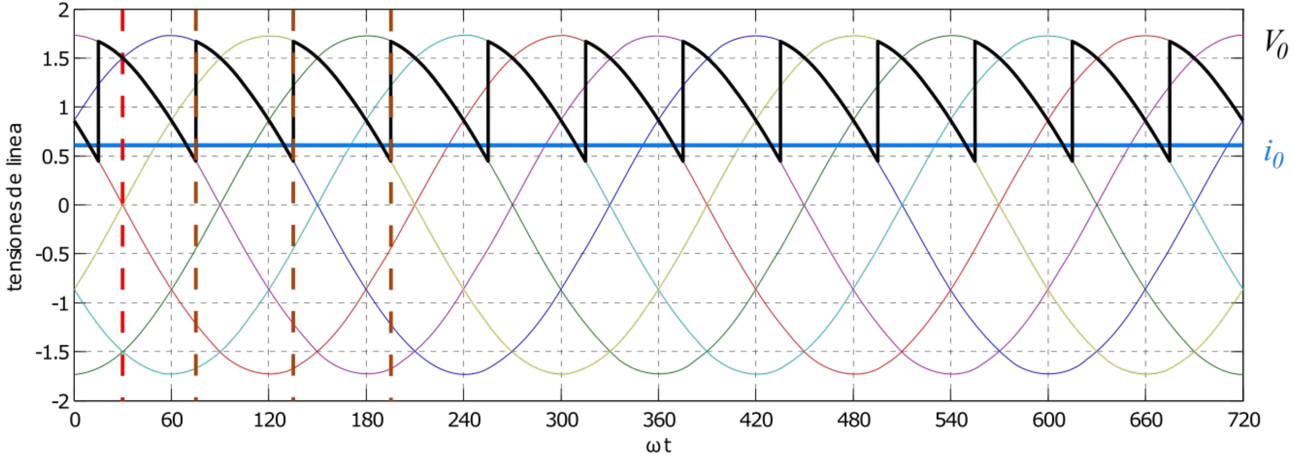

i_{T1}
i_o
V_P
i_a
v_{T1}
T_1
T_3
T_5
a
b
c
V_{an}
V_{bn}
V_{cn}
Carga
v_o
T_4
T_6
T_2
n
i_{T4}
V_N

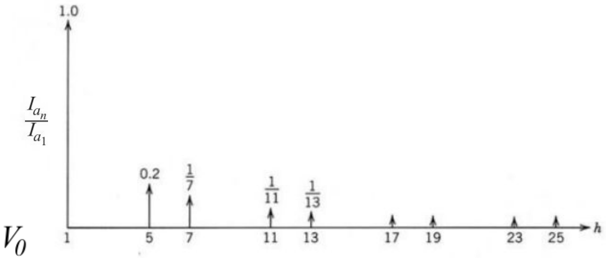

α
tensiones de fase
ωt
i_a

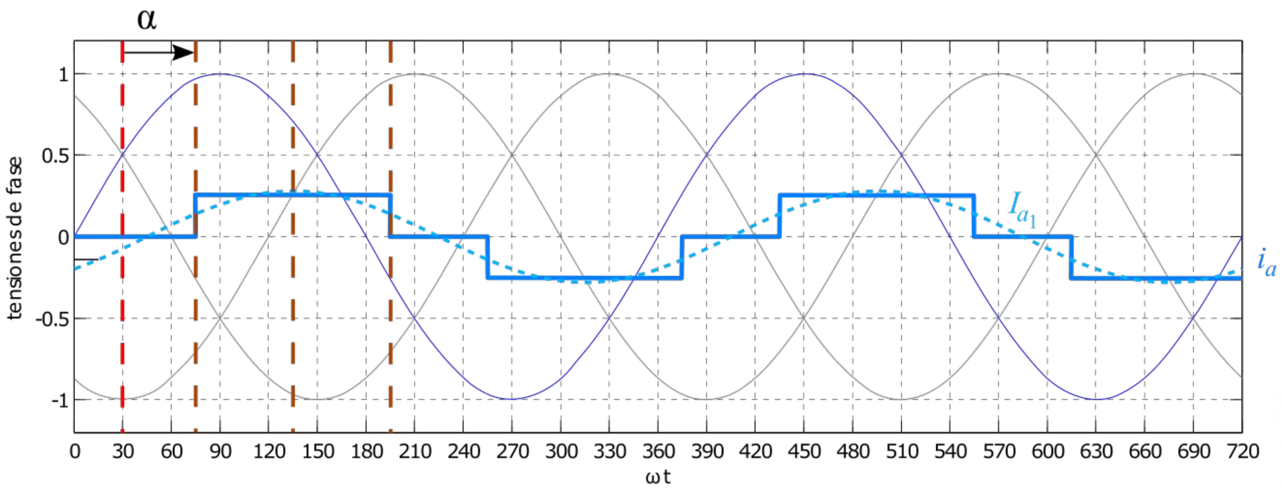

tensiones de linea
ωt
V_0
i_0


i_{T1}
i_o
V_P
v_{T1}
T_1
T_3
T_5
i_a
a
b
c
Carga
v_o
V_{an}
V_{bn}
V_{cn}
T_4
T_6
T_2
n
i_{T4}
V_N

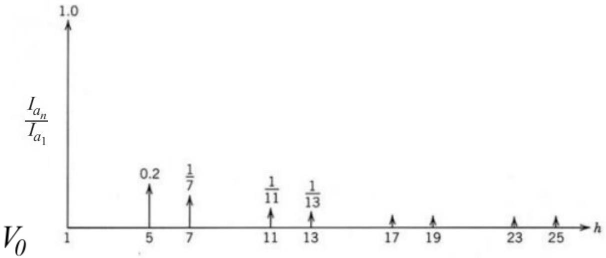

α
tensiones de fase
ωt
i_a


V_0
i_0
tensiones de linea
ωt


1.0
$\frac{I_{a_n}}{I_{a_1}}$
0.2
$\frac{1}{7}$
$\frac{1}{11}$
$\frac{1}{13}$
h



α
tensiones de fase
ωt
I_{a_1}
i_a



1.0
$\frac{I_{a_n}}{I_{a_1}}$
0.2
$\frac{1}{7}$
$\frac{1}{11}$
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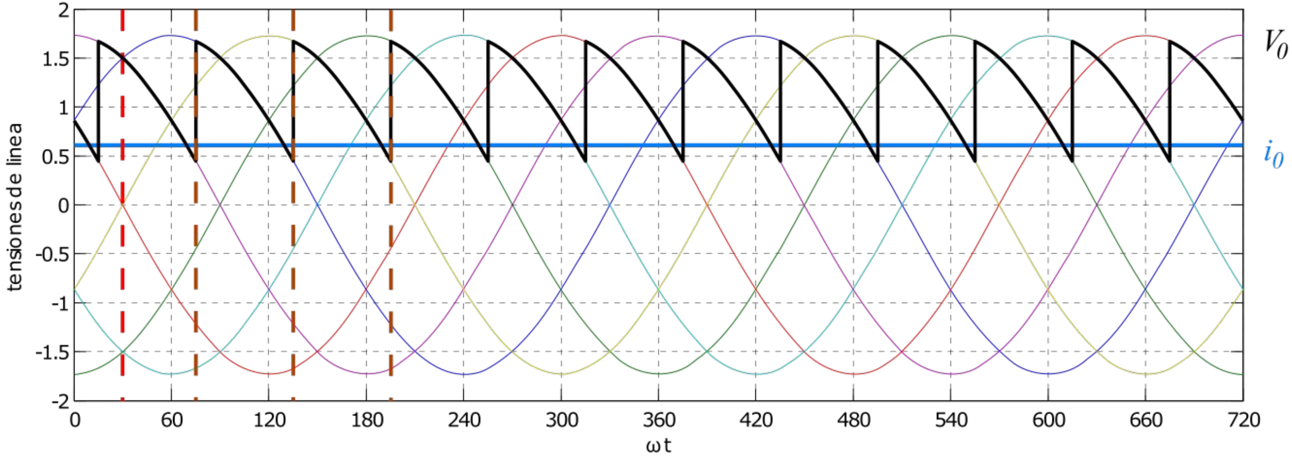
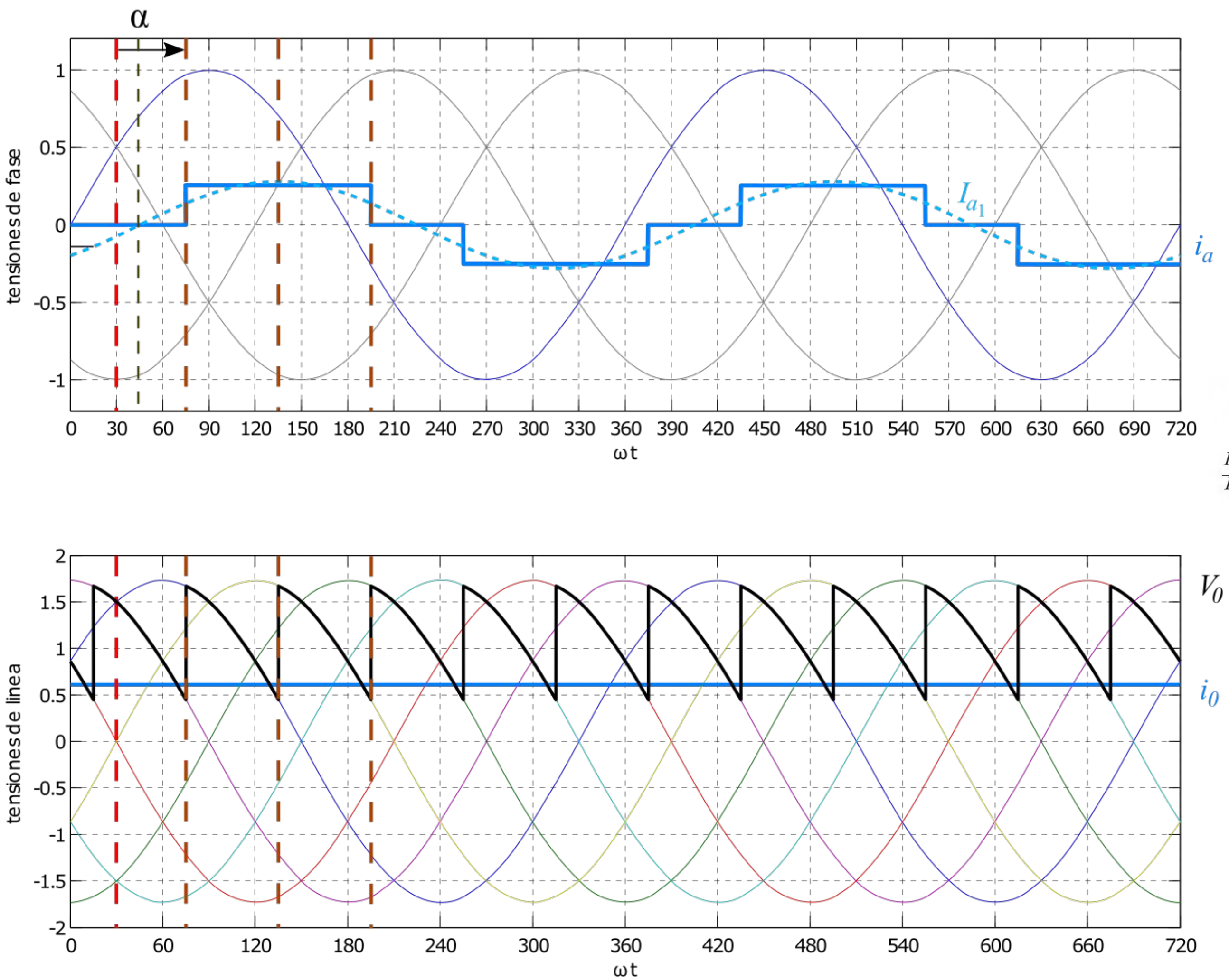
7
11
13
17
19
23
25
h



tensiones de linea
ωt
V_0
i_0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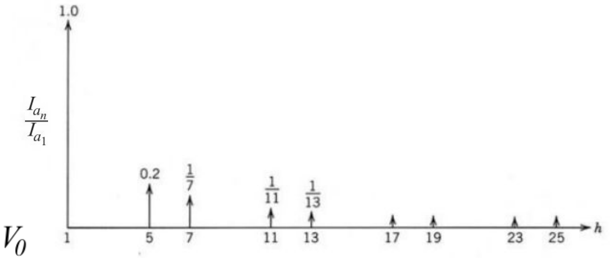
tensiones de fase
ωt
I_{a_1}
i_a
tensiones de linea
V_0
i_0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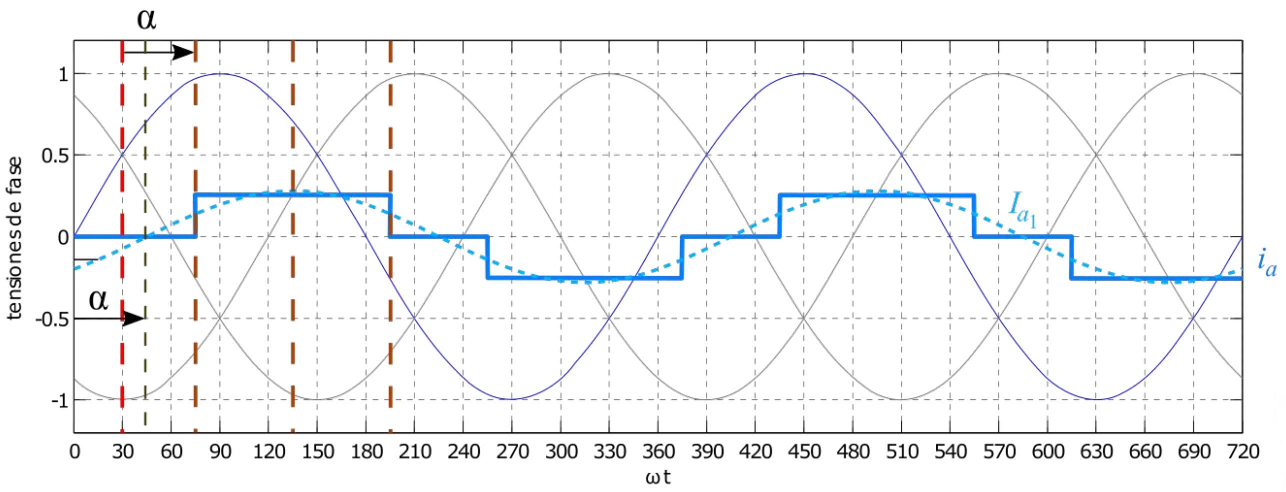
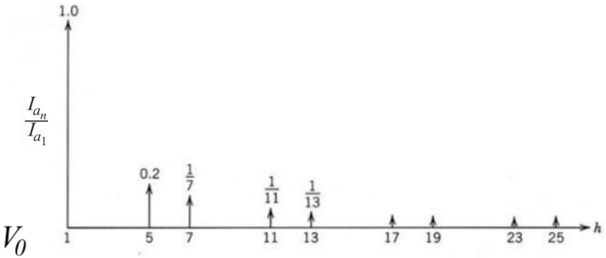
$\frac{I_{a_n}}{I_{a_1}}$
0.2
$\frac{1}{7}$
$\frac{1}{11}$
$\frac{1}{13}$
1 5 7 11 13 17 19 23 25
h


α
α
tensiones de fase
ωt
I_{a_1}
i_a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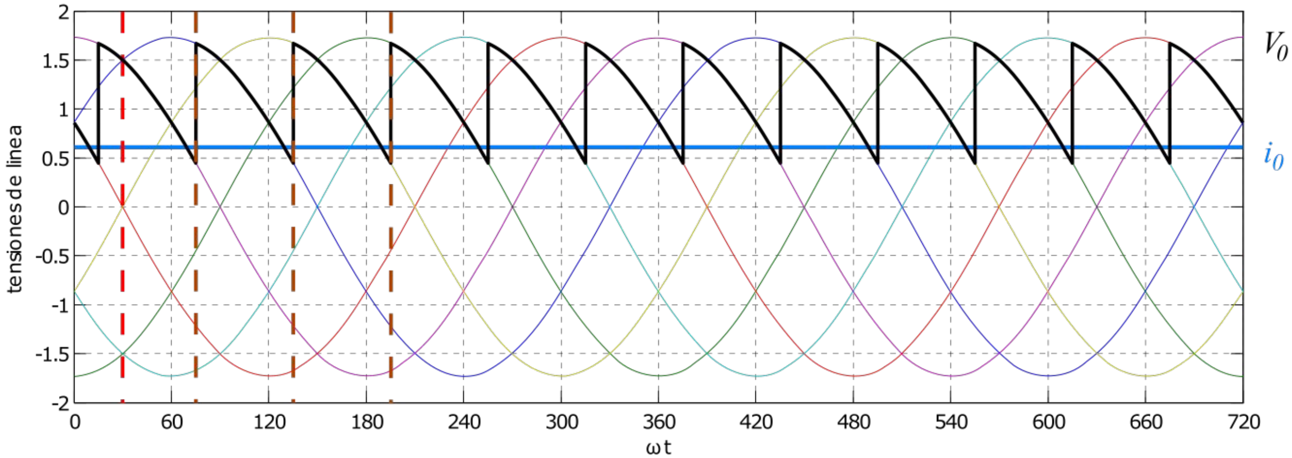
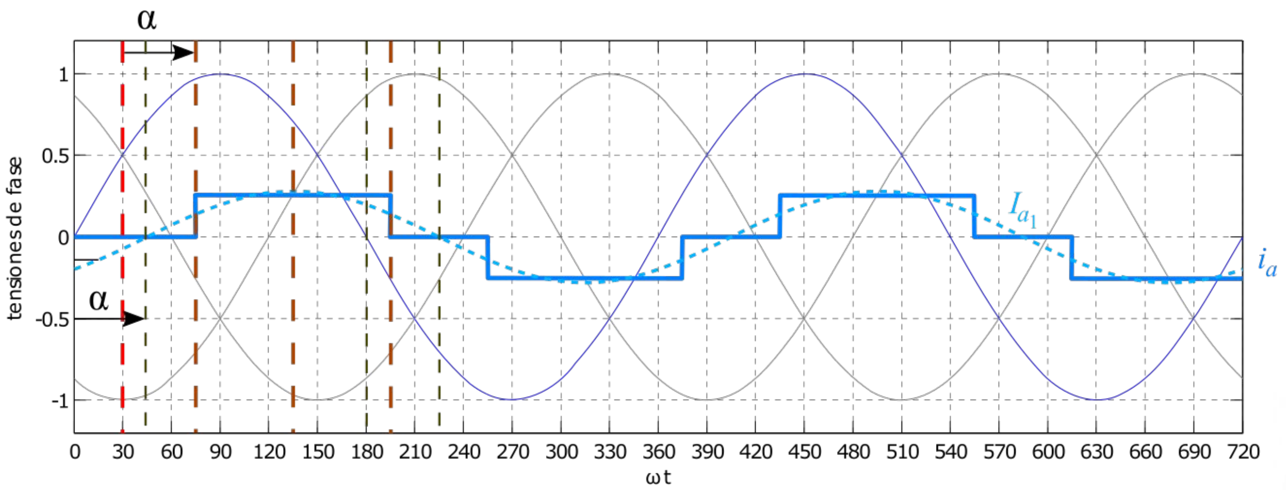
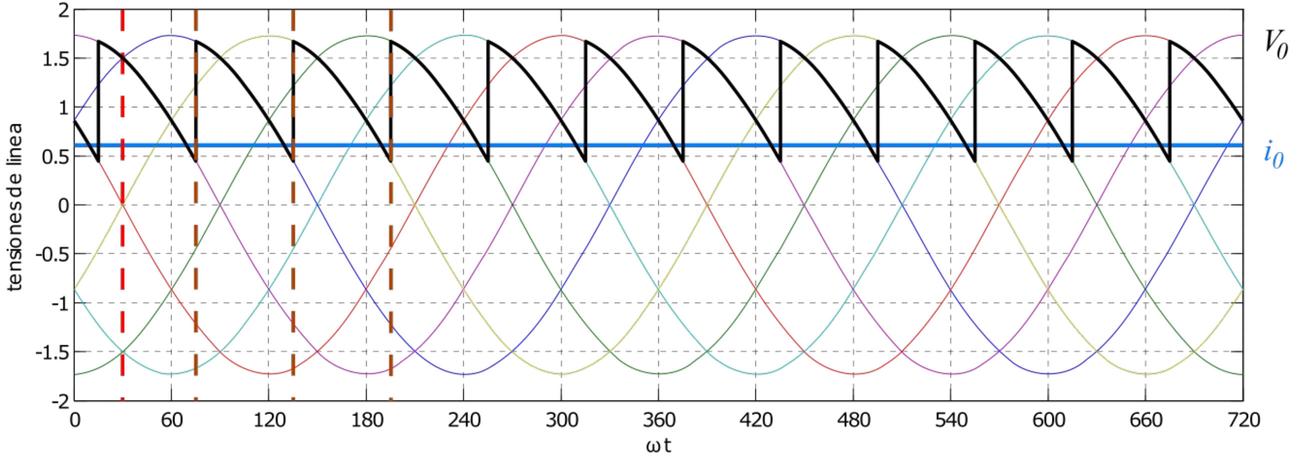
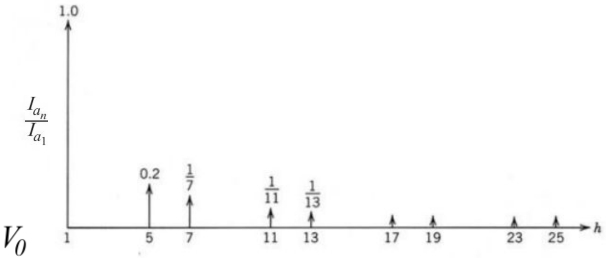
h


tensiones de linea
ωt
V_0
i_0


α
α
tensiones de fase
ωt
I_{a_1}
i_a
0 30 60 90 120 150 180 210 240 270 300 330 360 390 420 450 480 510 540 570 600 630 660 690 720
1 0.5 0 -0.5 -1


tensiones de linea
ωt
V_0
i_0
0 60 120 180 240 300 360 420 480 540 600 660 720
2 1.5 1 0.5 0 -0.5 -1 -1.5 -2


1.0
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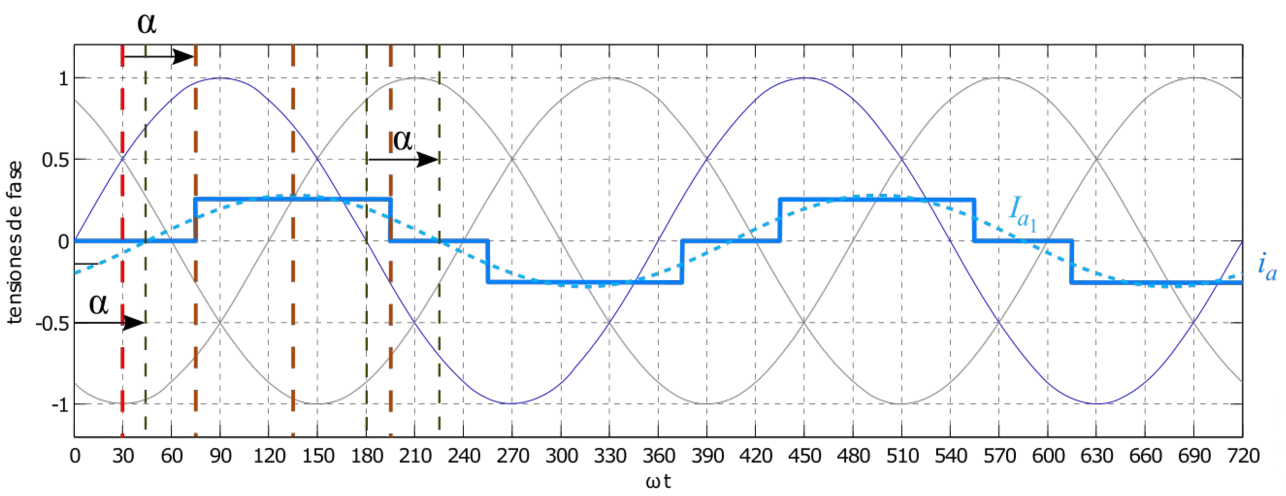
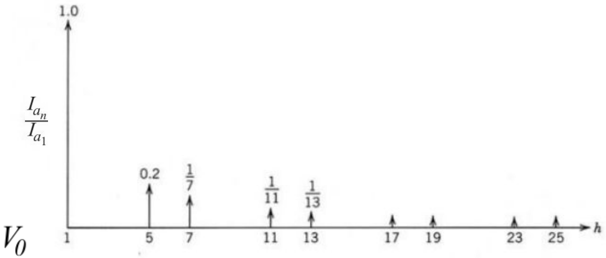
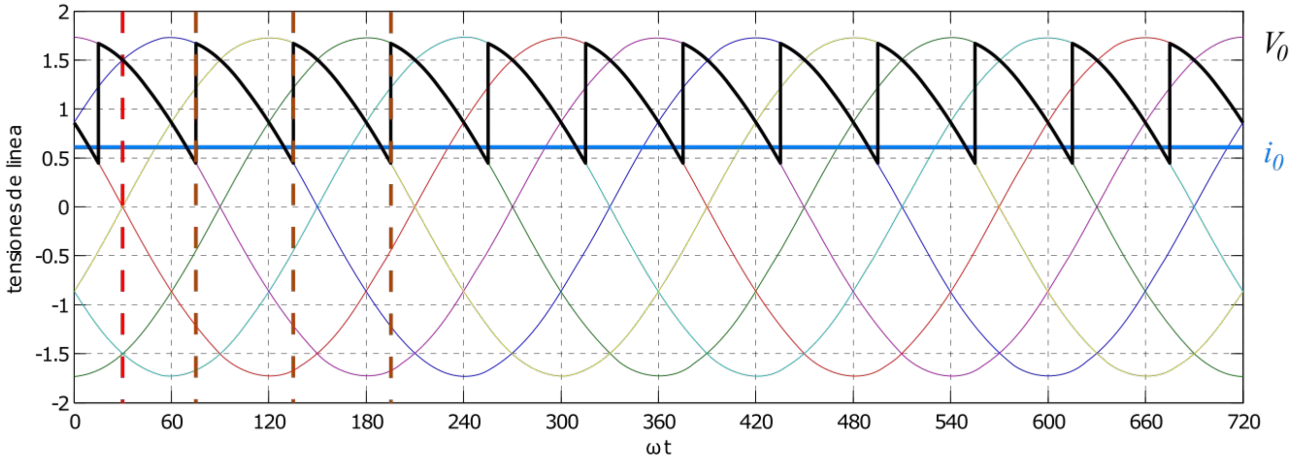
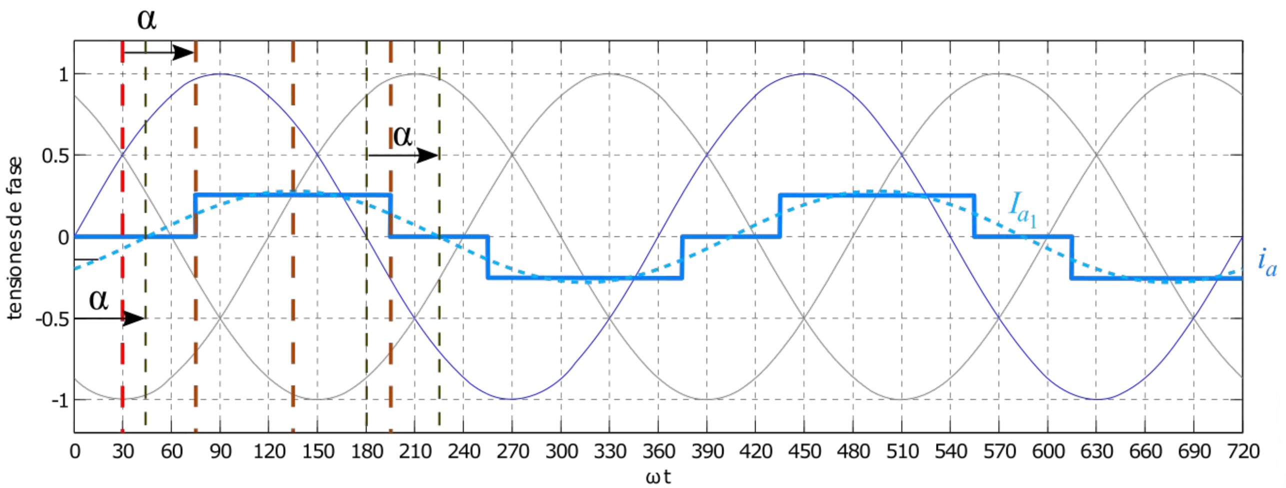
0.2
$\frac{1}{7}$
$\frac{1}{11}$
$\frac{1}{13}$
h
1 5 7 11 13 17 19 23 25

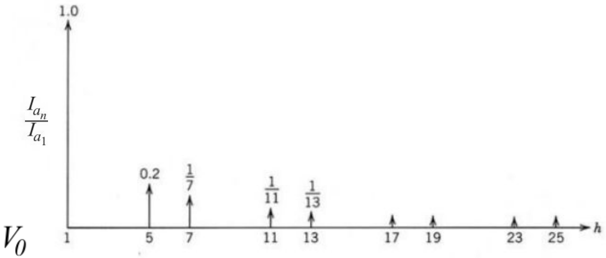

α
α
α
tensiones de fase
I_{a_1}
i_a
ωt

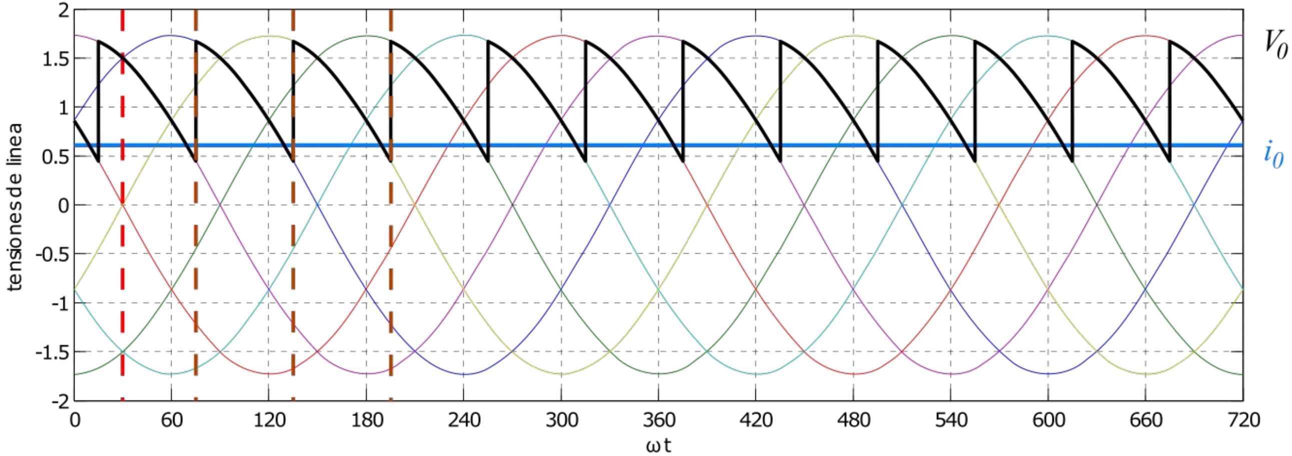

$\frac{I_{a_n}}{I_{a_1}}$
1.0
0.2
$\frac{1}{7}$
$\frac{1}{11}$
$\frac{1}{13}$
h

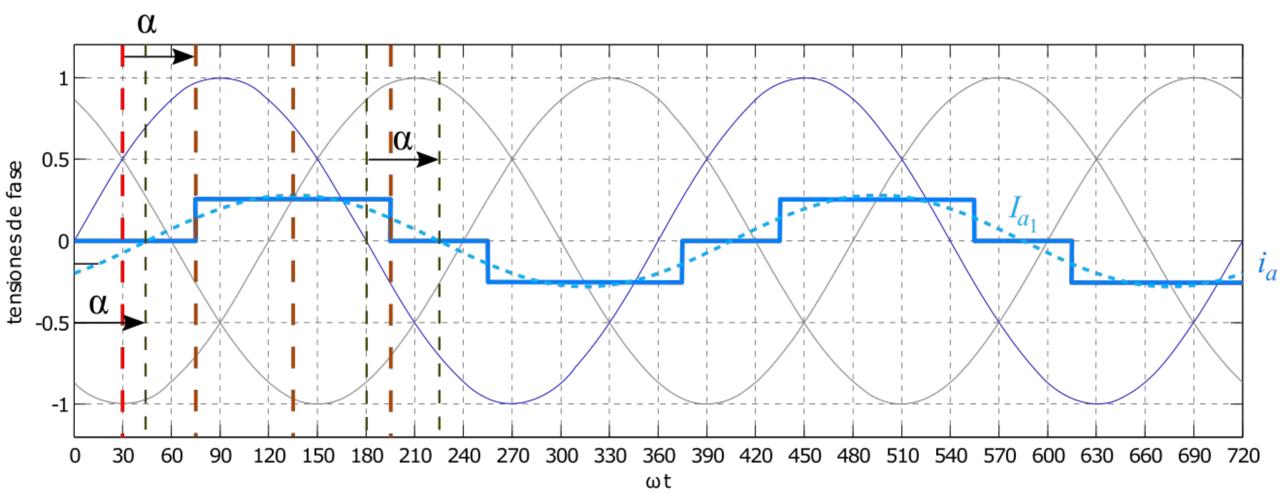

tensiones de linea
V_0
i_0
ωt



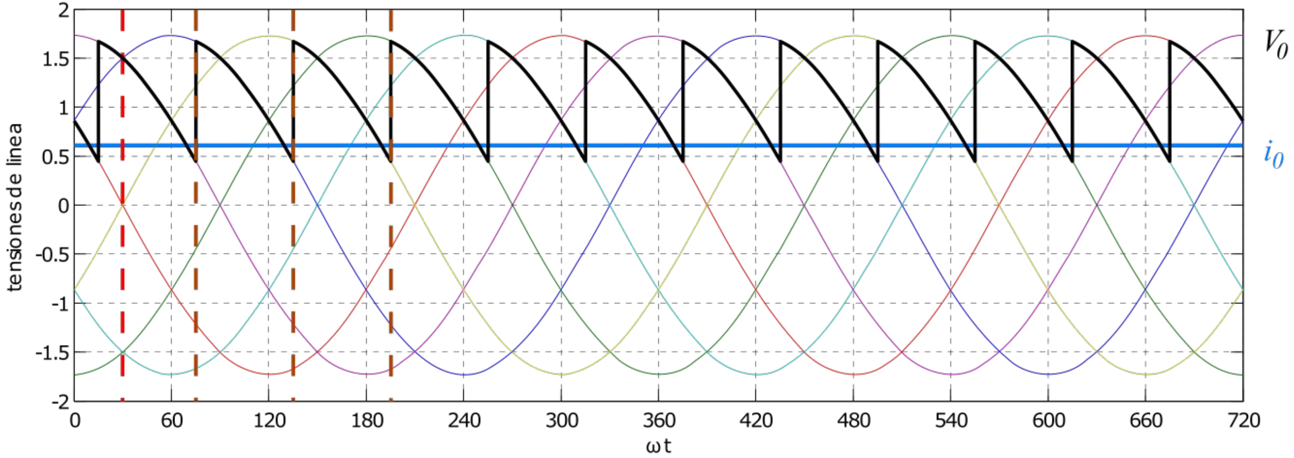
$$I_{a_1}(\omega t) = \frac{2\sqrt{3}}{\pi} I_0 \sin(\omega t - \alpha)$$

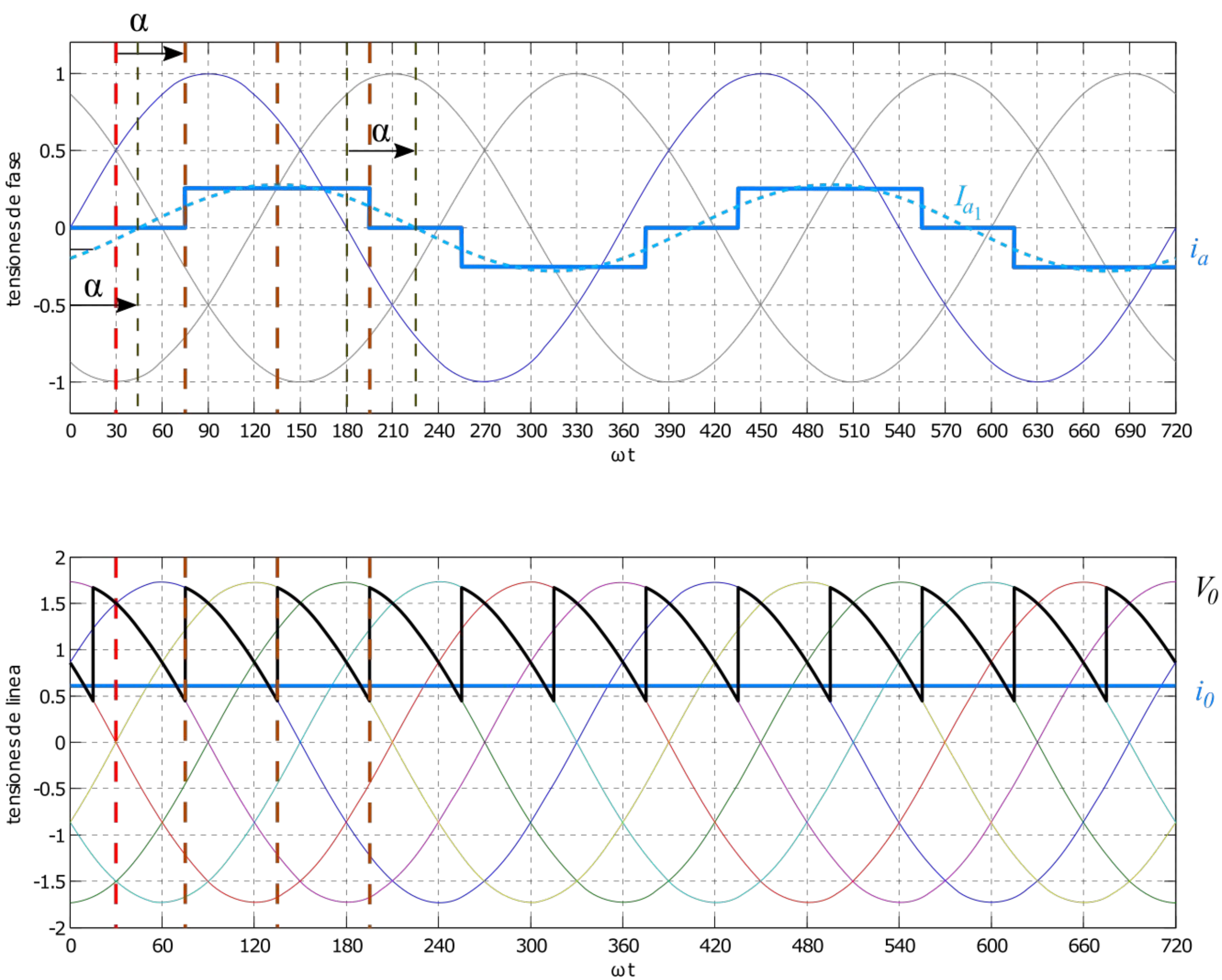






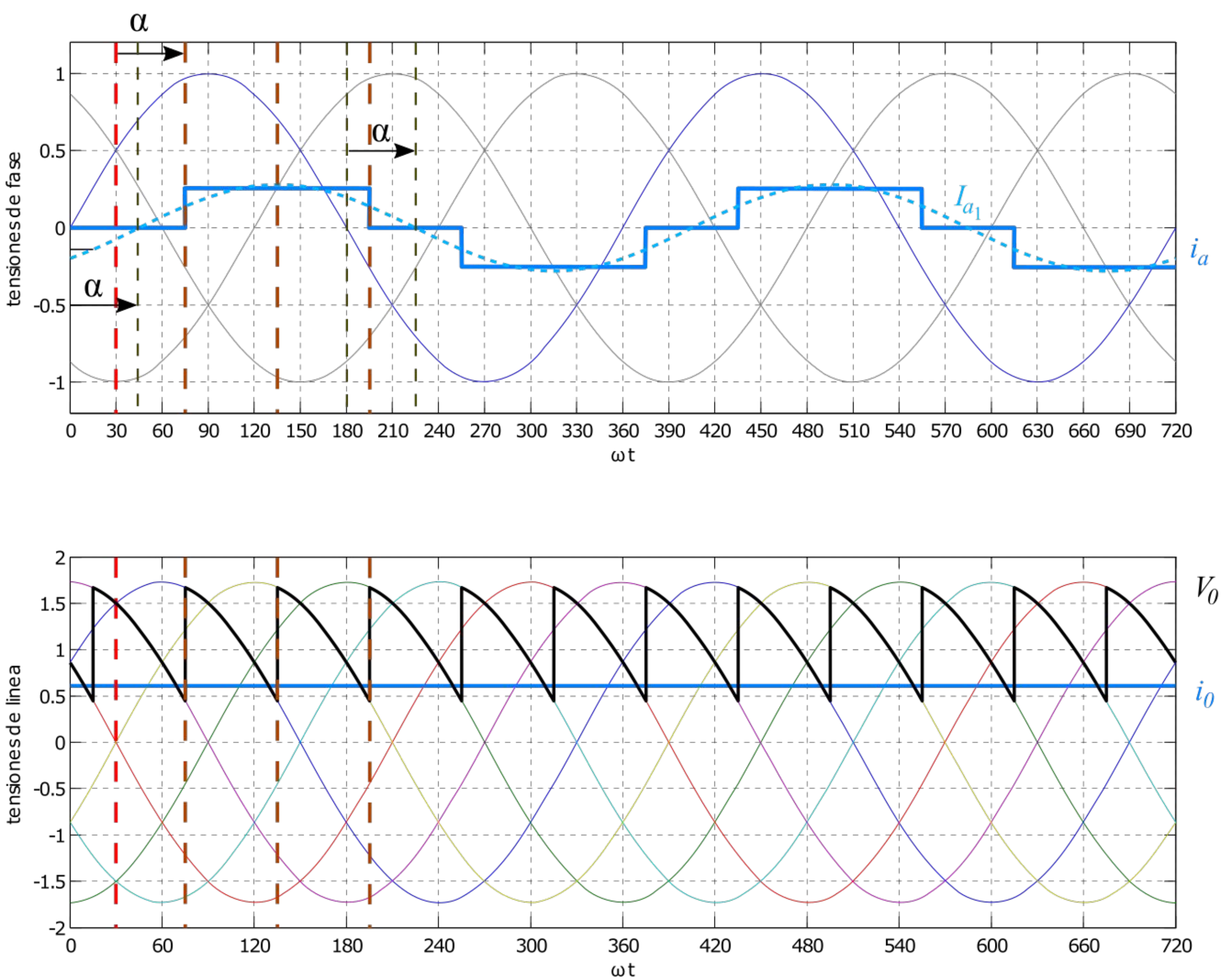
$$I_{a_1}(\omega t) = \frac{2\sqrt{3}}{\pi} I_0 \sin(\omega t - \alpha)$$





$$I_{a_1}(\omega t) = \frac{2\sqrt{3}}{\pi} I_0 \sin(\omega t - \alpha)$$

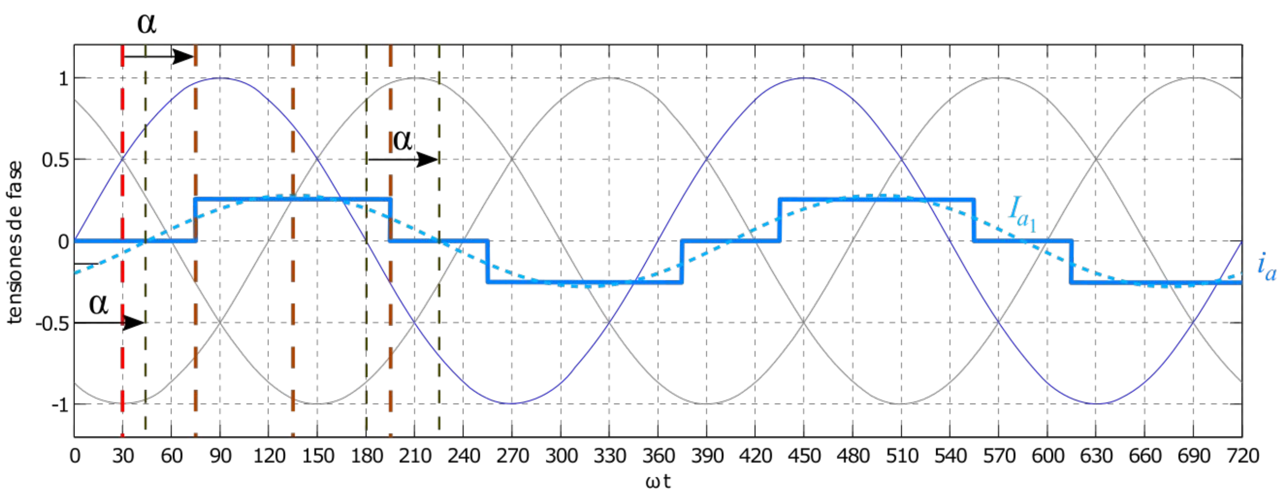
$$I_{a_{rms}} = \sqrt{\frac{2}{3}} I_0$$

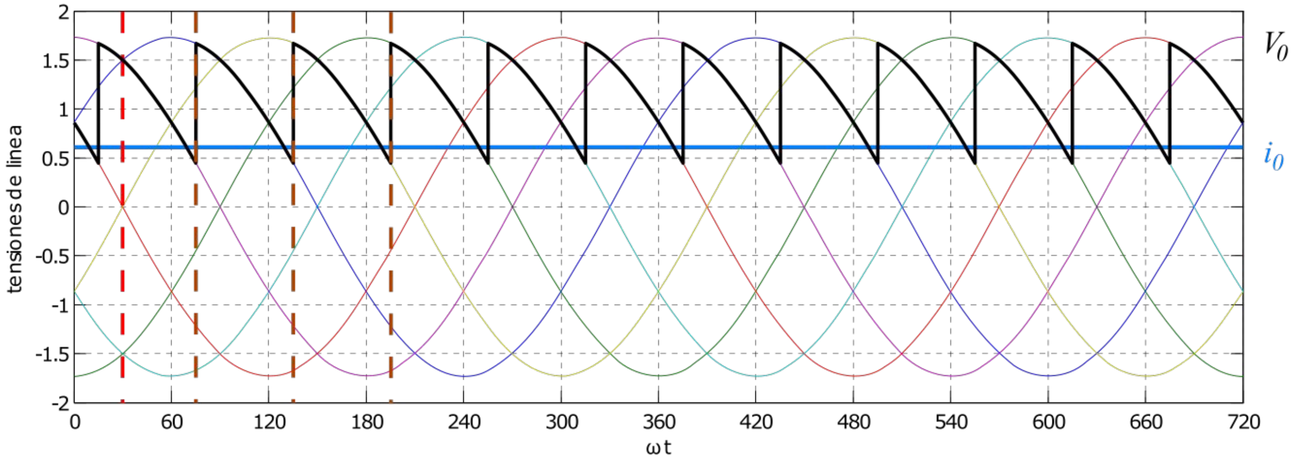


$$I_{a_1}(\omega t) = \frac{2\sqrt{3}}{\pi} I_0 \sin(\omega t - \alpha)$$

$$I_{a_{rms}} = \sqrt{\frac{2}{3}} I_0$$

$$\mathrm{FF} = \frac{3}{\pi} \approx 0{,}955$$





$$I_{a_1}(\omega t) = \frac{2\sqrt{3}}{\pi} I_0 \sin(\omega t - \alpha)$$

$$I_{a_{rms}} = \sqrt{\frac{2}{3}} I_0$$

$$\mathrm{FF} = \frac{3}{\pi} \approx 0{,}955$$

$$FP = \frac{3}{\pi} \cos\alpha \approx 0{,}955 \cdot \cos\alpha$$

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